Should I Stay or Should I Grow?

How Cities Affect Human Capital

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September 22, 2026

Abstract

Whilst the spatial concentration of skills begets regional disparities, it may increase aggregate human capital. To study this possible equity-efficiency tradeoff, I propose a theory in which cities affect learning by shaping whom individuals meet. Learning complementarities dictate whether the agglomeration of skills increases aggregate human capital. I estimate learning complementarities from a local projection of future skills on the skills of nearby individuals using French administrative data. I find that workers in skill-dense cities experience faster skill growth, especially skilled workers. In the aggregate, local human capital spillovers boost TFP by 5% but generate substantial learning gaps across birthplaces.

JEL Codes: E24, J24, J61, R11, R12, R23.

*Contact: hugo.lhuillier@columbia.edu. This article was born under the supervision of my advisors at Princeton University. I thank Gene Grossman, Richard Rogerson, Esteban Rossi-Hansberg and Gianluca Violante for their support and guidance, and Levi Crews, Jonathan Dingel, and Dávid Nagy for discussing the paper. This paper has also benefited from discussions with Mark Aguiar, Thomas le Barbanchon, Adrien Bilal, Zach Bleemer, Juanma Castro-Vincenzi, Théo Durandard, Mayara Felix, Cécile Gaubert, John Grigsby, Gregor Jarosch, Benny Kleinman, Moritz Lenel, Adrien Matray, Eduardo Morales, Ezra Oberfield, Steve Redding, Karthik Sastry, Andreas Schaab, John Sturm Becko, Derek Wenning, and Owen Zidar. I am grateful to Isaac Shon for excellent research assistant. This research benefited from financial support from the International Economics Section and the Simpson Center for Macroeconomics at Princeton University. Many thanks to Marine Doux and Médianes for hosting the data in France.

1 Introduction

New York, Paris, and Tokyo are epicenters of talent. As workers exchange ideas with, collaborate with, and potentially learn from their neighbors, the spatial concentration of skills may turn these cities into learning centers. At the same time, it may prevent workers in remote regions from accessing the best learning opportunities. Concerned about such regional disparities, governments spend a sizable share of their budget on place-based policies. Yet, little is known about the effects of such policies on human capital. How do cities shape human capital? To what extent are learning opportunities spatially concentrated, and how much does their concentration amplify regional disparities? Can place-based policies boost productivity while lowering geographical inequalities?

This paper offers answers to these questions with three contributions. First, I propose a theory that sheds light on how the spatial concentration of skills affects human capital accumulation. Second, I estimate the impact of cities' skill composition on human capital accumulation using French administrative data. Third, I use the estimated model to quantify the consequences of local human capital spillovers for aggregate productivity and spatial disparities.

I start by developing a theory of learning across space. Locations differ in productivity. Individuals enter the labor force with heterogeneous skills and work for two periods. Skills and local productivity are complements in production. Human capital spills over across workers when they interact. Interactions take place within cities. Individuals are forward-looking and choose where to work to maximize their lifetime utility.

How much learning follows from an interaction depend on the skills involved in the meeting. I impose minimal restrictions on the skill complementarities in the learning process. The learning technology may feature within-skill complementarities if skilled individuals disproportionately learn from other skilled peers, or between-skill complementarities if the marginal gains from meeting a high skill worker decrease in the individual's own skill.

I use the framework to underscore the importance of learning complementarities for the consequences of local human capital spillovers. First, learning complementarities determine the existence of a tradeoff between regional disparities and aggregate human capital. Cities' skill composition shapes local skill growth to the extent that workers learn from each other.

However, cities affect the aggregate stock of human capital only in the presence of learning complementarities. When human capital spills over uniformly across workers, geographic learning differences cancel out in the aggregate. By contrast, when within-skill complementarities prevail, a greater concentration of skills brings skilled workers closer to each other, thereby increasing the likelihood of the most productive interactions, and with that, aggregate human capital. If between-skill complementarities are strong however, such concentration hinders the ability of low- and high-skill workers to meet, and depresses aggregate human capital.

Second, learning complementarities govern whether local interactions accentuate the agglomeration of skills in productive cities. Workers choose between cities based on both the contemporaneous wages they pay and how their learning opportunities affect future earnings. Skilled workers concentrate in productive locations as they offer relatively higher wages. Within-skill learning

complementarities raise the skilled workers' incentives to co-locating with other skilled workers, thereby increasing the spatial concentration of skills.

Third, learning complementarities determine if the laissez-faire economy features too much spatial inequality. The equilibrium is inefficient as workers do not internalize their impact on others' learning. When high-skill workers learn relatively more from each other, low-skill workers crowd out the local learning opportunities in productive cities. Too little skill concentration follows. The reverse occurs when between-skill complementarities are relatively stronger, as skilled workers do not consider how their location choices help workers with other skills learn. In either case, the optimal allocation is decentralized by place-based policies that vary by skill.

I add three additional sources of heterogeneity to suit the model for empirical applications. First, workers face migration costs. Their birthplace thus conditions their lifetime learning opportunities. Second, workers are endowed with learning abilities that affect their human capital accumulation throughout their careers. Third, cities differ in learning fundamentals that shape the local skill growth independently of their skill composition (e.g., their industrial composition).

The second part of the paper uses French administrative data to estimate the model. I start by documenting several facts about the geography of wage growth. I use commuting zones (CZs) as the geographic unit. In the cross-section, within-worker wage growth is faster in CZs where wages are higher. The correlation with local wages is sizeably larger than the correlation with city size, and is stronger for workers who initially earn higher wages. Differences in industry and occupation composition do not explain this relationship. Likewise, the positive correlation remains when comparing workers employed in different CZs within the same firm, or when controlling for the worker composition of their establishment.

I use spatial variation in wage growth to identify learning complementarities. I estimate the learning technology using a model-consistent local projection of future skills on current skills and the skill composition of the city of employment. I allow the two to interact flexibly to capture within- and between-skill complementarities.

There are three identification challenges. First, skills are unobserved, while wages reflect skills and local productivity. I separately identify the two by focusing on the oldest workers. Workers nearing the end of their careers no longer learn. Conditional on where they previously lived, the sensitivity of their location choices to local wages informs the extent of sorting in the economy. I then recover local TFPs from wages after accounting for the selection of skilled workers, and infer workers' skills for all ages from wages net of local productivity.

Second, workers with greater learning abilities may sort into fast-learning locations where they reap greater benefits from the local opportunities. I address this selection problem by instrumenting workers' locations with their birthplaces. In the presence of migration costs, birthplace indeed predicts subsequent location choices. The exclusion restriction requires that learning abilities are on average equal across birthplaces.

Third, local skill growth reflects both human capital spillovers and unobserved learning fundamentals. I isolate the former by instrumenting local skill densities with variation in demand for managers and professionals across industries. I interact local industry employment shares with

aggregate industry-level managers employment shares. This shift-share instrument is valid to the extent that the industry demand for managers and professionals is orthogonal to the local learning fundamentals (Borusyak et al., 2022).

I estimate large local human capital spillovers and strong within-skill learning complementarities. Over five years, the average worker experiences skill growth 4 log points faster when working in the most skill-dense city, Paris, relative to an average skill growth of 5.7 log points. Workers in the top quintile of the skill distribution enjoy dynamic gains twice as large. Selection into fast-learning locations is quantitatively important. Without the birthplace instrument, the Paris skill growth premium for the average worker would be three times larger.

I assess the exclusion restriction by controlling for local school quality and average skills by birthplace, two proxies for systematic differences in learning abilities. The estimates are unchanged. I further show that higher moments of the learning ability distribution are only weakly correlated across workplaces and birthplaces, suggesting that the intergenerational transmission of learning abilities is of limited concern.

I test the model through two overidentification exercises. First, I show that the learning technology produces increasing and concave lifecycle profiles of average skills and skill dispersion, as in the data. Second, in the data, the location choices of new entrants are the most sensitive to cities' skill composition, but older workers account for the bulk of the between-city variation in skills. The model reconciles these patterns: while most sorting occurs upon entry into the labor force, faster learning in skill-dense places magnifies these differences throughout the lifecycle.

In the third part of this paper, I use the estimated model to quantify the consequences of local human capital spillovers on aggregate productivity and spatial inequalities. To do so, I compare the baseline equilibrium, in which individuals learn from their neighbors, with a counterfactual economy in which they learn from workers throughout the country. I find the existence of a steep equity-efficiency tradeoff.

Local spillovers explain the majority of the learning differences across CZs. Fast growth in high-wage locations may result from three factors: selection of fast learners, high learning fundamentals, or large spillovers. When shutting down the third channel, the between-city skill growth standard deviation halves. Local spillovers are disproportionately important for skilled locations and skilled workers due to strong within-skill complementarities. Absent learning complementarities, local spillovers explain 31.8% of the between-city dispersion in skill growth and overestimate the importance of local learning fundamentals.

Local spillovers have substantial positive consequences on aggregate human capital and productivity. The average skill is 2.7% higher under local spillovers. These spillovers benefit workers of all skills. Holding fixed the number of skilled workers in the economy, local spillovers slow the growth of low-skill workers as they lack access to fast-learning opportunities. Yet, in general equilibrium, their growth also accelerates because there are more high-skill workers from whom to learn. Aggregate TFP rises beyond aggregate human capital (5%) as local interactions strengthen the sorting of skilled workers into productive places. Within-skill learning complementarities explain close to all of the productivity gains: assuming uniform spillovers, the average skill increases by 0.8%.

Local spillovers also generate large learning, wage, and welfare disparities. The expected lifetime skill growth of workers born in Paris is 11 percentage point (p.p.) faster than that of the median worker. Furthermore, learning gaps follow gravity: doubling the distance from birthplace to Paris lowers lifetime skill growth by 1.1 p.p., as greater distance raises the cost of moving to the capital. The lack of skilled interactions explains the learning inequalities. Absent local spillovers, the Paris birth premium falls to 3 p.p., and the gravity factor to 0.1 p.p. Local interactions generate particularly strong spatial inequalities amongst skilled workers for whom the lack of access to skilled peers is highly detrimental. Large gaps in lifetime wages and welfare follow.

Altogether, the paper provides a quantified theory of local human capital spillovers, documenting their central importance for aggregate productivity and the steep tradeoff they generate between aggregate efficiency and regional inequalities.

Related literature. This paper adds to four strands of the literature. The first studies how human capital spillovers shape growth. Initiated by the seminal work of Lucas (1988) and Jovanovic and Rob (1989), and later developed by Lucas and Moll (2014), Perla and Tonetti (2014), and Akcigit et al. (2018), amongst many others, this literature initially abstracted from space.

The idea that cities facilitate knowledge diffusion is, of course, not novel (Marshall, 1890; Jacobs, 1969).[1] If knowledge diffuses through interactions and individuals primarily exchange with those around them, the spatial distribution of talent must matter for human capital accumulation. Two papers have recently quantified this insight. Martellini (2022) finds that local peer effects explain the majority of the faster lifecycle wage growth of large cities in the United States. Crews (2023) builds a dynamic model in which the vibrancy of cities, defined by their size and skill composition, shapes how workers learn. He concludes that relaxing land-use regulations in New York and San Francisco spurs growth.[2] I contribute to this literature by studying the importance of learning complementarities for how cities affect human capital. Theoretically, I show that the spatial segmentation of learning opportunities may not always be beneficial, an insight already noted by Benabou (1993). Quantitatively, I find that within-skill complementarities substantially amplify the aggregate productivity effects of local human capital spillovers.

The second strand of the literature estimates human capital spillovers between neighbors and coworkers. Across space, Glaeser and Maré (2001), Baum-Snow and Pavan (2012), de la Roca and Puga (2017), Bilal and Rossi-Hansberg (2021), and Eckert et al. (2022), establish that lifecycle wage profiles are steeper in larger cities. Within firms, Nix (2020) and Jarosch et al. (2021) document that workers employed in high-wage firms experience faster wage growth. I add to this literature in three ways. First and foremost, I estimate that skilled workers benefit disproportionately from local human capital spillovers. Second, I find that cities' skill composition, rather than their size, primarily shapes these spillovers. Third, I show that space matters for growth even after controlling for firm and industry composition.

[1] Since then, a long list of papers has documented how proximity shapes knowledge diffusion (e.g. Jaffe et al., 1993; Audretsch and Feldman, 2004; Moretti, 2021; Atkin et al., 2022).

[2] A related literature focuses on the joint evolution of city sizes and aggregate growth (Rossi-Hansberg and Wright, 2007; Herkenhoff et al., 2018; Hsieh and Moretti, 2019; Duranton and Puga, 2023).

Third, I quantify how learning within cities shapes the spatial distribution of economic activity. Glaeser (1999) establishes that a greater ability to learn by imitation increases urbanization, while Davis and Dingel (2019) shows that local learning interactions induce spatial sorting in the presence of within-skill learning complementarities. I quantify both channels and find them to be large. In particular, local human capital spillovers amplify the spatial concentration of skills, a pattern documented by Combes et al. (2008), Diamond (2016), and Card et al. (2025), among others.

Finally, this paper adds to the literature on spatial policies (see Kline and Moretti, 2014 for surveys). Rossi-Hansberg et al. (2019) and Fajgelbaum and Gaubert (2020) study optimal policy in the presence of worker sorting and static production spillovers, followed by Donald et al. (2025), which derive dynamic optimal spatial policies under static spillovers. I characterize optimal spatial policies when the spillovers are dynamic, and show that within-skill learning complementarities lead to a tradeoff between higher productivity and greater spatial skill concentration.

The rest of this paper is organized as follows. Section 2 presents the theory and characterizes the equilibrium. Section 3 turns to the French matched employer-employee data to document the geography of wage growth. Section 4 uses the French data to estimate the model. Section 5 quantifies the consequences of local spillovers, and Section 6 concludes.

2 A Theory of Learning across Space

2.1 Setup

The structure is an overlapping generations model. There are three types of agents: workers, firms, and landlords. Individuals work for two periods, denoted young (y) and old (o). Old workers are replaced by new entrants upon exit, and the total mass of workers is constant at two. Workers interact with each other within L locations, indexed by ℓ. Time is discrete. There are no aggregate shocks, and I focus on the steady-state equilibrium.

Preferences. Individuals consume a freely traded good taken as the numéraire and local housing. They have linear utility over the numéraire, while housing is a necessity good: workers must rent α housing units at price p_ℓ to live in ℓ. They supply one unit of labor inelastically per period, so that wages and incomes are interchangeable. They do not have access to a saving device, which is without loss given risk neutrality.

Individuals choose in which city to work throughout their lives subject to idiosyncratic location preferences, denoted $\boldsymbol{\varepsilon} = \{\varepsilon_\ell\}_{\ell=1}^{L}$. These preferences are drawn from independent Gumbel distributions with city-specific location parameters B_ℓ and scale parameter ϑ^{-1}. The location parameter B_ℓ reflects the amenities offered by city ℓ, which I take as given. The smaller ϑ, the more dispersed idiosyncratic preferences are and the more they matter for location choices. Location preferences are drawn in both periods and are independent of each other.

Consumption and location preferences are perfect substitutes: the flow utility of a worker with

income y and preferences $\boldsymbol{\varepsilon}$ when working in city ℓ is $y - \alpha p_\ell + \varepsilon_\ell$.[3] Over time, workers discount the future at rate β, and old workers do not express altruism towards future generations.

Human capital. Young workers are endowed with an initial skill $s_y \in \mathbb{R}_+$, drawn from the skill distribution N^y. The young skill distribution is a primitive of the model. The support of N^y is connected, positive, and bounded, $\mathcal{S}^y = [\underline{s}_y, \bar{s}_y]$, with $\underline{s}_y \geq 0$ and $\bar{s}_y < \infty$.

Workers learn in their youth, shaped by the interactions they encounter. A worker with skill s_y meeting a partner with skill s_p obtains new skill $s_o = e\gamma(s_y, s_p)$. The learning technology γ summarizes skill complementarities in the learning process. In addition, workers experience an idiosyncratic learning shock, e, which captures other sources of learning (e.g., enrollment in a training program). These shocks are drawn from the distribution F with support $\mathbb{R}_+$.[4] I impose minimal restrictions on the learning technology.

Assumption 1 (Learning technology)**.**
The learning technology γ is positive, bounded, and twice differentiable in (s_y, s_p).

Assumption 1 allows my framework to nest the prevalent theories of human capital accumulation. First, it encompasses cases where human capital does not spill over across workers: $\gamma(s_y, s_p) = g(s_y)$ for some function g. In this case, learning may still vary across skills, e.g., if skilled workers are better at learning-by-doing (Huggett et al., 2011). Second, human capital may spill over uniformly across workers: $\gamma(s_y, s_p) = g(s_y) + h(s_p)$, for another function h. Then, the learning technology does not feature any complementarity. By contrast, within-skill complementarities may be stronger than between-skill complementarities, in which case the learning technology is supermodular:

$$\frac{\partial^2 \gamma(s_y, s_p)}{\partial s_y \partial s_p} > 0, \quad \forall (s_y, s_p).$$

Supermodularity requires the gains from a skilled interaction to be greater for skilled individuals, which occurs when workers with similar skills are better able to learn from each other (e.g., Jovanovic, 2014). Alternatively, between-skill complementarities may dominate if knowledge flows from skilled to relatively less skilled individuals (Lucas and Moll, 2014). Then, the technology is submodular,

$$\frac{\partial^2 \gamma(s_y, s_p)}{\partial s_y \partial s_p} < 0, \quad \forall (s_y, s_p).$$

The simple learning technology

$$\gamma(s_y, s_p) = g_1 \times s_y + g_2 \times s_p + g_{12} \times s_y \times s_p \tag{1}$$

exemplifies the relative strength of within- and between-skill complementarities. While g_1 governs

[3] The combination of linear utility, housing necessity, and additive idiosyncratic preferences implies that marginal utilities are constant across locations *as if* markets were complete. Optimal spatial policies are thus not dictated by redistributive concerns (Mongey and Waugh, 2024).

[4] Learning shocks smooth the aggregate skill distribution of old workers and thus guarantee its existence. Regularity conditions on F and N^y are imposed in Assumption 2.

the impact of workers' own skill on their learning and g_2 determines whether spillovers stem from skilled workers, the complementarities are determined by g_{12}. The technology is supermodular if $g_{12} > 0$ and submodular if $g_{12} < 0$.

Learning interactions are governed by three assumptions. First, interactions are spatially segmented: they occur between individuals in the same location. Second, interactions are random, capturing the serendipity inherent in learning. Third, the matching function exhibits constant returns to scale. Together, these assumptions imply that the likelihood of interacting with a worker of skill s_p in city ℓ is given by the local density of this skill, $\pi^y_\ell(s_p)$. For now, I further assume that interactions only take place amongst young workers to simplify the exposition.

Local interactions microfound the link between a city's skill composition and its human capital spillovers. An alternative specification is to suppose that the future skill of a worker with skill s_y in location ℓ is $s_o = eZ_\ell(s_y)$, where $Z_\ell(s)$ summarizes the local spillovers:

$$Z_\ell(s) = \int \gamma(s, s_p)\pi^y_\ell(s_p)ds_p. \tag{2}$$

In this reduced-form approach, γ is a kernel that dictates how the local density of a particular skill feeds into the learning of others. This kernel may capture that workers with different skills are not as likely to interact, e.g., because of within-city segregation. The model's predictions and its estimation are robust to this alternative. As such, assuming random interactions is without loss of generality as long as the effective likelihood interactions is considered policy-invariant.

In both cases, learning takes time, and the new skills are used later in the lifecycle. The skill distribution of old workers, N^o, reflects the learning of every young worker.

Production. Production of the numéraire takes place within cities. Each city has a representative competitive firm with productivity T_ℓ, which reflects the industries and firms present there.[5] The firm combines the local workers $\boldsymbol{n}_\ell = \{n_\ell(s)\}_s$ according to the constant returns to scale technology

$$Y_\ell(\boldsymbol{n}_\ell) = T_\ell \int sn_\ell(s)ds.$$

This technology has two properties. First, TFP and skills are complements: skilled workers produce relatively more in productive cities than other workers. Second, skills are perfect substitutes. Production complementarities have been the focus of the spatial literature thus far (e.g., Eeckhout et al., 2014; Giannone, 2022). I abstract away from them to isolate the role of learning complementarities.

The labor markets are segmented by cities and skills. Free entry pins down wages, $w_\ell(s) = sT_\ell$, which inherit the complementarity between TFP and skill.

Landlords. In every location, landlords produce housing services with a Cobb-Douglas technology $H_\ell = K^\phi_\ell \bar{H}^{1-\phi}$, where K_ℓ is a numéraire input and $\bar{H}$ is a fixed factor, such as land. Landlords use their profits to consume the numéraire.

[5] For the equilibrium to be scale invariant, I normalize the dispersion of the idiosyncratic location preferences by the (unweighted) average city TFP, $\vartheta = \theta/\mathbb{E}[T_\ell]$, where θ denotes the scale-invariant dispersion.

Feasibility. The spatial allocation of workers is feasible if the number of workers across locations is not greater than the total number of workers available in the economy:

$$\sum_{\ell} n_{\ell}^{a}(s) \leq n^{a}(s), \quad \forall\, s,\ \forall\, a \in \{y, o\}, \tag{3}$$

where $n_{\ell}^{a}(s)$ is the measure of workers with age a and skill s in city ℓ, and $n^{a}(s) = dN^{a}(s)$ is the aggregate density of skill s for workers with age a. Accordingly, the density of skill s in city ℓ amongst workers with age a is

$$\pi_{\ell}^{a}(s) = \frac{n_{\ell}^{a}(s)}{N_{\ell}^{a}}, \tag{4}$$

where $N_{\ell}^{a} = \int n_{\ell}^{a}(s)ds$ is the mass of workers with age a employed in ℓ.

2.2 The Steady State Equilibrium

Five equations govern the steady-state equilibrium.

Old workers. Old workers exit the labor force in the next period and thus solve a static location choice problem. They choose a city that maximizes their present utility conditional on their skill s_o and location preferences $\boldsymbol{\varepsilon}$:

$$V^{o}(s_o, \boldsymbol{\varepsilon}) = \max_{\ell} \left\{ w_{\ell}(s_o) - \alpha p_{\ell} + \varepsilon_{\ell} \right\}.$$

Idiosyncratic preferences ensure that each skill is present in every location. The number of old workers with skill s_o employed in city ℓ is

$$n_{\ell}^{o}(s_o) = n^{o}(s_o) \left(\frac{\mathrm{e}^{\vartheta(w_{\ell}(s_o) - \alpha p_{\ell} + B_{\ell})}}{\sum_{\ell'} \mathrm{e}^{\vartheta(w_{\ell'}(s_o) - \alpha p_{\ell'} + B_{\ell'})}} \right). \tag{5}$$

The first term captures the aggregate skill supply amongst old workers. The second is the probability to work in city ℓ, derived from the Gumbel preferences. All else equal, old workers are more likely to work in cities that offer high wages, high amenities, and low housing prices.

Young workers. Young workers solve a dynamic location choice problem,

$$V^{y}(s_y, \boldsymbol{\varepsilon}) = \max_{\ell} \left\{ w_{\ell}(s_y) - \alpha p_{\ell} + \varepsilon_{\ell} + \beta \int \int \mathcal{V}^{o}[e\gamma(s_y, s_p)] dF(e) \pi_{\ell}^{y}(s_p) ds_p \right\},$$

where $\mathcal{V}^{o}(s) \equiv \mathbb{E}[V^{o}(s, \boldsymbol{\varepsilon}') \mid s]$ is the expected utility of an old worker with skill s integrated across the future taste shocks. In addition to the considerations of old workers, young workers also consider where the best learning opportunities are. The expected future utility from working in city ℓ aggregates the value of each possible interaction, $\mathcal{V}^{o}[e\gamma(s_y, s_p)]$, weighted by the likelihood of such an interaction, $\pi_{\ell}^{y}(s_p)$. A location constitutes an attractive learning center if it offers a large density

of individuals from whom to learn. When human capital does not spill over across workers, this dynamic motive disappears.

As for old workers, the number of young workers with skill s_y employed in city ℓ follows from the Gumbel location preferences and the feasibility condition (3),

$$n_\ell^y(s_y) = n^y(s_y)\left(\frac{\mathrm{e}^{\vartheta(w_\ell(s_y)-\alpha p_\ell+B_\ell+\beta O_\ell(s_y))}}{\sum_{\ell'}\mathrm{e}^{\vartheta(w_{\ell'}(s_y)-\alpha p_{\ell'}+B_{\ell'}+\beta O_{\ell'}(s_y))}}\right), \tag{6}$$

where $O_\ell(s) \equiv \int\int \mathcal{V}^o[e\gamma(s,s_p)]dF(e)\pi_\ell^y(s_p)ds_p$ is the learning option value offered by ℓ. This option value is an equilibrium object that depends on the location choices of the other young workers. When deciding where to work, individuals take the location choices of others as given.

Skill distribution. The skill distribution of old workers reflects the learning of young individuals. The fraction of old workers with skill less than s_o is given by

$$N^o(s_o) = \sum_\ell \int\int\int \mathbb{1}\left\{e\gamma(s_y,s_p) \le s_o\right\} dF(e)n_\ell^y(s_y)\pi_\ell^y(s_p)ds_p ds_y. \tag{7}$$

The spatial allocation of young workers determines the skill distribution of old workers, and thus the aggregate stock of human capital, by shaping the aggregate probability that two skills meet.

Housing markets. Housing markets clear in every location. The local demand for housing is $\alpha\sum_a N_\ell^a$, while the supply is $H_\ell = \bar{H}(p_\ell\phi)^{\frac{\phi}{1-\phi}}$. In equilibrium:

$$p_\ell = \frac{1}{\phi}\left(\frac{\alpha\sum_a N_\ell^a}{\bar{H}}\right)^{\frac{1-\phi}{\phi}}. \tag{8}$$

Housing acts as a congestion force: the larger the city, the more expensive housing is as landlords face diminishing marginal returns.

Equilibrium. Equations (4)–(8) describe the steady state. Given the spatial distribution of human capital spillovers, young workers choose where to work. In equilibrium, local spillovers must be consistent with the location choices of young workers. This allocation feeds into the aggregate skill distribution of old workers, and housing prices adjust to clear the housing markets.

Definition 1 (Steady state equilibrium)**.**
A steady state equilibrium is a spatial allocation of young and old workers, $\{n_\ell^y, n_\ell^o\}_\ell$, local young skill densities, $\{\pi_\ell^y\}_\ell$, a skill distribution of old workers, n^o, and housing prices $\{p_\ell\}_\ell$, such that:

1. *Given the local skill densities, the spatial allocation of young workers satisfies* (6)*;*
2. *Given the aggregate distribution of old workers, their spatial allocation satisfies* (5)*;*
3. *The local skill densities are consistent with the location decisions of young workers* (4)*;*
4. *The skill distribution of old workers satisfies* (7)*;*
5. *Housing prices satisfy the market clearing condition* (8)*.*

The steady state is given by an infinite-dimensional, nonlinear fixed point in the local skill densities and housing prices. Existence of an equilibrium is guaranteed by the idiosyncratic location preferences, which smooth out location choices. As is standard in spatial models, there exists a unique steady state if the dispersion forces dominate the agglomeration forces (Redding and Rossi-Hansberg, 2017). In the present framework, the dispersion forces scale with the taste shock dispersion ϑ^{-1}. Hence, there exists a unique steady state when ϑ is small enough.

Proposition 1 (Existence and uniqueness)**.**
A steady state equilibrium exists. If ϑ is small enough, there exists a unique steady state.

Proof. See Appendices D.1 and D.2. □

I characterize the steady-state equilibrium in three steps. First, I study how cities impact human capital accumulation. Second, I derive how local interactions generate spatial agglomeration. Third, I compare the laissez-faire equilibrium with the optimal allocation.

2.3 The Human Capital Consequences of Local Spillovers

Cities shape learning as they bring together workers who learn from one another. For a worker with initial skill s, the difference in their average skill growth rate between two locations is

$$\mathbb{E}_\ell \left[\frac{S'}{s} \mid s \right] - \mathbb{E}_{\ell'} \left[\frac{S'}{s} \mid s \right] = \int \int e \frac{\gamma(s, s_p)}{s} \left(\pi_\ell(s_p) - \pi_{\ell'}(s_p) \right) dF(e) ds_p. \tag{9}$$

Cities thus impact learning under two conditions. First, interactions must shape human capital accumulation. Second, cities must differ in their skill composition. Residents primarily interact with skilled peers in locations where they are relatively abundant. In these places, faster learning thus takes place to the extent that human capital flows from skilled workers. Spatial learning differences need not be uniform across workers depending on the relative importance of within-skill and between-skill learning complementarities. Either way, while skill composition matters, city size does not, since the matching function features constant returns to scale.

Do the geographic learning differences impact aggregate human capital? The average skill of old workers is

$$\mathbb{E}[S^o] = \sum_\ell \int \int \int e \gamma(s_y, s_p) n_\ell^y(s_y) \pi_\ell^y(s_p) dF(e) ds_p ds_y. \tag{10}$$

This stock of human capital can be compared to an economy in which cities do not shape interactions. In that counterfactual economy, the likelihood of learning from a worker with skill s_p is given by its aggregate skill density, $n^y(s_p)$. The average skill of old workers is then

$$\mathbb{E}[\bar{S}^o] = \int \int \int e \gamma(s_y, s_p) n^y(s_y) n^y(s_p) dF(e) ds_p ds_y. \tag{11}$$

The difference between (10) and (11) reveals the macroeconomic consequences of local human capital spillovers for aggregate human capital:

$$\mathbb{E}[S^o] - \mathbb{E}[\bar{S}^o] = \int\int \gamma_{12}(s, s_p)\mathbb{C}\text{ov}^N\left(P_\ell(s), P_\ell(s_p)\right) ds ds_p, \tag{12}$$

where $P_\ell(s) \equiv \int_s^{\bar{s}} \pi_\ell(\sigma) d\sigma$ is the survival function —the probability of interacting with skills greater than s. The covariance is taken across locations, weighted by the local employment share N_ℓ^y.

Cities therefore affect aggregate learning by shaping how often complementary interactions occur. Absent learning spillovers, $\gamma(s_y, s_p) = g(s_y)$, cities are trivially irrelevant. However, cities also have no impact on aggregate learning if human capital flows uniformly across skills, $\gamma(s_y, s_p) = g(s_y) + h(s_p)$. In that case, local spillovers are zero-sum. If faster learning takes place in some places thanks to exposure to fast learning interactions, workers elsewhere necessarily learn more slowly because they are deprived of those same interactions. The losses exactly offset the gains.

More generally, learning complementarities dictate how the spatial concentration of skills affects aggregate human capital. If workers sort across locations according to their skills, cities increase the chance that human capital flows between similar workers. When within-skill complementarities are strong, the spillovers are the strongest when they occur amongst peers. In such cases, the spatial concentration of skilled individuals boosts aggregate human capital. By contrast, if between-skill complementarities dominate, cities reduce the average skill of the economy by pushing low and high skill workers apart and preventing human capital from flowing between them.

Proposition 2 (Aggregate human capital)**.**
Suppose the local skill distributions can be ordered by first order stochastic dominance; that is, there exists a relabeling such that $\pi_1 \succeq_{\text{FOSD}} \pi_2 \succeq_{\text{FOSD}} \cdots \succeq_{\text{FOSD}} \pi_L$*. If the learning technology is supermodular,* $\mathbb{E}[S^o] > \mathbb{E}[\bar{S}^o]$*, while submodularity implies* $\mathbb{E}[S^o] < \mathbb{E}[\bar{S}^o]$*.*

Proof. Under the FOSD ordering, $\ell \to P_\ell(s)$ is monotone for all s. Hence, $\mathbb{C}\text{ov}^N\left(P_\ell(s), P_\ell(s_p)\right) \geq 0$, and $\mathbb{E}[S^o] - \mathbb{E}[\bar{S}^o]$ inherits the sign of γ_{12} from (12). □

Learning complementarities therefore determine the tradeoff, or lack thereof, between spatial inequalities and human capital accumulation. Consider a policy that reduces spatial inequalities by reallocating skilled workers from a skill-abundant location to a skill-scarce one, while keeping the size of both places constant. By bringing low and high skill workers closer to each other, the policy boosts aggregate human capital in the presence of strong between-skill complementarities. By contrast, by weakening the spatial agglomeration of skilled workers, the policy lowers aggregate human capital if within-skill complementarities are dominant.

Proposition 3 (Equity-efficiency tradeoff)**.**
Consider two feasible allocations $\cdot$ *and* $\cdot'$*. Suppose that skills are less spatially concentrated under the second: there exist two locations,* ℓ *and* ℓ'*, such that (i)* $N_l = N_l'$ *for* $l \in \{\ell, \ell'\}$*, (ii)* $P_{\ell'}(s) - P_\ell(s) \geq P'_{\ell'}(s) - P'_\ell(s) \geq 0$ *for all* s*, and (iii)* $(N_l, \pi_l) = (N_l', \pi_l')$ *for all* $l \notin \{\ell, \ell'\}$*. If the learning technology is supermodular, then* $\mathbb{E}[S^{o'}] < \mathbb{E}[S^o]$*. If the technology is submodular, the inequality is reversed.*

Proof. See Appendix A.1. □

Propositions 2 and 3 hold to the extent that some cities are more skill-abundant than others. Sorting need not be pure: low and high skill workers can coexist within each city. Likewise, sorting could arise from multiple forces. High skill workers may prefer the amenities offered by large cities à la Diamond (2016). Nonhomothetic demand could generate skill concentration (e.g., Hoelzlein, 2019). Alternatively, workers may sort according to productive comparative advantage. In that case, comparative advantage could be endogenous or exogenous, and the propositions hold regardless of the production technology.

Nonetheless, cities' skill composition is not exogenous and may itself be affected by the local learning opportunities. It thus remains an open question whether sorting is compatible with local human capital spillovers and, if so, how the two interact. Furthermore, while Proposition 3 establishes a possible equity-efficiency tradeoff, it is silent on the optimal policy. To answer these questions, I leverage the full structure of the model.

2.4 The Spatial Consequences of Local Human Capital Spillovers

I characterize analytically the spatial allocation by focusing on an approximate equilibrium. In the absence of restrictions on the learning technology, the steady state equilibrium depends on the entire skill distribution of every city. To gain tractability, I approximate the equilibrium around the point where cities have homogeneous TFP, $\log T_\ell \approx \log \bar{T}$ for all ℓ and some $\bar{T}$. When cities share a common TFP, the production complementarities vanish. Skilled workers no longer have an additional incentive to move to high TFP cities, and there exists an equilibrium in which all locations have the same skill composition (Proposition D.1). Approximating the equilibrium around homogeneous TFPs thus simplifies how to keep track of the local skill densities. For ease of exposition, I focus in the main text on an economy with homogeneous amenities. The general expressions are in the appendices, and the propositions apply to the general setup.

I start by characterizing the location decisions of old workers to contrast them with those of the young. To first order, the likelihood that an old worker with skill s_o works in city ℓ is

$$\frac{n^o_\ell(s_o)}{n^o(s_o)} \approx \underbrace{\frac{1}{L}}_{\text{Symmetric}} + \underbrace{\frac{\theta}{L}\log\left(\frac{T_\ell}{\bar{T}}\right)s_o}_{\text{Sorting for working}} - \underbrace{\frac{\theta\lambda}{L}\log\left(\frac{T_\ell}{\bar{T}}\right)}_{\text{Housing congestion}}, \tag{13}$$

where $\theta = \vartheta\bar{T}$, and $\lambda \equiv \frac{\vartheta\alpha\frac{1-\phi}{\phi}}{1+\vartheta\alpha\frac{1-\phi}{\phi}}$ is the TFP pass-through into housing costs.[6] Productive cities offer higher wages and thus attract more workers. Housing prices rise, which dampens agglomeration according to the pass-through λ. The between-city wage gaps are wider for skilled workers due to the production complementarities. High-skill workers are therefore relatively more present in productive cities: $T_\ell > T_{\ell'}$ implies $\pi^o_\ell \succ_{\text{FOSD}} \pi^o_{\ell'}$. Nevertheless, idiosyncratic location preferences imply that

[6]The first order approximation is detailed in Appendix D.3, and the second order approximation in Appendix D.5.

some low-skill workers also live there. The dispersion in idiosyncratic location preferences, θ, pins down the local labor supply elasticity and the amount of sorting.

The location decisions of young workers differ in that they also care about local learning opportunities. To first order, the probability that a young worker with skill s_y chooses location ℓ is

$$\frac{n_\ell^y(s_y)}{n^y(s_y)} \approx \underbrace{\frac{1}{L}}_{\text{Symmetric}} + \underbrace{\frac{\theta}{L}\log\left(\frac{T_\ell}{\bar{T}}\right)s_y}_{\text{Sorting for working}} + \underbrace{\frac{\theta^2\beta}{L}\log\left(\frac{T_\ell}{\bar{T}}\right)\Theta(s_y)}_{\text{Sorting for learning}} - \underbrace{\frac{\theta\lambda}{L}\log\left(\frac{T_\ell}{\bar{T}}\right)}_{\text{Housing congestion}}. \tag{14}$$

The third, new term captures the learning considerations. The function $\Theta(s_y)$ summarizes how much young workers value the learning opportunities of marginally more productive cities. When $\Theta(s_y) > 0$, these opportunities are attractive and therefore increase the supply of s_y in productive locations. When $\Theta'(s_y) > 0$, skilled workers disproportionately value these learning opportunities, intensifying their concentration in space. The total willingness of young workers to sort combines the working and learning motives, $\eta(s_y) \equiv s_y + \theta\beta\Theta(s_y)$.

How attractive the learning opportunities of productive cities are depends on who works there and how much individuals learn from them:

$$\Theta(s_y) = \int \underbrace{\left(\gamma\left(s_y, s_p\right) - \mathbb{E}\left[\gamma\left(s_y, S^y\right)\right]\right)}_{\text{Relative learning from } s_p} \underbrace{\left(s_p + \theta\beta\Theta(s_p)\right)}_{\substack{\text{Willingness to} \\ \text{sort of } s_p}} dN^y(s_p). \tag{15}$$

The probability of learning from skill s_p in a particular city depends on how willing that worker is to work there and, therefore, on how they value the local spillovers. Formally, $\Theta(s_y)$ solves an infinite dimensional fixed point problem. The first order approximation simplifies this fixed point by making it linear. I show in Lemma D.3 that the integral equation (15) has a unique solution when either the dispersion forces are large enough or the spillovers are not too strong.

The strength of the between-skill learning complementarities determines whether the young allocation also exhibits sorting. Absent learning, young workers sort across cities in the same manner as the old. Learning partially offsets the production complementarities if low-skill workers disproportionately value the learning opportunities of productive cities, but it need not offset them provided the learning incentives are not too strong.

Proposition 4 (Sorting)**.**
Suppose that $\theta\beta\mathbb{C}ov[\gamma_1(s, S^y), f(S^y)] > -1$ *for all* $s \in \mathcal{S}^y$ *and every function* f *weakly increasing and satisfying* $\mathbb{S}d[f(S^y)] \leq \mathbb{S}d[\eta(S^y)] < \infty$*. Then,* η *is increasing, and* $T_\ell > T_{\ell'}$ *implies* $\pi_\ell^y \succ_{FOSD} \pi_{\ell'}^y$*.*

Proof. See Appendix A.2 for the proof and an explicit bound on $\mathbb{S}\text{d}[\eta(S^y)]$. □

Proposition 4 effectively bounds the strength of between-skill learning complementarities. For instance, when the learning technology is given by (1), Proposition 4 restricts how submodular the technology can be: $g_{12} \geq -\kappa$, $\kappa > 0$. In particular, submodular learning is compatible with sorting. I assume that Proposition 4 holds for the remainder of the section, consistent with the empirical

evidence found in this paper and in the literature (e.g., Combes et al., 2008; Card et al., 2025).

Under sorting, human capital disproportionately spills over from skilled workers in productive cities. These locations are therefore attractive learning centers to whomever benefit from such interactions. To understand how these interactions shape agglomeration, I return to the comparison with the counterfactual economy in which spillovers are dictated by the aggregate skill density.

Local spillovers shape space in three ways. First, equation (9) implies that skill growth is faster in productive cities whenever workers learn disproportionately from skilled peers. Second, cities that offer faster growth grow in size. In particular, if human capital flows from skilled workers, local spillovers make productive cities larger. Third, local spillovers shape how spatially concentrated skills are. To first order, the average skill of young workers in location ℓ is

$$\mathbb{E}[S_\ell^y] \approx \underbrace{\mathbb{E}[S^y]}_{\text{Symmetric}} + \underbrace{\theta \log\left(\frac{T_\ell}{\bar{T}}\right) \mathbb{V}\text{ar}[S^y]}_{\text{Sorting for working}} + \underbrace{\theta^2 \beta \log\left(\frac{T_\ell}{\bar{T}}\right) \mathbb{C}\text{ov}[S^y, \Theta(S^y)]}_{\text{Sorting for learning}}. \tag{16}$$

When within-skill complementarities dominate, skilled workers have greater incentives to interact with their peers. They therefore place a higher value on the learning opportunities of productive cities, and sorting intensifies. Likewise, skilled workers agglomerate further in productive cities when learning incentives become relatively more important, as captured by a higher discount factor. In both cases, greater between-city wage inequality follows.

Proposition 5 summarizes the spatial consequences of local human capital spillovers.

Proposition 5 (Spatial inequality)**.**
Suppose that $\log T_\ell \approx \log \bar{T}$ *for all* ℓ*. If* $\gamma(s_y, \cdot)$ *is increasing, then* $\Theta(s_y) > 0$*, while supermodularity implies* $\Theta'(s_y) > 0$*. The converse holds when* $\gamma(s_y, \cdot)$ *is decreasing and* γ *is submodular. As a result, comparing economies with and without local human capital spillovers:*

1. *If* $\gamma(s_y, \cdot)$ *is increasing for all* s_y*, skill growth is on average faster in productive cities. The most productive cities are larger. With homogeneous amenities, this relationship extends to all productive cities. The converse holds when* $\gamma(s_y, \cdot)$ *is decreasing.*
2. *If* γ *is supermodular, the average skill of young workers is higher in productive cities, leading to greater between-city wage variance. The converse holds when* γ *is submodular.*

Proof. See Appendix A.3. □

Workers' location choices thus reinforce the sorting patterns favorable to aggregate human capital. Within-skill complementarities induce a greater spatial concentration of skills, which itself fosters aggregate learning. Conversely, low- and high-skill workers are more likely to work in the same city under between-skill complementarities, thereby mitigating the detrimental effects of cities on human capital. Nevertheless, private incentives may still fall short of what is socially desirable.

2.5 Optimal Policies

I consider a utilitarian planner. I assume the planner has homogeneous Pareto weights across skills and cities to abstract from redistributive concerns. As in Rossi-Hansberg et al. (2019) and Donald et al. (2023), I suppose that the planner cannot condition its allocation on idiosyncratic location preferences, either because they are unobserved or because doing so is not politically feasible.

The planner chooses a sequence of spatial allocations of workers, $\{n_{t\ell}^{y\star}, n_{t\ell}^{o\star}\}_{t,\ell}$, together with a sequence of consumption allocations for workers and landlords, $\{c_{t\ell}^{y\star}, c_{t\ell}^{o\star}, c_{t\ell}^{L\star}\}_{t,\ell}$, aggregate skill densities $\{n_{t+1}^{o\star}\}_t$, and intermediate inputs in the housing sector, $\{K_{t\ell}^{\star}\}_{t,\ell}$, to maximize the discounted sum of each cohort's expected lifetime utility,

$$\sum_{\ell} \int \mathbb{E}_\ell[V_0^{o\star}(s_o, \boldsymbol{\varepsilon}) \mid s_o] n_{0\ell}^{o\star}(s_o) ds_o + \sum_{t \geq 0} \beta^t \sum_{\ell} \left\{ \int \mathbb{E}_\ell[V_t^{y\star}(s_y, \boldsymbol{\varepsilon}) \mid s_y] n_{t\ell}^{y\star}(s_y) ds_y + c_{t\ell}^{L\star} \right\}. \quad (17)$$

The function $V_0^{o\star}(s_o, \boldsymbol{\varepsilon})$ is the utility of old workers with skill s_o and preferences $\boldsymbol{\varepsilon}$ in the initial period under the planner's allocation. Likewise, $V_t^{y\star}(s_y, \boldsymbol{\varepsilon})$ is the lifetime utility of young workers in cohort t. The planner does not condition on idiosyncratic preferences and thus maximizes expected utility conditional on location choices. At the same time, the spatial allocation of workers must be consistent with the local supply of each skill given their idiosyncratic preferences. These preferences act as an incentive constraint: to increase the supply of workers in a particular city, the planner must compensate them for their foregone location preferences. Finally, the consumption allocation must not exceed aggregate output, and housing production must adjust to satisfy local housing demand.[7] I focus on the steady state allocation, referred to as the optimal allocation.

Old workers do not exert externalities as they do not affect the learning of the young. The laissez-faire and planner allocations thus coincide conditional on the same young allocation.

In contrast, the allocation of young workers is generally inefficient. Young workers fail to internalize how their location choices impact others' learning. As a result, their incentives to live in specific cities fall short of what is socially desirable. Specifically, there exists a wedge between the optimal and laissez-faire consumption allocations,

$$c_\ell^{y\star}(s_y) - c_\ell^{y}(s_y) = \tau_\ell^{\star}(s_y) - X(s_y), \quad (18)$$

where X is some function independent of ℓ.[8] The wedge $\tau_\ell^{\star}(s_y)$ is

$$\tau_\ell^{\star}(s_y) = \beta \int \int \left(\mathcal{V}^{o\star}[e\gamma(\sigma, s_y)] - \mathbb{E}[\mathcal{V}^{o\star}(e\gamma(\sigma, S_\ell^{y\star}))] \right) dF(e) \pi_\ell^{y\star}(\sigma) d\sigma, \quad (19)$$

where $\mathcal{V}^{o\star}(s) = \mathbb{E}[V^{o\star}(s, \boldsymbol{\varepsilon}) \mid s]$ is the expected utility of an old worker under the optimal allocation. The wedge is positive if workers in location ℓ learn relatively more from skill s_y than from the average local worker. In that case, the laissez-faire equilibrium features too few workers with this skill in that location.

[7] Appendix D.6 presents the planner's problem and derives the first order conditions.

[8] The function X reflects that transfers across skills are not pinned down since utilities are linear in consumption.

Equation (18) implies that the laissez-faire equilibrium is inefficient as long as there are human capital spillovers. Characterizing the spatial misallocation amounts to solving for a nonlinear, infinite dimensional fixed point defined by (18), (19), the local labor supplies, and housing feasibility. As for the laissez-faire equilibrium, I simplify this fixed point by studying the optimal allocation when TFP gaps are small, $\log T_\ell \approx \log \bar{T}$. Appendix D.7 details the approximation.

Local human capital spillovers generate spatial misallocation through two channels. The first affects the entire local labor supply. Conditional on learning from skilled individuals, young workers agglomerate in productive cities. In doing so, they forego their idiosyncratic preferences for less productive places. However, high TFP cities do not offer *ex ante* better learning opportunities. Holding aggregate productivity constant, the planner thus desires to decentralize learning centers and reallocate workers from high to low TFP locations independently of their skills.

The second channel operates through the relative skill supply. When within-skill complementarities are strong, skilled workers benefit disproportionately from their peers. Internalizing these spillovers, the planner increases the supply of skilled workers in productive cities, thereby increasing spatial inequalities. In contrast, low skill workers experience too few spillovers from skilled workers in the laissez-faire equilibrium when between-skill complementarities prevail. The planner corrects this misallocation by reallocating skilled workers towards low TFP cities where low skill are abundant.

Proposition 6 (Optimal allocation)**.**
The laissez-faire equilibrium is efficient if interactions do not shape learning, $\gamma(s, s_p) = g(s)$. Otherwise, the laissez-faire equilibrium is inefficient. Furthermore, when $\log T_\ell \approx \log \bar{T}$ for all ℓ:

1. *If $\gamma(s_y, \cdot)$ is increasing for all s_y, cities that grow from local human capital spillovers are too large in the laissez-faire equilibrium: $N_\ell > \bar{N}_\ell \iff N_\ell^\star < N_\ell$. If amenities are homogeneous, productive cities are too large. The converse holds when $\gamma(s_y, \cdot)$ is decreasing.*
2. *If γ is supermodular, there is too little skill concentration in productive cities: $\mathbb{E}[S_\ell^{y\star}] > \mathbb{E}[S_\ell^y] \iff T_\ell > \bar{T}$. The converse holds if γ is submodular.*

Proof. See Appendix A.4. □

The optimal allocation is decentralized through transfers that vary by place and skill, given by (19). To first order, transfers are decomposed into two components: a city specific tax that corrects the size distortion, and skill-by-city subsidies that generate the optimal amount of sorting.

Taking stock, learning complementarities dictate the consequences of local human capital spillovers for aggregate productivity, geographic disparities, and optimal policies. To quantitatively evaluate the consequences of these spillovers, I add several sources of heterogeneity to the framework.

2.6 Quantitative model

I add three key elements that improve the empirical relevance of the model. I introduce time invariant learning abilities and exogenous local learning fundamentals as other sources of between-city differences in learning. In addition, workers face migration costs, frequently invoked to justify the need for local policies.

Workers. Workers live for A periods, indexed by a. They are born in a location, ℓ_b, inherited from their parents. In addition to their initial skill, workers are endowed with a learning ability, x. The joint density of skills and abilities is denoted χ_b and is identical across birthplaces. Learning abilities are constant over time and shape human capital accumulation: all else equal, the skill growth rate is proportional to x. Learning interactions occur between workers independently of their age and are dictated by the local skill density π_ℓ. Workers give birth to a newborn with probability b_a at age a and retire in $A+1$.

Geography. Workers face migration costs $\kappa_{al\ell}$ to move from l to ℓ at age a. Migration costs vary across the lifecycle, for instance, if older workers have accumulated more local social capital. Moving costs generate ex ante spatial inequality: where workers are born shapes their future workplace, and therefore their human capital, income, and lifetime utility.[9]

Locations also differ in an exogenous learning fundamental, μ_ℓ. These fundamentals capture a range of forces: a location's access to outside knowledge, an industry composition favorable to fast growth, amenities that facilitate interactions, etc. As with learning abilities, the skill growth of a worker in ℓ is proportional to μ_ℓ, all else equal.

Finally, I let local amenities differ across the lifecycle, e.g., because younger and older workers value nightlife activities differently. Age specific amenities are denoted $B_{a\ell}$. I impose no relationship between the various local fundamentals, $\{T_\ell, \mu_\ell, B_{1\ell}, \ldots, B_{A\ell}\}_\ell$.

Functional forms. I impose the following functional forms. First and foremost, I assume a semiparametric learning technology:

$$\log \gamma(s, s_p) = \xi(s) + \varphi(s) s_p. \tag{20}$$

The function ξ determines nonparametrically how a worker's own skill shapes their future growth. The second function, $\varphi(s)$, governs how human capital spillovers vary across the skill distribution. If $\varphi(s) > 0$, a worker with skill s learns relatively more from skilled workers, whereas $\varphi'(s) > 0$ implies that the returns to skilled interactions are greater for skilled workers themselves.[10]

Following Dingel and Tintelnot (2026), I reduce overfitting concerns in the estimation of migration costs by assuming a semiparametric gravity structure,

$$\kappa_{al\ell} = 1\{l \neq \ell\} \zeta_a \left(\tau_l + \delta \log \text{dist}_{l\ell}\right),$$

where $\text{dist}_{l\ell}$ is the distance between l and ℓ. The parameter δ is the distance semi-elasticity, τ_l measures the cost of moving out of l, and ζ_a lets costs vary proportionally across the lifecycle. I normalize $\zeta_A = 1$ for the oldest workers.

Finally, I suppose that the joint distribution of skills and learning abilities is log-normal. I normalize the mean of learning abilities to zero, while the average skill of new entrants is μ_s. Their

[9] Given linear utility, the migration costs can be interpreted in utils or in numéraire units.

[10] The condition $\varphi'(s) > 0$ for all s implies that γ is log supermodular. A sufficient condition for supermodularity is $\varphi(s) \geq 0$, $\varphi'(s) \geq 0$, and $\xi'(s) \geq 0$ for all s. In particular, supermodularity holds if $\varphi'(s) = 0$, $\varphi(s) > 0$, and $\xi'(s) > 0$.

respective standard deviations are denoted σ_x and σ_s, and their correlation by ρ. Likewise, the distribution of learning shocks is log-normal with zero mean and standard deviation σ_e.

I solve the model numerically and globally. Appendix E formally describes the quantitative model and the numerical algorithm. I estimate the model on French data, which I now introduce.

3 The Geography of Wage Growth

3.1 Data

I estimate the model on French matched employer-employee data (DADS Postes).[11] The data consist of repeated two-year panels covering the near universe of employed workers. They provide information on wages, hours worked, and the locations where individuals work and live, along with various demographic characteristics. I restrict the sample to the 2002-2019 window.[12]

I construct a long panel out of this dataset using the matching algorithm of Babet et al. (2025). The resulting dataset has several advantages over other sources used in the literature. First, its long dimension allows me to track workers for more than fifteen years and therefore measure long-term wage growth. Second, outcomes at granular geographical levels are measured precisely thanks to the data's universal coverage, including rare events such as migrations. Third, it is one of the few matched employer-employee datasets that provides information on where individuals work (firm, establishment, and neighborhood) and live.

I focus my analysis on full-time employed workers aged 25 to 55 in metropolitan France. Workers employed in the public sector in France have their wages determined nationally by their tenure, and as such, their wages may not fully reflect their productivity. I therefore keep workers employed in the private sector and exclude the education and health industries due to their large fraction of public servants. I deflate earnings by the aggregate consumer price index. I drop observations with hourly wages below the minimum wage and in the top 0.1% of the hourly wage distribution to minimize measurement errors in hours worked. Finally, I exclude individuals unemployed for more than three consecutive years.

I define a city as a commuting zone (CZ). A commuting zone is a statistical area defined by the French statistical agency. It consists of contiguous municipalities clustered together to reduce commuting flows across them. There are 297 such areas in metropolitan France, which span the whole territory. Commuting zones are the natural geographical unit when thinking of local labor markets. In particular, their relatively large size is well suited to capture all the human capital spillovers a worker may experience. The Paris region is divided into several CZs along administrative borders due to its large size. I aggregate back these CZs and end up with 285 cities. I use the terms commuting zones, locations, and cities interchangeably.

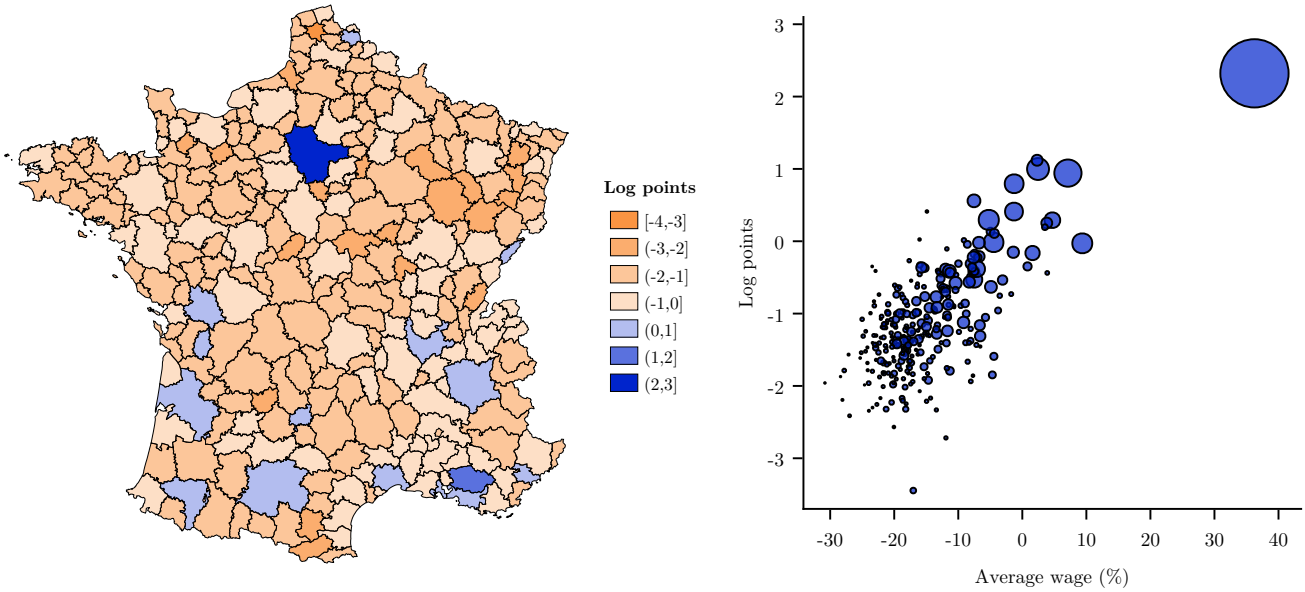
[11] DADS stands for "*Declarations Annuelles de Données Sociales*". The dataset is constructed from employer tax records, which are compiled by the French statistical agency. The Postes file corresponds to the version of the data at the level of employee, year, and employment spell. Appendix H contains more information on the dataset.

[12] Before 2002, the yearly files do not provide a worker identifier to link job spells within a year, and the matching algorithm of Babet et al. (2025) is not applicable. I stop in 2019, before the COVID-19 pandemic.

Figure 1: The geography of wage growth

(a) 5-year wage growth, by CZ

(b) 5-year wage growth, by CZ wage



Data source: DADS Postes, cleaned and aggregated as detailed in Appendix H. Left: average log wage growth at the five-year horizon by CZ. Right: average log wage growth at the five-year horizon against the relative CZ average wage. The size of the markers is proportional to the city size.

Finally, I set a period in the model to five years. I aggregate the data accordingly at the employee and age group level. I compute wages within a period as the ratio of total pre-tax labor earnings to total hours worked. For categorical variables, I keep the value associated with the highest paying job within each five-year window. I restrict the sample to workers observed for at least two periods, and I weight observations to obtain a uniform age distribution.

The final sample contains 13 million workers observed on average for 2.8 periods. The largest location is Paris, with 2.6 million workers, while the median CZ has 17,596 workers. Table I.1 provides further summary statistics.

3.2 Reduced-form evidence

In the model, wage growth is informative about human capital accumulation as it partially tracks skill growth. This section documents several facts about how wage growth varies across space.

There are large differences in lifecycle wage growth across CZs in France. Figure 1 depicts relative five-year wage growth by CZ in France. In Paris, the average wage growth is 2.3 log points faster than the national average. Meanwhile, 94% of cities, containing 63% of the French workforce, feature slower wage growth than in the aggregate. Fast growth is concentrated in cities where wages are high. Figure 1b displays average wage growth against the average city wage. In the cross-section, a one standard deviation increase in wages is associated with 0.6 log points faster growth. These differences capture faster growth along the lifecycle rather than cities diverging from one another. Figure J.1a indeed shows that wage gaps across CZ are the same in the late 2010s as

at the beginning of the 2000s.

I assess whether the local differences in wage growth are explained by the composition of cities by estimating the local projection

$$\log w_{it+1} = \alpha + \beta \bar{w}_{\ell_{it}} + \boldsymbol{\gamma}' \boldsymbol{X}_{it} + u_{it+1}, \tag{21}$$

where i is an individual, ℓ_{it} is the CZ where i works in period t, $\bar{w}_\ell$ is the average wage of location ℓ, and $\boldsymbol{X}_{it}$ is a vector of controls that includes the present log wage of worker i.[13] As such, β captures the conditional correlation between local wages and subsequent lifecycle wage growth.

Table 2 reports this correlation under various sets of controls. First and foremost, wage growth remains faster in high-wage places after controlling for city size, the focus of the literature thus far (column 2). In fact, the correlation between city size and wage growth is three times smaller than that between local wages and wage growth.

Second, growth disparities persist when comparing workers within the same industry and occupation (column 3), as well as workers employed at the same firm but in different CZs (column 4). Column 6 further suggests that space matters beyond the boundary of the firm: the correlation between city wages and future wage growth is of the same order of magnitude as the correlation between establishment wages and future wage growth. Finally, I find a similar correlation when exploiting variations from workers who live in the same neighborhood but are employed in different CZs, suggesting that where individuals work matters more than where they live (column 5).[14]

Third, there is greater variation in wage growth across CZs for skilled workers with high wages. I interact local wages in (21) with dummy variables for the wage quintile in which workers are. I report the estimates in Figure 7a. For workers in the bottom 20% of the wage distribution, the predicted wage growth gap between Paris and the median CZ is 3.3 log points. It rises to 5 log points for workers in the top 20%. Likewise, the predicted wage growth gaps are larger for managers and professionals than for middle- and low-skill occupations (Figure 7b).

Lastly, local wages are positively correlated with other proxies for human capital accumulation. I estimate (21), replacing the dependent variable with a dummy variable that tracks whether workers upgrade their occupation. I find that the likelihood of switching to a higher-paying occupation is 0.9 p.p. higher in locations whose wages are one standard deviation higher (column 7).

There are therefore large differences in wage growth across space, particularly for workers with high wages. I now show that these variations are informative about local human capital spillovers.

4 Estimation

The estimation of the model is block recursive and takes three steps. In the first, I identify local TFPs, along with the dispersion in idiosyncratic preferences and other local fundamentals, from the spatial allocation of the oldest workers. Then, I identify the learning technology from spatial

[13] This local projection closely aligns with the design used to estimate the model. Its main advantage over a specification that projects wage growth on local wages is that it allows for mean reversion in workers' wages.

[14] I define a neighborhood as a municipality, which is comparable to a ZIP code in the United States.

variation in skill growth. Lastly, I set the discount factor exogenously, and I calibrate internally the remaining age-specific parameters by indirect inference. The parameters are reported in Table 3.

4.1 Local fundamentals

Local TFPs and the dispersion in idiosyncratic location preferences are jointly identified from between-city wage gaps and the location choices of the oldest workers. In contrast to standard spatial frameworks, local TFPs cannot be read from wages as wages also reflect workers' skills. Furthermore, accounting for the selection of workers across locations is challenging as workers sort based on local learning opportunities and their ability to benefit from them, both of which are unobserved. I address this challenge by focusing on the oldest workers. As these workers near the end of their careers, they have little incentive to accumulate further human capital, and their location decisions are fundamentally static. The location choices of the oldest workers inform the taste shock dispersion, which lets me pin down TFPs from wages net of predicted skill selection.

Identification. Let $p^A_{l\ell}(s,x)$ denote the probability that an individual with skill s and learning ability x is in location ℓ in their fifties conditional on working in l in the previous period. Since old workers have no incentive to learn, their learning ability is irrelevant: $p^A_{l\ell}(s,x) \equiv p^A_{l\ell}(s)$. Comparing the probability that a worker moves to ℓ relative to staying in their origin location gives

$$\log\left(\frac{p_{Al\ell}(s)}{p_{All}(s)}\right) = \vartheta\left(w_\ell(s) - w_l(s)\right) + \mathrm{FE}_{l\ell}, \tag{22}$$

where

$$\frac{\mathrm{FE}_{l\ell}}{\vartheta} = B_{A\ell} - \alpha p_\ell - (B_{Al} - \alpha p_l) - \tau_l - \delta \log \mathrm{dist}_{l\ell}.$$

To build intuition, suppose skills are observed, and thus so are the log odds ratios. Conditional on an origin location, the propensity of high skill workers to move towards high-wage regions relative to that of low skill workers identifies the dispersion in idiosyncratic preferences, ϑ. The distance elasticity of migration flows pins down δ. Out-migration probabilities identify τ_l. Finally, amenities are obtained from the probability that workers move to a particular destination.[15]

Wage gaps net of worker selection identify local TFPs. The average wage amongst the oldest workers in ℓ is

$$\mathbb{E}\left[\log W^A_\ell\right] = \log T_\ell + \mathbb{E}\left[\log S^A_\ell\right]. \tag{23}$$

Given ϑ and the other local fundamentals, the model predicts how strongly workers sort across space, and thus $\mathbb{E}[\log S^A_\ell]$. Equation (23) then identifies local TFPs as a wage residual.

This argument establishes identification of the local fundamentals conditional on observing skills. In particular, local TFPs are identified independently of local human capital spillovers. As such,

[15]The housing cost α is identified from the housing expenditure share, as explained in Section F.

the dynamic spillovers documented in this paper complement any static agglomeration forces that may explain why some cities are more productive than others.[16]

When skills are unobserved, equations (22) and (23) define a fixed point. Given a guess for TFPs, skills are inferred from wages. I then update the preference dispersion, amenities, and migration costs based on workers' location choices. Finally, equation (23) updates the vector of TFPs. This fixed point has a unique solution when the variance of TFPs is not too large.

Proposition 7 (Local fundamentals)**.**
Suppose that $\log T_\ell \approx \log \bar{T}$ *for all* ℓ*. Then, equations* (22) *and* (23) *identify* ϑ *and* $\{T_\ell\}_\ell$ *given worker-level data on old workers with information on their wages and past and present locations. Given any* $\{T_\ell\}_\ell$ *and the same dataset, equation* (22) *identifies* $\{B_{A\ell}\}$*,* $\{\tau_\ell\}$*, and* δ*.*

Proof. See Appendix D.8. □

Since local productivities may be highly dispersed, I implement the fixed point numerically. To confirm uniqueness, I initialize the algorithm from two extreme starting points. In the first, productivities are uniform across locations, $\log T_\ell \propto 1$. In the second, TFPs explain the entire spatial variation in wages, $\log T_\ell \propto \mathbb{E}[\log W_\ell^A]$.

Results. Table 3 and Figure J.6 report the parameters. The estimates are similar across the starting points. I estimate a taste shock dispersion of 1.15. Expressed as a Fréchet elasticity, it is 1.02, within the range of estimates in Fajgelbaum et al. (2019).[17]

Worker sorting explains the bulk of the wage differences across cities. I quantify how much of the between-city wage gaps is explained by local skill composition through the exact decomposition

$$\mathbb{V}\text{ar}\left[\mathbb{E}\left(\log W_\ell^A\right)\right] = \mathbb{V}\text{ar}\left[\log T_\ell\right] + \mathbb{V}\text{ar}\left[\mathbb{E}\left(\log S_\ell^A\right)\right] + 2\mathbb{C}\text{ov}\left[\log T_\ell, \mathbb{E}\left(\log S_\ell^A\right)\right]. \tag{24}$$

30% of the between-city wage variation reflects TFP gaps. Of the remaining 70% caused by cities' skill composition, 69% arises from the sorting of high skill workers into high TFP cities, as captured by the covariance term in (24). These numbers broadly align with Lhuillier (2022), who computes a similar decomposition using a two-way fixed effects design à la Abowd et al. (1999).

Skills are thus highly concentrated in space. I now study how this concentration affects local human capital accumulation.

4.2 Learning technology

I identify local human capital spillovers from spatial variation in skill growth, instrumenting workers' locations with their birthplaces and their exposure to skilled workers with industry-wide demand for managers and professionals.

[16] Local TFPs are assumed to be exogenous to focus on dynamic spillovers. For any vector of TFPs, static spillovers could be estimated given an instrument for city size or other local characteristics.

[17] The estimates are aligned despite using different sources of variation. While Fajgelbaum et al. (2019) relies on within-state variation over time in employment, I use differences in location choices across skills.

Identification. Tracking workers over time, their skills evolve according to

$$\tilde{s}_{it+1} = \xi(s_{it}) + \varphi(s_{it})\bar{s}_{\ell_{it}} + \tilde{x}_i + \tilde{\mu}_{\ell_{it}} + \tilde{e}_{it+1} + \nu_{it+1}, \tag{25}$$

where $\tilde{z}$ denotes the log of z, $\bar{s}_\ell = \mathbb{E}[s_{jt} \mid \ell_{jt} = \ell]$ is the average skill in ℓ, and e_{it+1} is the idiosyncratic learning shock, $e_{it+1} \perp (s_{it}, x_i, \ell_{it})$. The residual $\nu_{it+1} = \varphi(s_{it})\,(s_{p_{it}} - \bar{s}_{\ell_{it}})$ captures the difference between the skill of the worker with whom i interacted and the average skill in the location. Random interactions within a location imply that the average skill is a valid proxy for local human capital spillovers: $\nu_{it+1} \perp (s_{it}, x_i, e_{it+1}) \mid \bar{s}_{\ell_{it}}$.

It is worth repeating the rationale behind the random interaction assumption. Local random interactions serve as a microfoundation for the estimating equation. They do not impose a substantive restriction on the source of human capital gains. The function $\varphi(s)$ summarizes the dynamic return to working near skilled workers. Whether these gains arise through local interactions or other sources of human capital spillovers does not affect (25). In fact, the skill law of motion continues to hold under the reduced-form spillovers in (2).

A local projection of future skills onto local skills thus identifies local spillovers when flexibly controlling for workers' own skills, their learning abilities, and local learning fundamentals.

Two challenges underpin the identification of these spillovers. First, workers may self-select across locations according to their learning abilities. In particular, fast-learning workers may sort into fast-learning cities as they reap greater benefits, biasing the estimates of φ upward. Second, cities may vary in their learning fundamentals. Lille, for instance, a major economic hub in the north of France, may provide fast growth to its workers because of its proximity to other European cities like London, Brussels, and Amsterdam. Here as well, estimates would be biased upward. I address these two challenges sequentially.

I control for unobserved learning abilities by instrumenting workers' future workplaces with their birthplaces. This instrument is relevant as birthplaces shape workers' future location choices in the presence of migration costs. The instrument satisfies the exclusion restriction to the extent that learning abilities are homogeneously distributed across birthplaces.[18]

Consider the expected future skill of a worker born in ℓ_i^b who has skill s_{it} today. By (25):

$$\mathbb{E}[\tilde{s}_{it+1} \mid s_{it}, \ell_i^b] = \xi(s_{it}) + \mathbb{E}[\tilde{x}_i \mid s_{it}, \ell_i^b] + \sum_{\ell'} \Pr[\ell_{it} = \ell' \mid s_{it}, \ell_i^b]\,[\varphi(s_{it})\bar{s}_{\ell'} + \tilde{\mu}_{\ell'}]\,, \tag{26}$$

Homogeneous learning abilities across birthplaces imply $\mathbb{E}[\tilde{x}_i \mid s_{it}, \ell_i^b] = \mathbb{E}[\tilde{x}_i \mid s_{it}]$. Let $C(s) \equiv \xi(s) + \mathbb{E}[\tilde{x} \mid s]$ denote how a worker's own skill affects their learning, either directly or through their average learning ability. Provided the migration matrix $\boldsymbol{P}(s) = \{\Pr[\ell_{it} = \ell' \mid \ell_i^b = \ell, s]\}_{\ell,\ell'}$ has full rank (i.e., birthplaces shape future workplaces), (26) can be inverted to obtain expected future skills

[18] Learning abilities are time invariant. A common approach for dealing with time invariant unobserved characteristics is to rely on movers, as in the original Abowd et al. (1999), or in de la Roca and Puga (2017). However, if the unobserved characteristic loads onto skill growth, movers are selected: workers with greater learning abilities are more likely to reallocate to skill-abundant locations over time. By contrast, my design allows controlling for static and dynamic unobserved characteristics to the extent that they are uniformly distributed across birthplaces.

by workplace net of sorting on unobserved learning abilities:

$$\boldsymbol{P}(s_{it})^{-1}\mathbb{E}[\tilde{s}_{it+1} \mid s_{it}, \ell_i^b = \cdot] = C(s_{it}) + \varphi(s_{it})\bar{\boldsymbol{s}} + \tilde{\boldsymbol{\mu}}, \tag{27}$$

where bold letters refer to vectors and matrices. Absent local learning fundamentals, (27) identifies local spillovers from spatial differences in expected skill growth for workers with similar skills. Since (27) holds for each skill, it also identifies how the spillovers vary across skills.[19]

To address the second endogeneity concern, I use an external instrument that generates quasi-random variation in local skill densities from aggregate variation in the demand for managers and professionals across industries. Specifically, let $\omega_{m\ell}$ denote the employment share of industry m in location ℓ, and D_m the aggregate employment share of managers and professionals in that industry. I define the predicted local demand for managers in location ℓ given its industrial composition as

$$Q_\ell = \sum_m \omega_{m\ell} D_m, \tag{28}$$

Q_ℓ is a shift-share instrument. Following Borusyak et al. (2022), it satisfies the exogeneity restriction if the relative demand for managers and professionals across sectors is orthogonal to local learning fundamentals. This condition is satisfied, for instance, if the prevalence of managers in the insurance sector is not correlated with the geographic position of Lille.

Proposition 8 (Local spillovers)**.**
Suppose that (i) average learning abilities are identical across birthplaces, and (ii) the relative demand for managers and professionals across industries is orthogonal to local learning fundamentals. Then, the local spillover function φ is identified given panel data with information on wages, birthplace, present workplaces, and local TFPs.

Proposition 8 highlights the block recursivity of estimation. As for Section 2.3, the identification of the learning technology requires few elements of the model. In fact, given local TFPs, the only assumptions are the functional form for the skill law of motion, (25), and the exclusion restrictions behind the instruments. Identification does not rely on what drives location choices, or how correlated learning abilities are with skills.

Proposition 8 departs from the literature on dynamic city gains in several ways. The closest paper is Eckert et al. (2022), who uses the random dispatchment of refugees in Denmark to control for worker sorting. My design differs from theirs in three ways. First, the birthplace design lets me focus on the universe of workers. Second, I estimate how the spillovers vary across the skill distribution. Third, I allow for unobserved local learning fundamentals.

Results. I operationalize Proposition 8 by dividing the skill distribution into quintiles. I compute the migration matrix and the birthplace-specific expected skills for each quintile.[20] I define the

[19]The unit of observation in (27) is different from (25). While the latter is at the worker level, (27) is at the city-skill level. If spillovers were homogenous across the skill distribution, (27) would be a cross-city regression.

[20]The migration matrices are noisy in finite samples, which introduces measurement error in the left-hand side of (26). I use Tikhonov regularization to reduce the measurement error (see Appendix F for details).

Figure 2: Human capital spillovers by skill

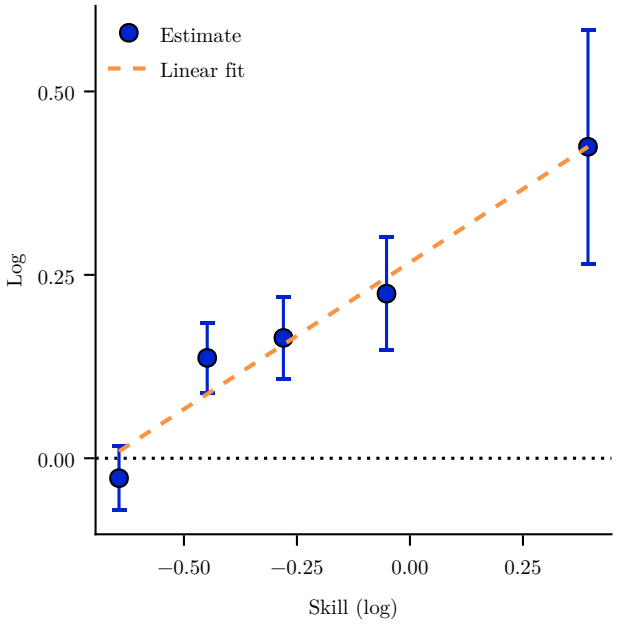


Table 1: Average local human capital spillovers

	(1)	(2)	(3)	(4)	(5)
$\mathbb{E}[\varphi(s_{it})]$	0.25	0.71	0.25	0.16	0.13
	(0.04)	(0.01)	(0.03)	(0.03)	(0.08)
Instruments					
Birthplace inst.	✓	·	✓	✓	✓
Industry inst.	✓	·	·	✓	✓
Controls					
School controls	·	·	·	✓	·
W/in industry	·	·	·	·	✓
F-stat	241.3	·	·	219.5	241.3

Both panels: coefficients estimated from (27). Left: coefficients by skill quintile against the average log skill of each quintile. Horizontal bars refer to 95% confidence intervals. Orange dashed line shows the linear projection of $\hat{\varphi}$ onto log skill. Right: average coefficients. Birthplace instrument refers to (27). Industry instrument is (28). School controls include all standardized test scores taken between 1st and 12th grade, aggregated at the CZ level (cf. Appendix H). W/in industry refers to industry fixed effects. School controls and industry fixed effects are included in (27) as explained in Appendix F. Skills are computed as wages deflated by TFP, as estimated in Section 4.1. Heteroskedastik standard errors reported in both panels.

shift-share instrument at the 3-digit industry level. Finally, I estimate (27) by two-stage least squares with skill quintile fixed effects and quintile-specific slopes. Table 1 shows the average spillovers, and Figure 2 displays the function $\varphi(s_{it})$ across skill quintiles.

Local human capital spillovers are large, and disproportionately so for high-skill workers. On average, skills grow by 5.7 log points every five years in France. When working in Paris, the local spillovers boost growth by an additional 4 log points for the average worker. Most workers enjoy the spillovers, although the effects are nil for the bottom 20% of skills. Meanwhile, workers in the top 20% benefit twice as much as the average worker. The learning technology exhibits strong log supermodularity and is well approximated by the linear model $\varphi(s) = \varphi_0 + \varphi_1 \log s$, displayed by the orange dashed line.

Failing to account for sorting on unobserved learning abilities substantially overstates the strength of local spillovers. Column 2 of Table 1 shows that, absent the birthplace instrument, the estimated spillovers are three times larger.[21] By contrast, local learning fundamentals explain little of the positive association between skill growth and local skill density (column 3).

Proposition 8 highlights the two assumptions required to identify local spillovers. I assess the exclusion restrictions through several robustness exercises. First, while learning abilities are unobserved, skills are observed. If the two are correlated, birthplace differences in learning abilities

[21]The cross-sectional relationship between skill density and skill growth is steeper than the relationship between wage growth and wages. This occurs because, upon moving from high TFP, skill dense places, workers necessarily move to workplaces with lower TFP, and vice versa for workers in low TFP places. Accordingly, not accounting for TFP gaps biases downward the estimated effect of cities on human capital accumulation.

should materialize in workers' initial skills. Figure J.2a plots the average skill of new entrants by birthplace. Initial skills vary little across birthplaces and are only weakly correlated with local skills.

Second, learning abilities may be transmitted across generations. Newborns in skill-dense cities would then be endowed with greater learning abilities to the extent that fast-learners locate there. The exclusion restriction behind Proposition 8 requires average learning abilities to be identical across birthplaces. I thus assess whether the intergenerational transmission of learning abilities violates the exclusion restriction by looking at higher moments of the learning abilities distribution.[22] Figure J.3 plots the second and third moments of the learning ability distribution, by birthplace and workplace. Transmission of abilities would imply a strong positive correlation between the two. While learning abilities across birthplaces and workplaces are indeed positively related, the correlation is weak, and much lower than that predicted by full transmission.

Third, local school quality may also generate heterogeneity in learning abilities across birthplaces. Figure J.2b plots average test scores upon entry into high school against the average skill of workers in each CZ. Although test scores vary across locations, the two are uncorrelated. I account directly for this heterogeneity by controlling for school quality in the estimation.[23] The exclusion restriction then requires learning abilities to be homogeneous across birthplaces conditional on school quality. The estimates are not statistically different from the baseline at the 5% level (column 4).

Fourth, the shift-share instrument may capture industry differences in skill growth rather than local spillovers. Although Section 3.2 shows that industry composition explains little of the between-city variation in wage growth, I add industry fixed effects to (26). The spillovers are then identified from workers in the same industry employed in different locations. Here as well, the estimates are statistically indistinguishable from the baseline at the 5% level (column 5).

4.3 Remaining Parameters

I briefly discuss the estimation of the remaining parameters here and provide further details in Section F. The initial skill distribution is estimated from the skill distribution of new entrants. The skill law of motion (25) lets me estimate the remaining parameters governing learning. The function ξ is identified from changes in skill growth over time and how they relate to changes in workers' skills. Given the two functions ξ and φ, I recover learning abilities from worker-level average skill growth. This allows me to estimate the dispersion in learning abilities, σ_x, together with their correlation with initial skills, ρ. Finally, I recover the learning shocks as residuals and estimate their dispersion, σ_e. All else equal, I find that skills mean revert over time.[24] Meanwhile, I estimate a strong correlation (0.62) between skills and learning abilities.

It remains to estimate the parameters governing the location choices of younger workers, together with the local learning fundamentals. I calibrate these parameters jointly by indirect inference.

[22] Learning abilities are recovered given an estimated learning technology (cf. Section 4.3 and Appendix 4.3).

[23] I include test scores taken between the 1st and 12th grade. Appendix H describes the school data, and Appendix F.2 explains how columns 4 and 5 are implemented.

[24] As for φ, the own skill function ξ is well approximated by a log-linear function. Figure J.4a depicts the non-parametric estimates together with the log-linear approximation used in the model.

Amenities determine the attractiveness of each location, while age specific migration costs govern how easily workers of different ages can relocate. Learning fundamentals, in turn, govern between-city gaps in skill growth. I therefore calibrate these parameters by targeting the age specific employment shares of each CZ, age specific migration rates, and average local skill growth. Figures J.7, J.9a, and J.10a show that the model fits these targets perfectly, and Figures J.8, J.9b, and J.10b display the estimated parameters. Overall, I find that amenities are higher in larger cities throughout the lifecycle, while younger workers face relatively lower migration costs.

4.4 Over-identification exercises

The model is overidentified. The estimation uses little information on how skills and locations vary over the lifecycle. I use these untargeted moments to evaluate the model's quantitative performance.

Lifecycle skill profile. The estimation targets average skill growth and its dispersion. It does not use how these moments vary over the lifecycle. Figures 8 plot the average skill and skill standard deviation by age. The orange squares and blue circles correspond to the moments in the data and the model. Consistent with the literature on human capital, average skill and skill dispersion rise over time and plateau toward the end of the lifecycle. The model generates the same profiles.

The model generates rising and concave lifecycle skill profiles through a subtle interplay between the learning technology and learning abilities. Figure 8c plots the learning technology against the skill of the partner: $s_p \to \log\left(\frac{\gamma(s,s_p)}{s}\right)$. Young workers start their careers with relatively low skills and initially catch up with the rest of the labor force. Meanwhile, workers endowed with higher learning abilities experience consistently faster skill growth, increasing skill dispersion. As workers become more skilled, mean reversion makes further human capital accumulation increasingly difficult, generating the concave profiles.

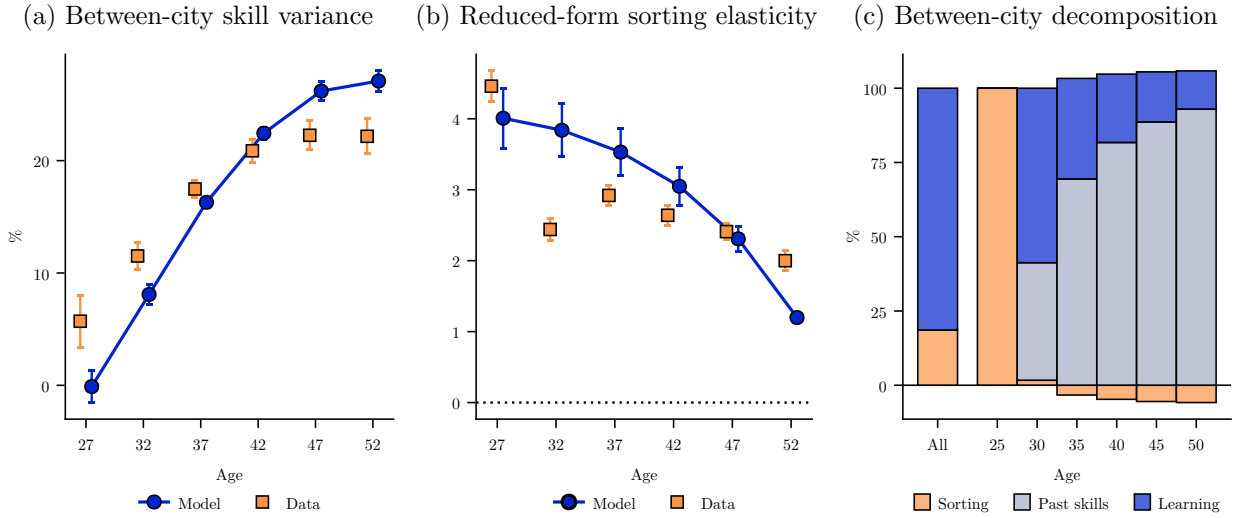
Sorting across the lifecycle. The estimation of the taste shock dispersion relies on the sorting of the oldest workers. It does not target how sorting varies over the lifecycle. Yet, given the estimated learning technology, skilled workers benefit disproportionately from the learning opportunities available in high-TFP locations. Since younger workers have stronger incentives to learn, local spillovers should amplify their incentives to locate there. The lifecycle profile of sorting therefore provides a strong overidentification test of the learning technology.

I begin by quantifying how the spatial dispersion of skills varies across ages. By the law of iterated expectations, the average skill in location ℓ satisfies

$$\mathbb{E}[\log S_\ell] = \sum_{a=1}^{A} \frac{N_\ell^a}{N_\ell} \mathbb{E}[\log S_\ell^a]. \tag{29}$$

Projecting each term on the right-hand side onto the local average skill quantifies the contribution of each age group to the between-city skill variance. Figure 3a reports this decomposition in the data (orange squares) and the model (blue circles). In both cases, older workers account for the larger share of the spatial concentration of skills.

Figure 3: Skill sorting across the lifecycle



Left: between-city skill variance decomposition (29). Each marker represents the coefficient from a cross-sectional projection of $\mathbb{E}[\log S_\ell^a]N_\ell^a/N_\ell$ onto $\mathbb{E}[\log S_\ell]$. Markers sum to 100%. 95% confidence intervals are plotted. Middle: reduced-form sorting elasticity (30) by age group. Probabilities $p_{al\ell}(s)$ are computed by skill quintile in the model and the data. 95% confidence intervals are plotted. In both panels, blue circles and orange squares are moments computed in the model and the data. In the data, skills are computed as wages deflated by TFP, as estimated in Section 4.1. Right: between-city skill variance decomposition (31), implemented in the model. The decomposition is obtained by projecting each term on the right-hand side onto $\mathbb{E}[\log S_\ell^a]$, separately for each age. The aggregate decomposition is obtained by combining (29) and (31). Orange, blue, and grey bars display the contributions of sorting, learning, and inherited skills.

Two opposing forces underpin Figure 3a in the model. On the one hand, skilled workers have stronger incentives to locate in high-TFP cities, and skills rise over the lifecycle. On the other hand, conditional on their skills, younger workers have stronger incentives to choose high-TFP cities because of local learning opportunities.

I isolate and test the second force through the reduced-form regression

$$\log \frac{p_{al\ell}(s)}{p_{all}(s)} = \mathrm{FE}_{al} + \mathrm{FE}_{a\ell} + \mathrm{FE}_{l\ell} + \mathrm{FE}_a(s) + \beta^a s \log T_\ell + u_\ell^a(s). \tag{30}$$

This regression extends (22) across ages. While (22) is exact, (30) is not, as younger workers also value local learning opportunities. The latter are unobserved in the data but correlated with local TFP. Accordingly, the parameter β^a is a reduced-form sorting elasticity. It measures the relative propensity of high-skill workers to work in high-TFP locations across the lifecycle. I cluster skills into quintiles and estimate (30) at the quintile level.

Holding skills constant, younger workers sort more into high-TFP locations. Figure 3b plots the sorting elasticities. In both the data and the model, the elasticity declines over the lifecycle. This age gradient provides strong indirect support for the presence of within-skill learning complementarities.

At face value, Figures 3a and 3b contradict each other. If older workers are less willing to sort, how can they account for most of the between-city skill differences? To understand this apparent contradiction, I decompose the average skill of workers with age a in location ℓ as

$$\mathbb{E}[\log S_\ell^a] = \underbrace{\mathbb{E}[\log S_\ell^{a-1}]}_{\text{Past skills}} + \underbrace{\mathbb{E}[\log S_\ell^a \mid \ell^{a-1}] - \mathbb{E}[\log S_\ell^{a-1}]}_{\text{Learning}} + \underbrace{\mathbb{E}[\log S_\ell^a] - \mathbb{E}[\log S_\ell^a \mid \ell^{a-1}]}_{\text{Sorting}}, \tag{31}$$

where $\mathbb{E}[\log S_\ell^a \mid \ell^{a-1}]$ denotes the average skill holding workers' location choices fixed to their previous age. The first term is the average skill inherited from the past. The second captures skill growth before workers re-optimize their location choices and therefore isolates learning. The third measures how the local skill composition changes once workers update their locations. By construction, spatial skill differences amongst the youngest workers arise entirely from sorting. Plugging (31) into (29), the first term cancels out, and the decomposition quantifies the relative contributions of learning and sorting to total between-city skill differences.

Faster learning in skill-dense cities is the main reason why skills are spatially concentrated and reconciles Figures 3a and 3b. Figure 3c implements the decomposition in the model. Learning accounts for 81% of the between-city skill differences, while sorting explains the remaining 19%. Workers sort primarily upon entering the labor market. Thereafter, skill gaps widen not because skilled workers continue to move into high-TFP cities, but because those who initially selected into these locations accumulate human capital faster.

The model thus matches well how workers accumulate human capital over the lifecycle and the implications for the spatial concentration of skills. Having validated the model, I use it to quantify the consequences of local spillovers for aggregate human capital and learning inequalities.

5 The Consequences of Local Human Capital Spillovers

I quantify the consequences of local human capital spillovers by comparing the baseline economy with a counterfactual world in which spillovers are global. In this alternative economy, learning interactions take place, but geography does not constrain them, and they depend on the aggregate skill distribution. I compare the two economies in steady state. I isolate the role of learning complementarities by reproducing the exercise under homogeneous spillovers: $\log \gamma(s, s_p) = \xi(s) + \bar{\varphi} s_p$.[25] To do so, I fully re-estimate the model under that assumption, and then turn off local spillovers.[26] Table 4 summarizes the results of these two experiments.

5.1 Local Growth

Local human capital spillovers explain half of the variation in human capital accumulation across commuting zones.

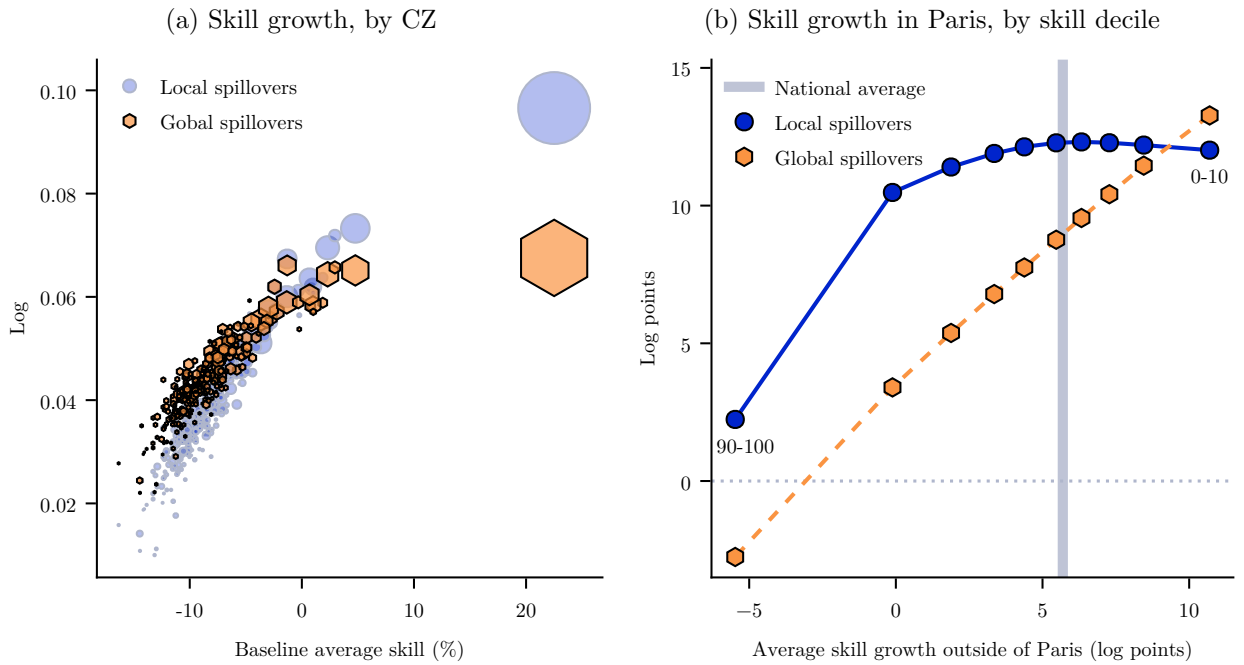
Figure 4a plots the average five-year skill growth rate by CZ when workers learn from their neighbors (light blue circles). In France, the average growth rate is 5.7 log points. Individuals in skill-dense cities learn faster: a one standard deviation increase in the local average skill is associated with a skill growth rate 1 log points higher.

Several factors contribute to local skill growth. Fast learners select into skill-dense cities. These places also tend to offer better learning fundamentals. Finally, they generate greater human capital

[25] This technology in level reads $\gamma(s, s_p) = \mathrm{e}^{\xi(s)} s_p^{\bar{\varphi}}$. It thus features some complementary between $\mathrm{e}^{\xi(s)}$ and local interactions. It is very similar to the vibrancy metric of Crews (2023) once abstracting from the size margin.

[26] Block recursivity implies that the local fundamentals $\{\boldsymbol{T}, \boldsymbol{B}_A, \vartheta, \boldsymbol{\tau}, \alpha, \delta\}$ are the same as in the baseline estimation. The other parameters are estimated internally as in Sections 4.2 and 4.3, and are presented in Table 3.

Figure 4: The consequences of local spillovers for local growth



Left: average skill growth by location of work. Light blue circles represent an economy with local spillovers, where the likelihood of an interaction is given by the local skill density. Orange hexagons represent an economy with aggregate spillovers, where the likelihood is given by the aggregate skill density. The counterfactual equilibrium is computed holding the spatial allocation fixed at that of the local spillovers economy. Both series are plotted against the relative average skill of the workplace. The size of the markers is proportional to city size. Right: average skill growth of individuals working in Paris by decile of the aggregate skill distribution. Each marker represents a decile. Bottom and top deciles are indicated on the figure. Blue circles and orange hexagons depict the growth rates under local and global spillovers. Both series are plotted against the average skill growth of each skill decile for workers outside of Paris in the economy with local spillovers. The grey vertical area refers to the aggregate skill growth rate in the local spillovers economy. All moments are computed in the model in both panels.

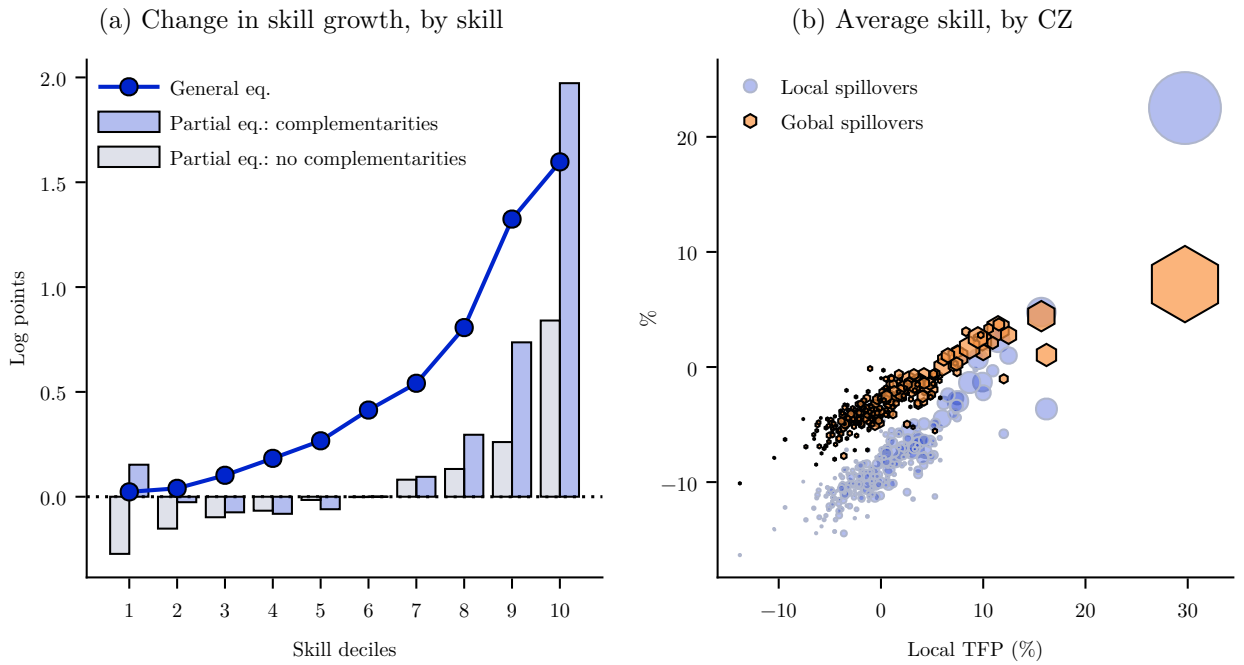
spillovers. I isolate the spillover channel by comparing skill growth rates between the baseline and counterfactual economies. In the latter, I hold the spatial allocation of workers fixed at baseline, keeping the selection margin constant across economies. The darker orange hexagons in Figure 4a display the counterfactual growth rates.

The majority of the cross-city differences in learning disappear absent local human capital spillovers. When human capital spills over freely across space, a one standard deviation increase in the local average skill is associated with a skill growth rate 0.6 log point higher. Local spillovers are particularly strong in very skilled places: the average five-year skill growth rate in Paris falls from 9.7 to 6.8 log points. Meanwhile, 95% of cities, containing 63% of the French workforce, enjoy faster growth when the human capital present in Paris also flows to them. Overall, the between-city skill growth standard deviation halves.

The learning interactions that take place in skill-dense cities are particularly beneficial for skilled workers. Figure 4b shows the average skill growth of workers in Paris by skill decile, contrasting it with the skill-specific growth outside of the capital on the x-axis. On average, skilled workers accumulate human capital at a relatively slower pace as they face diminishing marginal returns to their own skill. Working in a city filled with other skilled individuals allows them to accumulate further human capital and, in fact, outpace the average French worker.

Within-skill learning complementarities explain why local human capital spillovers are dis-

Figure 5: The consequences of local spillovers for aggregate productivity



Left: change in average skill growth between economies with local and global spillovers. In the former, the likelihood of an interaction is given by the local skill densities; in the latter, by the aggregate density. Changes in skill growth computed by decile of the skill distribution. Blue circles show the change between the two equilibria. Light blue bars depict the change in growth when the aggregate skill distribution is kept constant at its local spillovers level. The grey bars show the same partial equilibrium change assuming homogeneous spillovers ($\log \gamma(s, s_p) = \xi(s) + \bar{\varphi} s_p$). Right: average skill of each CZ relative to the national average, under local (blue circles) and global (orange hexagons) spillovers. Average skills are plotted against local TFP. The size of the markers is proportional to city size under local spillovers. All moments are computed in the model.

proportionately strong in productive locations, and particularly so for skilled workers. The blue hexagons in Figure 9a show the change in local skill growth between the local and global spillovers economies. The light orange circles depict the same counterfactual assuming homogeneous spillovers. The estimation guarantees that the baseline local growth rates are the same with and without complementarities. However, were human capital to spill over uniformly to skilled and unskilled workers, local spillovers would reduce the average skill growth in Paris by only 1.4 log points. Furthermore, in such a case, the human capital gains offered by Paris would be spread uniformly across the skill distribution (Figure 9b). In total, local interactions would explain only 31.8% of the between-city skill growth dispersion.

Learning complementarities are key to accounting for the importance of human capital spillovers for local growth. They are also paramount for the consequences of these spillovers for productivity.

5.2 Aggregate Productivity

Local human capital spillovers substantially boost aggregate productivity. The average skill is 2.7% higher when learning interactions are local, and aggregate TFP rises by 5%.[27]

Local spillovers foster a greater stock of human capital by enabling skilled workers to learn faster without slowing down others. The blue circles in Figure 5a depict the change in average skill growth

[27] CRS and competitive labor markets imply that aggregate TFP equals aggregate output and wages.

when local spillovers are active relative to when they are not. These changes are computed for each decile of the skill distribution. Every five years, the median worker enjoys skill growth 0.3 log points faster when they learn from their neighbors. These gains rise to 1.6 log points for workers in the top 10% of skills. At the other end of the distribution, workers neither win nor lose on average.

To understand why all skills benefit, I consider two additional counterfactuals. In both, I keep the aggregate skill distribution constant at its local spillover level. In the first exercise, depicted by the light blue bars in Figure 5a, I maintain the same learning technology, whereas in the second, I further assume homogeneous spillovers (grey bars).

Within-skill learning complementarities are essential for generating large aggregate gains. Uniform spillovers underestimate how much skilled workers gain from concentrating in skill-abundant locations. The increase in skill growth for workers in the top 10% of skills is half as large as with complementarities. Meanwhile, local spillovers reduce the aggregate learning of low skill workers, who are hurt by reduced access to interactions with skilled peers. Together, the aggregate stock of human capital rises by 0.3%, compared with 1.5% in partial equilibrium with complementarities.

The increase in the aggregate stock of human capital ensures that workers throughout the skill distribution gain from local spillovers. In partial equilibrium, local spillovers do not produce benefits for the median worker or those with fewer skills. For these workers, who are relatively more present in smaller cities, slower learning in remote regions cancels out the gains in skill-dense places. In general equilibrium, local spillovers increase the number of skilled workers in the economy, enabling workers throughout France to learn faster. This multiplier effect explains why the median worker learns faster and boosts the increase in the stock of human capital by an additional 1.2 p.p.

Local human capital spillovers generate productivity gains beyond their impact on human capital by agglomerating workers in productive locations. The change in aggregate TFP can be written as

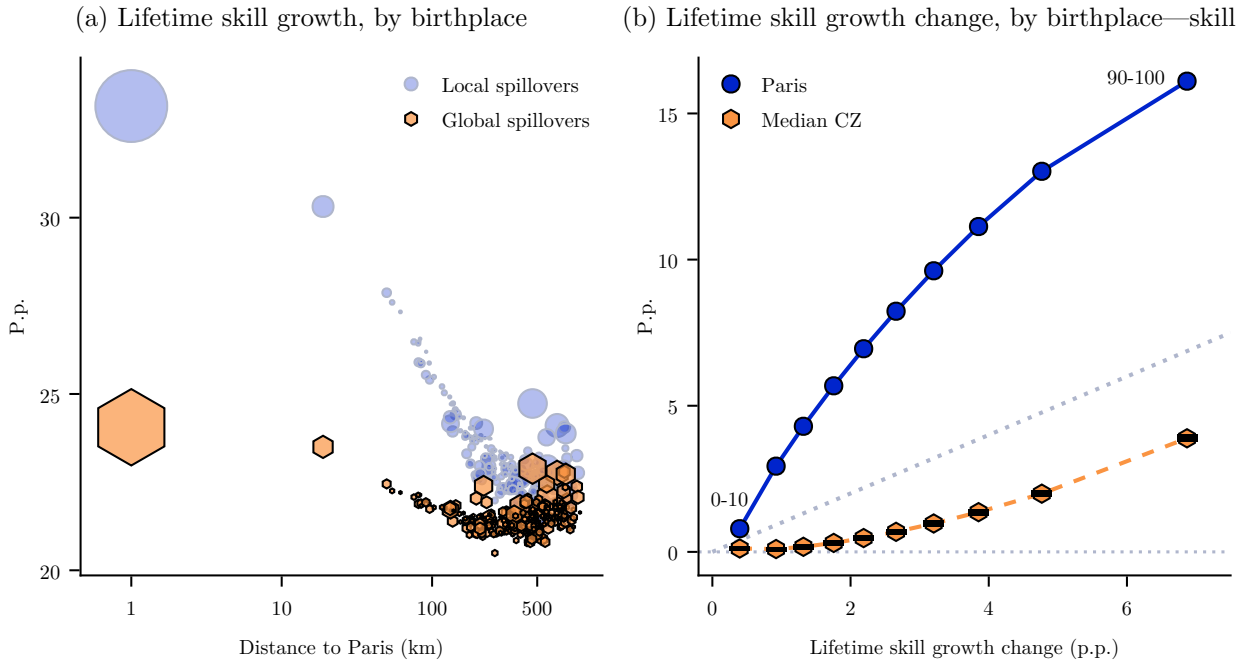
$$d\log \text{TFP} = d\log \mathbb{E}[S] + \mathbb{C}\text{ov}^N\left[\frac{T_\ell}{\text{TFP}}, \mathbb{E}[S_\ell] d\log N_\ell\right] + \mathbb{C}\text{ov}^N\left[\frac{T_\ell}{\text{TFP}}, \mathbb{E}[S_\ell] d\log \frac{\mathbb{E}[S_\ell]}{\mathbb{E}[S]}\right]. \tag{32}$$

Besides the stock of human capital, TFP depends on how large productive cities are and how many skilled individuals work there.

The agglomeration effects are large. Half of the TFP increase is attributed to higher human capital. At the same time, productive locations grow as workers must locate there to benefit from the human capital spillovers generated by skilled workers. The employment share of Paris, for instance, rises from 16% to 21%. Combined with the higher productivity offered by local firms, this agglomeration effect explains 29% of the TFP gains.

The remainder of the TFP growth occurs as local interactions induce a greater concentration of skilled workers in productive places. Figure 5b depicts the average skill of each CZ against local productivity, in the presence of local (dimmed blue circles) and global (orange hexagons) spillovers. Productive places become more skill abundant as skilled workers benefit relatively more from the spillovers taking place there. While the average worker in the median city is 3.3% less skilled, the average skill in Paris rises by 17.4%. Here as well, uniform local spillovers would underestimate the sorting margin, and TFP would be estimated to rise by only 1.6%.

Figure 6: The consequences of local spillovers for learning inequalities



Left: expected lifetime skill growth by location of birth, under local (blue circles) and global (orange hexagons) spillovers. In the former, the likelihood of an interaction is given by the local skill densities $\pi_\ell(s_p)$. In the latter, it is given by the aggregate density $n(s_p)$. Both series are plotted against the (log) kilometer distance between the birthplace and Paris. The size of the markers is proportional to the number of workers born in each CZ under local spillovers. Right: change in lifetime skill growth between the local and global spillovers economies. The change is computed for each decile of the new entrants' skill distribution. Bottom and top deciles are indicated on the figure. Blue circles refer to the change in skill growth for workers born in Paris. Orange hexagons plot the predicted change in skill growth for a CZ 404 kilometers away from Paris. The prediction is computed from gravity models estimated separately for each skill decile. The 95% confidence interval of the prediction is also depicted. Both series are plotted against the aggregate change in lifetime skill growth between local and global spillovers by skill decile. All moments are computed in the model.

5.3 Spatial Inequalities

While local human capital spillovers generate substantial productivity gains, they also engender large learning, wages, and welfare disparities across space.

First, local interactions spur greater wage gaps across commuting zones. The intensified sorting of skilled workers in Paris increases average wages there, and the between-city wage standard deviation increases starkly from 0.13 to 0.19.

Second, and perhaps more importantly, local spillovers engender vast learning inequalities rooted in where workers are born. Learning inequalities are consequential in France. Figure 6a displays expected lifetime skill growth as a function of birthplace, plotted against the distance of each CZ from Paris. The lifetime skill growth of workers born in Paris is 11 p.p. higher than that of workers born in the median city. Furthermore, learning gaps follow gravity: the closer workers are born to Paris, the easier their access to the French capital, and the higher their human capital.

The vast majority of the learning gaps across birthplaces arise because workers born in remote places lack access to skilled workers. Indeed, when the human capital concentrated in productive places flows to other cities, the difference in lifetime skill growth between Paris and the median city shrinks to 3 p.p., and the gravity semi-elasticity falls from -1.56 to -0.16. The remaining differences are caused by local learning fundamentals.

The productivity gains generated by local human capital spillovers are thus very unevenly distributed across places and people. The blue circles in Figure 6b depict the extra lifetime skill growth enjoyed by workers born in Paris when spillovers are local. Each circle refers to a particular skill decile. Meanwhile, the orange hexagons represent the change in lifetime skill growth as predicted by a gravity equation for a commuting zone 404 kilometers away from Paris, the median distance. For reference, 42% of workers are born in this commuting zone or farther away from Paris. Both series are plotted against the aggregate change in lifetime skill growth by skill decile.

The learning inequalities generated by local spillovers are particularly strong for skilled workers. Workers who start their careers with the median skill experience lifetime skill growth 2.2 p.p. faster on average. However, most of these gains accrue to workers born in Paris: for those born 404 kilometers away, lifetime skill growth improves by a modest 0.5 p.p. Workers born there with very high skill do learn faster, but their gains are nowhere close to those enjoyed by workers born in Paris (4 p.p. versus 16.1 p.p.). Indeed, while strong within-skill complementarities mean that skilled workers gain on average the most from the spatial concentration of human capital, they also imply that lacking access to such capital is highly detrimental to learning.

Learning inequalities feed into income and welfare disparities. Workers born in the median-distance CZ earn lifetime income 17% lower than workers born in Paris, a gap 6 p.p. greater than in the absence of local spillovers (Figure 10a). Likewise, while the spillovers generate welfare gains equivalent to a 4.5% increase in lifetime income for workers born in Paris, the welfare of workers in the median CZ is close to unchanged (0.6%, Figure 10b).[28] In particular, the welfare gains of workers born in remote locations, or the lack thereof, are more muted than their wage gains, as they also account for the migration costs that workers must pay to relocate to skill-dense places when learning opportunities are concentrated there.

6 Conclusion

This paper documents that local human capital spillovers have large and positive consequences for aggregate productivity. Theoretically, I develop a framework of learning in space wherein cities shape who interacts with whom, and establish that within-skill learning complementarities are necessary for the clustering of skilled workers to benefit aggregate human capital. Empirically, I estimate strong within-skill learning complementarities from between-city variation in skill growth, controlling for selection on learning abilities by instrumenting current location choices with workers' birthplaces. Quantitatively, I show that local human capital spillovers explain the majority of the spatial variation in skill growth and foster aggregate learning.

At the same time, the gains produced by local human capital spillovers are concentrated among workers in skill-abundant locations. For an inequality-averse planner, local spillovers thus generate a steep equity-efficiency tradeoff. While it is not the purpose of this paper to characterize optimal redistributive transfers under various social welfare functions, the paper sheds light on the

[28] See Appendix G for details on welfare measurement. A single good and linear utility implies that this welfare metric is consistent with that of Baqaee and Burstein (2026).

distortionary potential of such policies. Strong within-skill learning complementarities imply that policies reallocating skilled workers to remote regions have large negative productivity consequences. By contrast, targeted subsidies to workers born in these areas can reduce inequalities without efficiency losses.

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A Proofs

A.1 Proof of Proposition 3

Take the difference in (10) between allocation $\cdot'$ and $\cdot$. Integrating by parts twice implies:

$$\mathbb{E}[S^{o'}] - \mathbb{E}[S^{o}] = \int\int \gamma_{12}(s, s_p) \left[\mathbb{C}\text{ov}^{N'}\left(P_l'(s), P_l'(s_p)\right) - \mathbb{C}\text{ov}^{N}\left(P_l(s), P_l(s_p)\right)\right] ds\, ds_p. \tag{33}$$

Define the population-weighted average survival function, $\bar{P}_{\ell\ell'}(s) \equiv \frac{N_\ell P_\ell(s) + N_{\ell'} P_{\ell'}(s)}{N_\ell + N_{\ell'}}$. Feasibility requires $\bar{P}'_{\ell\ell'}(s) = \bar{P}_{\ell\ell'}(s)$. Consider the contribution of locations ℓ and ℓ' to the integrand in (33). For any $\{s, s_p\}$,

$$\begin{aligned} &N_\ell \left[P_\ell(s) - P(s)\right]\left[P_\ell(s_p) - P(s_p)\right] + N_{\ell'}\left[P_{\ell'}(s) - P(s)\right]\left[P_{\ell'}(s_p) - P(s_p)\right] = \\ &\quad (N_\ell + N_{\ell'})\left[\bar{P}_{\ell\ell'}(s) - P(s)\right]\left[\bar{P}_{\ell\ell'}(s_p) - P(s_p)\right] + \frac{N_\ell N_{\ell'}}{N_\ell + N_{\ell'}} D_{\ell\ell'}(s) D_{\ell\ell'}(s_p), \end{aligned}$$

where $D_{\ell\ell'}(s) \equiv P_{\ell'}(s) - P_\ell(s)$. The first term is the same under allocations $\cdot$ and $\cdot'$. All other locations are also unchanged. Therefore,

$$\mathbb{C}\text{ov}^{N'}\left(P_l'(s), P_l'(s_p)\right) - \mathbb{C}\text{ov}^{N}\left(P_l(s), P_l(s_p)\right) = \frac{N_\ell N_{\ell'}}{N_\ell + N_{\ell'}}\left[D'_{\ell\ell'}(s)D'_{\ell\ell'}(s_p) - D_{\ell\ell'}(s)D_{\ell\ell'}(s_p)\right].$$

The proposition then follows from (33) and the definition of super and submodularity.

A.2 Proof of Proposition 4

Omit the y superscript. First, note that the approximated allocation of young workers satisfies $\pi^y_\ell \succ_{\text{FOSD}} \pi^y_{\ell'} \iff T_\ell \geq T_{\ell'}$ if $\eta(s) \equiv s + \theta\beta\Theta(s)$ is weakly increasing. Indeed, $\pi^y_\ell \succ_{\text{FOSD}} \pi^y_{\ell'} \iff T_\ell \geq T_{\ell'}$ if, for any non-decreasing function g, $\mathbb{E}[g(S_\ell)] - \mathbb{E}[g(S_{\ell'})] \geq 0 \iff T_\ell \geq T_{\ell'}$. To first order, the local skill densities are given by $\pi^y_\ell(s) \approx n^y(s)\left(1 + \theta\left[\eta(s) - \mathbb{E}[\eta(S^y)]\right]\mathcal{T}_\ell\right)$. Hence, the difference in expectation writes $\mathbb{E}[g(S_\ell)] - \mathbb{E}[g(S_{\ell'})] \approx \theta \log\left(\frac{T_\ell}{T_{\ell'}}\right)\mathbb{C}\text{ov}[g(S^y), \eta(S^y)]$. If η is weakly increasing, both g and η are weakly increasing functions of the same variable, and $\mathbb{C}\text{ov}[g(S^y), \eta(S^y)] \geq 0$. Therefore the first-order approximation satisfies $\pi^y_\ell \succ_{\text{FOSD}} \pi^y_{\ell'} \iff T_\ell \geq T_{\ell'}$.

I now show that η is increasing provided the between-skill learning complementarities are not too strong. Define the operator $T : L^2(dN^y) \to L^2(dN^y)$ by

$$Th(s) = \int K(s,t)h(t)dN^y(t), \tag{34}$$

and the class of functions

$$\mathcal{F} = \left\{ f \in L^2(dN^y) : f \text{ is weakly increasing and } \|f - \mathbb{E}[f(S^y)]\| \leq \frac{\mathbb{S}\text{d}[S^y]}{1 - \theta\beta\|K\|} \right\}.$$

Define the sequence $\eta_0(s) = s$ and $\eta_{m+1}(s) = s + \theta\beta T\eta_m(s)$. By Lemma D.4, $\|\eta_m - \mathbb{E}[\eta_m(S^y)]\| \leq \frac{\mathbb{S}\text{d}[S^y]}{1-\theta\beta\|K\|}$ for every m. I show by induction that every η_m is strictly increasing. For $m = 0$, $\eta_0(s) = s$ is strictly increasing. Suppose that η_m is weakly increasing. Since γ is continuously differentiable in its first argument, differentiation under the integral sign gives $\frac{d}{ds}T\eta_m(s) = \mathbb{C}\text{ov}[\gamma_1(s, S^y), \eta_m(S^y)]$. Therefore $\eta'_{m+1}(s) = 1 + \theta\beta\mathbb{C}\text{ov}[\gamma_1(s, S^y), \eta_m(S^y)] > 0$, where the strict inequality follows from the condition on γ assumed in Proposition 4. Hence η_{m+1} is strictly increasing. Since $\theta\beta\|K\| < 1$, the sequence converges in $L^2(dN)$ to the unique solution $\eta \in \mathcal{F}$. Differentiating (52) returns $\eta'(s) = 1 + \theta\beta\mathbb{C}\text{ov}[\gamma_1(s, S^y), \eta(S^y)] > 0$. Hence η is strictly increasing.

A.3 Proof of Proposition 5

Let e_ℓ be the employment share of ℓ under the symmetric equilibrium, given by the fixed point

$$e_\ell = \frac{\exp\{\vartheta(B_\ell - \alpha\bar{p}_\ell)\}}{\sum_{\ell'}\exp\{\vartheta(B_{\ell'} - \alpha\bar{p}_{\ell'})\}} \quad \text{and} \quad \bar{p}_\ell = \frac{1}{\phi}\left(\frac{2\alpha e_\ell}{\bar{H}}\right)^{(1-\phi)/\phi}.$$

Define relative TFP as $\mathcal{T}_\ell \equiv \log T_\ell - \sum_{\ell'} e_{\ell'}\log T_{\ell'}$. Throughout, let $\bar{\cdot}$ denote variable $\cdot$ when there is no local human capital spillovers.

We start with city sizes. To first order, the spatial allocation of old workers under local and global spillovers is $n^o_\ell(s) \approx \bar{n}^o(s)e_\ell\left(1 + \theta\left[s\,\mathcal{T}_\ell - \mathcal{P}_\ell\right]\right)$ and $\bar{n}^o_\ell(s) \approx \bar{n}^o(s)e_\ell\left(1 + \theta\left[s\,\mathcal{T}_\ell - \bar{\mathcal{P}}_\ell\right]\right)$. That of young workers is $n^y_\ell(s) \approx n^y(s)e_\ell\left(1 + \theta\left[(s + \theta\beta\Theta(s))\mathcal{T}_\ell - \mathcal{P}_\ell\right]\right)$ and $\bar{n}^y_\ell(s) \approx n^y(s)e_\ell\left(1 + \theta\left[s\mathcal{T}_\ell - \bar{\mathcal{P}}_\ell\right]\right)$. Integrating across skills and taking the difference: $N_\ell - \bar{N}_\ell = e_\ell\left(\theta^2\beta\mathbb{E}[\Theta(S^y)]\mathcal{T}_\ell - 2\theta\left[\mathcal{P}_\ell - \bar{\mathcal{P}}_\ell\right]\right)$.

Meanwhile, the difference in housing prices is $\mathcal{P}_\ell - \bar{\mathcal{P}}_\ell = \frac{\theta\beta\mathbb{E}[\Theta(S^y)]}{2}\left(\lambda_\ell \mathcal{T}_\ell + (1-\lambda_\ell)\sum_{\ell'}\omega_{\ell'}\mathcal{T}_{\ell'}\right)$, for $\lambda_\ell \in (0,1)$, and ω_ℓ well-defined weights. Combined:

$$N_\ell - \bar{N}_\ell \approx \theta^2\beta e_\ell(1-\lambda_\ell)\left(\mathcal{T}_\ell - \sum_{\ell'}\omega_{\ell'}\mathcal{T}_{\ell'}\right)\mathbb{E}[\Theta(S^y)].$$

Since η is increasing (Proposition 4), $\Theta(s) = \mathbb{C}\text{ov}[\gamma(s, S^y), \eta(S^y)]$ is positive when $\gamma(s,\cdot)$ is increasing, and negative when $\gamma(s,\cdot)$ is decreasing. Focusing on the first case, $N_\ell > \bar{N}_\ell \iff \mathcal{T}_\ell > \sum_{\ell'}\omega_{\ell'}\mathcal{T}_{\ell'}$, which is necessarily true for the most productive location. When amenities are homogeneous, $\lambda_\ell \propto 1$ and $\omega_\ell \propto 1$, and $N_\ell > \bar{N}_\ell \iff T_\ell > \bar{T}$.

I conclude with the average skill. To first order, the difference in the local skill densities of young workers between economies with local and global spillovers is $\pi^y_\ell(s) - \bar{\pi}^y_\ell(s) \approx n^y(s)\theta\left[\Theta(s) - \mathbb{E}[\Theta(S^y)]\right]\mathcal{T}_\ell$. Accordingly, the difference in average skill is $\mathbb{E}[S^y_\ell] - \mathbb{E}[\bar{S}^y_\ell] \approx \theta^2\beta\mathcal{T}_\ell\mathbb{C}\text{ov}[S^y, \Theta(S^y)]$. Likewise, to second order, the difference in between-city wage standard deviation is $\mathbb{S}\text{d}[\mathbb{E}(\log W^y_\ell)] - \mathbb{S}\text{d}[\mathbb{E}(\log \bar{W}^y_\ell)] \approx \mathbb{S}\text{d}_{e_\ell}[\log T_\ell]\theta^2\beta\mathbb{C}\text{ov}[\log S^y, \Theta(S^y)]$ (Section D.5). For both, if γ is supermodular, $\Theta'(s) = \mathbb{C}\text{ov}[\gamma_1(s, S^y), \eta(S^y)]$, and therefore $\mathbb{C}\text{ov}[f(S^y), \Theta(S^y)]$ is positive for any strictly increasing function f. The signs are reversed under submodularity.

A.4 Proof of Proposition 6

I use similar notations as in Section A.3. First, I prove that the laissez-faire equilibrium is efficient if interactions do not impact learning. Suppose $\gamma(s, s_p) = g(s)$. From (19), $\tau_\ell(x) = 0$, and (18) implies $c^{y\star}_\ell(x) = xT_\ell - \alpha p^\star_\ell$, where $\boldsymbol{p}^\star$ are the shadow housing prices. Hence, the optimal allocation of young workers coincides with the laissez-faire if $\boldsymbol{p}^\star = \boldsymbol{p}$. But the shadow housing price (62) follows the same expression as housing prices (8). Hence, the laissez-faire and optimal allocation coincides.

I turn to optimal city size. To first order, the optimal measure of young workers is given by $n^{\star y}_\ell(s) \approx n^y(s)e_\ell\left(1 + \theta\left[\eta^\star(s)\mathcal{T}_\ell - \mathcal{P}^\star_\ell\right]\right)$, where $\eta^\star$ is given by the fixed point (69). Integrating across skills and using the expression for the shadow housing price similar to the laissez-faire housing prices, the optimal population in ℓ is $N^\star_\ell \approx 2e_\ell + 2\theta e_\ell\mu^\star\left(\mathcal{T}_\ell - \mathcal{P}_\ell\right)$, where $\frac{1}{2}\left(\mathbb{E}[S^o] + \mathbb{E}[\eta^\star(S^y)]\right)$. By contrast, the laissez-faire population is $N^\star_\ell \approx 2e_\ell + 2\theta e_\ell\mu\left(\mathcal{T}_\ell - \mathcal{P}_\ell\right)$. Taking the difference and using $\mathbb{E}[\eta^\star(S^y)] = \mathbb{E}[S^y]$ (Lemma D.6): $N_\ell - N^\star_\ell \approx \theta^2\beta e_\ell\left(\mathcal{T}_\ell - \mathcal{P}_\ell\right)\mathbb{E}[\Theta(S^y)]$. Using the same proof as for Proposition 5 concludes.

Lastly, I turn to the local skill compositions. The optimal skill density in city ℓ is $\pi^{y\star}_\ell(s) \approx n^y(s)\left(1 + \theta\left[\eta^\star(s) - \mathbb{E}[\eta^\star(S^y)]\right]\mathcal{T}_\ell\right)$. Hence, the difference in the local average skill compared to the laissez-faire allocation is. $\mathbb{E}[S^{y\star}_\ell] - \mathbb{E}[S^y_\ell] \approx \theta\mathcal{T}_\ell\left(\mathbb{C}\text{ov}[S^y, \eta^\star(S^y)] - \mathbb{C}\text{ov}[S^y, \eta(S^y)]\right)$. It remains to compare the two covariances. In the laissez-faire, we know that $\eta(s) = s + \theta\beta\mathbb{C}\text{ov}[\gamma(s, S^y), \eta(S^y)]$. Therefore,

$$\mathbb{C}\text{ov}[S^y, \eta^\star(S^y)] = \mathbb{C}\text{ov}[\eta(S^y), \eta^\star(S^y)] - \theta\beta\mathbb{C}\text{ov}\left[\mathbb{C}\text{ov}[\gamma(S^y, \widetilde{S}^y), \eta(\widetilde{S}^y) \mid S^y], \eta^\star(S^y)\right]. \tag{35}$$

Likewise, in the optimal allocation, $\eta^\star(s) = s + \theta\beta\mathbb{C}\text{ov}[\gamma(s, S^y), \eta^\star(S^y)] + \theta\beta\mathbb{C}\text{ov}[\gamma(S^y, s), \eta^\star(S^y)] + c$, where c is a constant independent of s. Accordingly,

$$\begin{aligned}\mathbb{C}\text{ov}[S^y, \eta(S^y)] = \mathbb{C}\text{ov}[\eta^\star(S^y), \eta(S^y)] - \theta\beta\mathbb{C}\text{ov}\left[\mathbb{C}\text{ov}[\gamma(S^y, \widetilde{S}^y), \eta^\star(\widetilde{S}^y) \mid S^y], \eta(S^y)\right] - \\ \theta\beta\mathbb{C}\text{ov}\left[\mathbb{C}\text{ov}[\gamma(\widetilde{S}^y, S^y), \eta^\star(\widetilde{S}^y) \mid S^y], \eta(S^y)\right].\end{aligned} \tag{36}$$

Since S^y and $\widetilde{S}^y$ are independent and identically distributed,

$$\mathbb{C}\mathrm{ov}\left[\mathbb{C}\mathrm{ov}[\gamma(S^y,\widetilde{S}^y),\eta(\widetilde{S}^y)\mid S^y],\eta^\star(S^y)\right]=\mathbb{C}\mathrm{ov}\left[\mathbb{C}\mathrm{ov}[\gamma(\widetilde{S}^y,S^y),\eta^\star(\widetilde{S}^y)\mid S^y],\eta(S^y)\right].$$

Subtracting (36) from (35) therefore yields

$$\mathbb{C}\mathrm{ov}[S^y,\eta^\star(S^y)]-\mathbb{C}\mathrm{ov}[S^y,\eta(S^y)]=\theta\beta\mathbb{C}\mathrm{ov}\left[\mathbb{C}\mathrm{ov}[\gamma(S^y,\widetilde{S}^y),\eta^\star(\widetilde{S}^y)\mid S^y],\eta(S^y)\right]. \tag{37}$$

By Lemmas 4 and D.7, η and $\eta^\star$ are strictly increasing. Moreover, $\partial_x\mathbb{C}\mathrm{ov}[\gamma(x,\widetilde{S}^y),\eta^\star(\widetilde{S}^y)]=\mathbb{C}\mathrm{ov}[\gamma_1(x,\widetilde{S}^y),\eta^\star(\widetilde{S}^y)]$. If γ is supermodular, $s\mapsto\gamma_1(x,s)$ is increasing. Hence, $x\mapsto\mathbb{C}\mathrm{ov}[\gamma(x,\widetilde{S}^y),\eta^\star(\widetilde{S}^y)]$ is increasing. Since η is also increasing, the covariance on the right-hand side of (37) is positive, and $\mathbb{C}\mathrm{ov}[S^y,\eta^\star(S^y)]>\mathbb{C}\mathrm{ov}[S^y,\eta(S^y)]$. If γ is submodular, the same function is decreasing, and the inequality is reversed.

B Tables

Table 2: Wage growth differences across commuting zones

	Future wage (log)						Switch occ.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Wage (log)	0.979	0.979	0.889	0.937	0.971	0.924	
	(0.003)	(0.003)	(0.005)	(0.001)	(0.001)	(0.001)	
CZ wage	0.074	0.056	0.060	0.066	0.049	0.030	0.116
	(0.004)	(0.005)	(0.006)	(0.001)	(0.006)	(0.002)	(0.003)
CZ size (log)		0.002					
		(0.001)					
Establishment wage						0.051	
						(0.002)	
Firm wage						0.044	
						(0.003)	
Fixed effects							
Age	✓	✓	✓	✓	✓	✓	✓
Industry × occupation	·	·	✓	·	·	·	·
Firm	·	·	·	✓	·	·	·
Residence	·	·	·	·	✓	·	·
Occupation	·	·	·	·	·	·	✓
Obs. (thousands)	22,584	22,584	22,583	22,584	22,583	22,584	22,584

Data source: DADS Postes, cleaned and aggregated as detailed in Appendix H. Elasticities estimated from (21). "Future wage" and "wage": log wage in the next 5-year period and log wage in the past 5-year period. Periods do not overlap. "CZ wage", "establishment wage", and "firm wage": average wages at the CZ, establishment, and firm level. "CZ size": log CZ employment share. "Switch occ.": dummy if the worker switches 2-digit occupation within the next 5 years. Industry and occupation fixed effects at the two-digit level. Residence defined as the municipality. Standard errors clustered by CZ.

Table 3: Parameter estimates

	Parameter	Value			Moment		
		Baseline	Asymmetric	Homogeneous	Target	Data	Model
	A. External calibration						
β	5-year discount rate		0.60				
	B. Estimation						
α	Housing costs		46.20				
ϕ	Fixed factor housing share		0.90				
$\bar{H}$	Fixed factor		3.80×10^{22}				
ϑ	Taste shock dispersion	1.15	1.07	1.15			
δ	Mig. cost distance elas.	1.48	1.59	1.48			
$\boldsymbol{T}$	TFP		Fig. J.6a				
$\boldsymbol{B}_A$	Amenities		Fig. J.6b				
$\boldsymbol{\tau}$	Migration costs		Fig. J.6c				
μ_s	Entrant avg. skill	0.69	·	0.69			
σ_s	Entrant skill st. dev.	0.28	·	0.28			
σ_x	Learning abilities st. dev.	0.10	·	0.11			
ρ_{sx}	Correlation skill – abilities	0.62	·	0.59			
ξ'	Learning tech. own-skill slope	0.35	·	0.67			
φ	Learning tech. others-skill level	0.27	·	0.17			
φ'	Learning tech. others-skill slope	0.40	·	0.00			
	C. Indirect inference						
ξ	Learning tech. own-skill level	-0.26	·	-0.18	Avg. skill	0.91	0.91
σ_ϵ	Learning shock st. dev.	0.18	·	0.21	Skill st. dev.	0.17	0.17
$\{\boldsymbol{B}_a\}_{a<A}$	Amenities		Fig. J.8		City-age population	Fig. J.7	
$\{\zeta_a\}_{a<A}$	Age-specific mig. costs		Fig. J.9b		Age-specific mig. rate	Fig. J.9a	
$\boldsymbol{\mu}$	Learning fundamentals		Fig. J.10b		City skill growth rate	Fig. J.10a	

Asymmetric: local fundamentals obtained with an asymmetric TFP vector for initial guess. Homogeneous: uniform spillovers assumption $\log \gamma(s, s_p) = \phi(s) + \bar{\varphi} s_p$.

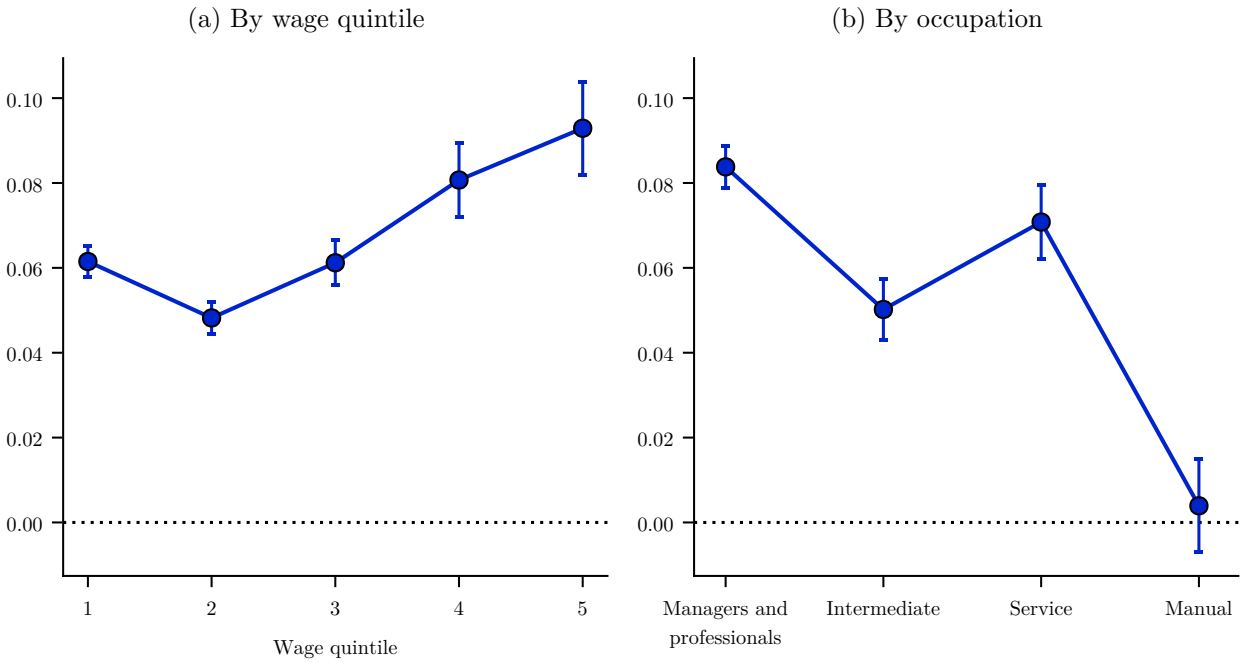
Table 4: The consequences of local human capital spillovers

	Baseline	Change from global spillovers	
		Heterogeneous	Homogeneous
Average skill (%)	0.91	2.70	0.81
TFP (%)	1.00	4.96	1.64
... Human capital (%)	·	55.76	49.62
... Agglomeration (%)	·	29.53	36.04
... Sorting (%)	·	14.71	14.34
5-year skill growth (p.p.)	5.69	0.46	0.16
Workplace wage st. dev. (log)	0.19	0.06	0.02
Workplace skill growth st. dev. (p.p.)	2.35	1.28	0.79
Birthplace lifetime skill growth st. dev. (p.p)	4.37	3.35	1.50

First column describes the economy under local and heterogeneous spillovers. Second and third columns show the change from global to local spillovers. Second column with learning complementarities, third column without. Units in parenthesis refer to the changes in the second and third columns. Third to fifth rows refer to the decomposition (32).

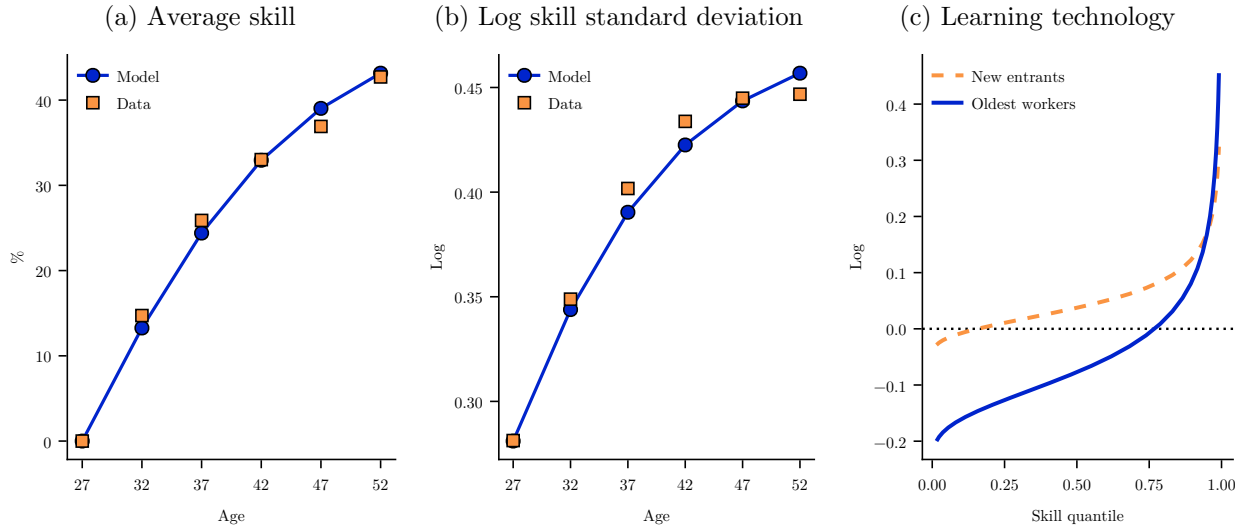
C Figures

Figure 7: Wage growth differences across commuting zones and workers



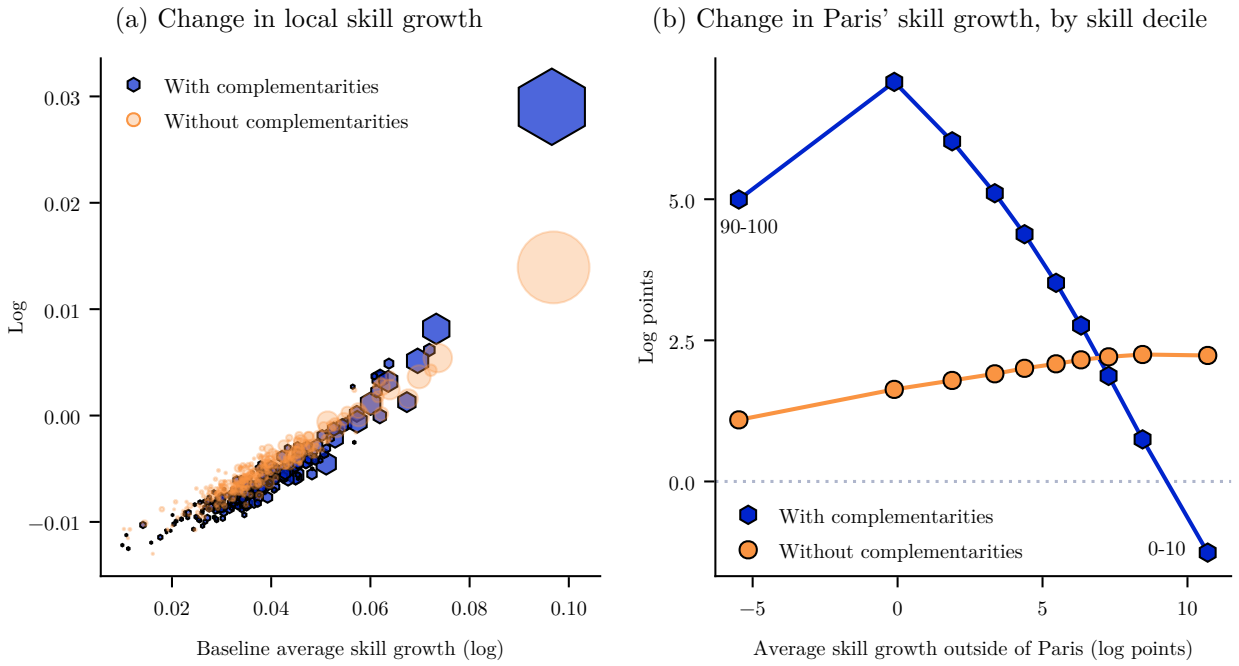
Data source: DADS Postes, cleaned and aggregated as detailed in Appendix H. In both panels, local wage elasticity of future wages as estimated by (21). Left: elasticities by quintile of the wage distribution. Right: elasticities by one-digit occupation. 95% confidence intervals plotted, with standard errors clustered at the city level.

Figure 8: Lifecycle skill profile



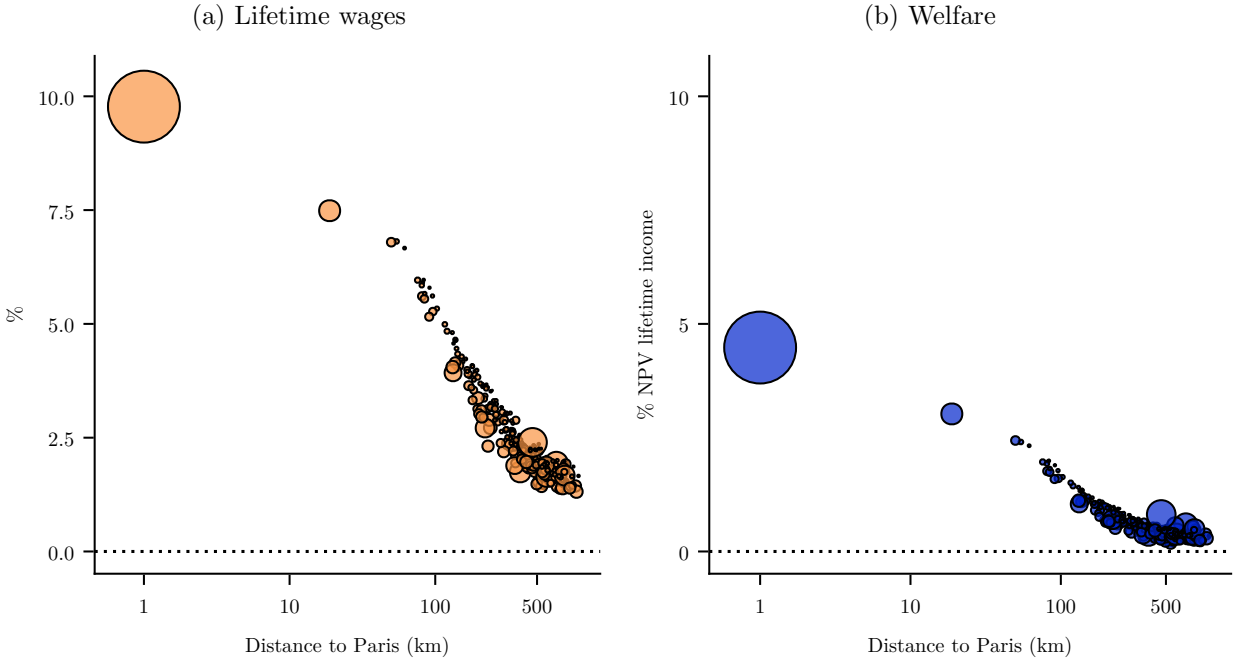
Left: average skill by age group, relative to the first group. Middle: log skill standard deviation by age group. Blue circles and orange squares are the moments in the model and the data. In the data, skills computed as wages deflated by TFP, where TFPs are estimated as explained in Section 4. Right: predicted skill growth by the learning technology against partner's skill, $s_p \to \log \gamma(s, s_p)/s$, plotted for the average skill of new entrants (dashed orange) and the average skill of old workers (solid blue).

Figure 9: The consequences of local spillovers without complementarities



Both panels: change in skill growth between economies with local and global spillovers. In the former, the likelihood of an interaction given by the local skill densities. In the latter, it is given by the aggregate density. Counterfactual equilibrium computed holding fixed the spatial allocation at the local spillovers economy. Left: change in average growth by CZ. Right: change in skill growth for workers in Paris by decile of the aggregate skill distribution. Each marker represents a decile. Change in skill growth plotted against the average skill growth of workers outside of Paris. Both panels: blue hexagons show changes with learning complementarities, and orange hexagons changes under homogeneous spillovers.

Figure 10: The consequences of local spillovers across birthplaces



Left: change in lifetime wages by location of birth between economies with local and global spillovers. Right: change in welfare by location of birth between economies with local and global spillovers. Welfare change computed as in Appendix G and expressed relative to the net present value of aggregate lifetime income. In both panels, the likelihood of an interaction is given by the local skill densities $\pi_\ell(s_p)$ under local spillovers and the aggregate density $n(s_p)$ under global spillovers. Both series plotted against the (log) kilometer distance between the birthplace and Paris. The size of the markers is proportional to the number of workers born in each CZ under local spillovers. All moments computed in the model.

Online Appendix

D Additional Derivations and Proofs

D.1 Proof of Proposition 1 (existence)

We prove existence of the equilibrium using Schauder's fixed-point theorem. To simplify the proof, we impose the following regularity conditions on the young skill and learning shock distributions.

Assumption 2 (Regularity conditions)**.**
The young skill distribution admits a density n^y on $\mathcal{S}^y = [\underline{s}, \bar{s}]$, with n^y continuous and bounded away from zero: $0 < \underline{n}^y \leq n^y(s) \leq \bar{n}^y < \infty$. F has a continuous density f and has finite first moment.

Let $E = C(\mathcal{S}^y, \mathbb{R}^L)$ be the space of continuous functions from $\mathcal{S}^y = [\underline{s}, \bar{s}]$ to $\mathbb{R}^L$ endowed with the sup norm. Define further

$$\mathcal{K} = \left\{ \eta \in E : \underline{\eta} \leq \eta_\ell(s_y) \leq \bar{\eta} \text{ for all } s_y \in \mathcal{S}^y \text{ and all } \ell = 1, \ldots, L, \quad \sum_{\ell=1}^{L} \int_{\mathcal{S}^y} \eta_\ell(s_y) ds_y = 1 \right\}.$$

The constants $\underline{\eta}$ and $\bar{\eta}$ are chosen so that $\mathcal{K}$ is nonempty. Meanwhile, define $\mathcal{P} = [0, \bar{p}]^L$, where

$$\bar{p} = \frac{1}{\phi} \left(\frac{2\alpha}{\bar{H}} \right)^{(1-\phi)/\phi}.$$

The set $\mathcal{K} \times \mathcal{P}$ is nonempty, closed, bounded, and convex in $E \times \mathbb{R}^L$.

Define the operator $\mathcal{A} : \mathcal{K} \times \mathcal{P} \to E \times \mathbb{R}^L$ by $\mathcal{A}[\eta, p] = (\mathcal{M}[\eta, p], \mathcal{Q}[\eta, p])$. The first component is the young allocation operator:

$$\mathcal{M}_\ell[\eta, p](s_y) = n^y(s_y) \frac{\exp\left\{\vartheta \left(w_\ell(s_y) - \alpha p_\ell + B_\ell + \beta O_p(s_y, \eta_\ell) \right)\right\}}{\sum_{\ell'=1}^{L} \exp\left\{\vartheta \left(w_{\ell'}(s_y) - \alpha p_{\ell'} + B_{\ell'} + \beta O_p(s_y, \eta_{\ell'}) \right)\right\}},$$

where

$$O_p(s_y, \eta_\ell) = \int_{\mathcal{S}^y} H_p(s_y, s_p) \pi_\ell^\eta(s_p) ds_p, \tag{38}$$

$$H_p(s_y, s_p) = \int \mathcal{V}_p^o(e\gamma(s_y, s_p))\, dF(e).$$

$$\mathcal{V}_p^o(s_o) = \mathbb{E}\left[\max_\ell \left\{ w_\ell(s_o) - \alpha p_\ell + \varepsilon_\ell \right\} \mid s_o \right]. \tag{39}$$

and I have defined $\pi_\ell^\eta(s) = \frac{\eta_\ell(s)}{\int_{\mathcal{S}^y} \eta_\ell(\sigma) d\sigma}$. The second component is the housing price operator. Given $\eta \in \mathcal{K}$, define the old skill distribution by

$$N_\eta^o(s_o) = \sum_{\ell=1}^{L} \int_{\mathcal{S}^y} \int_{\mathcal{S}^y} \eta_\ell(s_y) \pi_\ell^\eta(s_p) F\left(\frac{s_o}{\gamma(s_y, s_p)} \right) ds_p ds_y.$$

This distribution is well-defined. Since F admits a continuous density (Assumption 2) and γ is continuous and bounded away from zero on $(\mathcal{S}^y)^2$ (Assumption 1), N_η^o is continuously differentiable

with density denoted $n^o_\eta(s_o)$. Moreover, $F(0) = 0$ and $\lim_{x\to\infty} F(x) = 1$ imply $N^o_\eta(s_o) = 0$ as $s_o \to 0^+$ and $N^o_\eta(s_o) \to 1$ as $s_o \to \infty$, where the second limit follows from $\eta \in \mathcal{K}$. Given (η, p), define the old allocation as

$$n^o_{\ell,\eta,p}(s_o) = n^o_\eta(s_o)P^o_{\ell,p}(s_o) \quad ; \quad P^o_{\ell,p}(s_o) = \left(\frac{\mathrm{e}^{\vartheta(w_\ell(s_o)-\alpha p_\ell+B_\ell)}}{\sum_{\ell'}\mathrm{e}^{\vartheta(w_{\ell'}(s_o)-\alpha p_{\ell'}+B_{\ell'})}}\right).$$

Total population in city ℓ is $N_\ell(\eta, p) = \int_{\mathcal{S}^y} \eta_\ell(s_y)ds_y + \int_0^\infty n^o_{\ell,\eta,p}(s_o)ds_o$. The price operator is

$$\mathcal{Q}_\ell[\eta, p] = \frac{1}{\phi}\left(\frac{\alpha N_\ell(\eta, p)}{\bar{H}}\right)^{(1-\phi)/\phi}.$$

Schauder's fixed-point demands that (i) $\mathcal{A}(\mathcal{K}\times\mathcal{P}) \subseteq \mathcal{K}\times\mathcal{P}$, (ii) $\mathcal{A}$ is continuous, and (iii) $\mathcal{A}(\mathcal{K}\times\mathcal{P})$ is relatively compact in $E \times \mathbb{R}^L$. We prove the three ingredients in turn.

We start by showing $\mathcal{A}(\mathcal{K}\times\mathcal{P}) \subseteq \mathcal{K}\times\mathcal{P}$. $\mathcal{V}^o_p$ is continuous from (39). Meanwhile, $\mathcal{V}^o_p$ is bounded by $|\mathcal{V}^o_p(s_o)| \le C_0 + C_1 s_o$ for all $p \in \mathcal{P}$. Accordingly, the integrand of H_p is dominated by $e \to C_0 + C_1 e \max_{s_y,s_p}\gamma(s_y, s_p)$, which is F-integrable by Assumptions 1 and 2. Dominated convergence implies continuity of H_p on $(\mathcal{S}^y)^2$, and since $(\mathcal{S}^y)^2 \times \mathcal{P}$ is compact, H_p is bounded and uniformly continuous in (s_y, s_p, p). For any $\eta \in \mathcal{K}$, $O_p(s_y, \eta_\ell)$ is well-defined since $\eta_\ell(s) > 0$. Furthermore, since H_p is bounded and continuous, $s_y \to O_p(s_y, \eta_\ell)$ is also bounded and continuous. It follows that the utilities entering the young logit probabilities are bounded and continuous for all $(\eta, p) \in \mathcal{K}\times\mathcal{P}$. Hence the logit probabilities are bounded, continuous, and uniformly bounded away from zero. Since n^y is continuous and bounded away from zero, $\mathcal{M}_\ell[\eta, p]$ satisfies the same pointwise bounds for $\underline{\eta}$ small enough and $\bar{\eta}$ large enough. Moreover, $\sum_{\ell=1}^L \mathcal{M}_\ell[\eta, p](s_y) = n^y(s_y)$ for every $s_y \in \mathcal{S}^y$. Since n^y integrates to one, $\sum_{\ell=1}^L \int_{\mathcal{S}^y} \mathcal{M}_\ell[\eta, p](s_y)ds_y = 1$. Therefore $\mathcal{M}(\mathcal{K}\times\mathcal{P}) \subseteq \mathcal{K}$. It remains to show that $\mathcal{Q}(\mathcal{K}\times\mathcal{P}) \subseteq \mathcal{P}$. This follows immedialty from the bound on $\mathcal{P}$ defined since $N_\ell(\eta, p) \in (0, 2)$.

We then show continuity. Suppose $(\eta^m, p^m) \to (\eta, p)$ uniformly. Starting with $\mathcal{M}$, since $\eta_\ell(s) > 0$ for all (ℓ, s), we have $\frac{\eta^m_\ell}{\int_{\mathcal{S}^y}\eta^m_\ell} \to \frac{\eta_\ell}{\int_{\mathcal{S}^y}\eta_\ell}$ uniformly. Moreover, $\mathcal{V}^o_{p^m} \to \mathcal{V}^o_p$ uniformly from (39), and so does $H_{p^m} \to H_p$. Since H_p is bounded, $O_{p^m}(\cdot, \eta^m_\ell) \to O_p(\cdot, \eta_\ell)$ uniformly. The logit map is smooth on bounded sets, and n^y is continuous. Hence $\mathcal{M}[\eta^m, p^m] \to \mathcal{M}[\eta, p]$ uniformly. We turn to $\mathcal{Q}$. We need to prove convergence of the housing demand, and thus N_ℓ. N^y_{ℓ,η^m} converges uniformly by construction. For old workers, we have

$$N^o_{\ell,\eta^m,p^m} = \sum_{l=1}^L \int_{\mathcal{S}^y}\int_{\mathcal{S}^y}\int P^o_{\ell,p^m}(s_o)\eta^m_l(s_y)\pi^{\eta^m}_l(s_p)\frac{1}{\gamma(s_y, s_p)}f\left(\frac{s_o}{\gamma(s_y, s_p)}\right)ds_o ds_p ds_y$$

The logit probabilities of old workers converge pointwise because they are continuous in p. γ is bounded away from zero and f is continuous. Lastly, $\pi^{\eta^m}_\ell \to \pi^\eta_\ell$ uniformly. Hence, the integrand converges pointwise. Moreover, it is bounded by an integrable function, and dominated convergence implies $N^o_{\ell,\eta^m,p^m} \to N^o_{\ell,\eta,p}$. Therefore $\mathcal{Q}[\eta^m, p^m] \to \mathcal{Q}[\eta, p]$. Accordingly, $\mathcal{A}$ is continuous.

It remains to establish compactness of the image. Since $\mathcal{S}^y$ is compact, Arzelà–Ascoli implies that $\mathcal{M}(\mathcal{K}\times\mathcal{P})$ is relatively compact in E if it is uniformly bounded and equicontinuous. Uniform boundedness follows from $\mathcal{M}(\mathcal{K}\times\mathcal{P}) \subseteq \mathcal{K}$. Equicontinuity also follows from the uniform continuity of H_p and n^y, together with the smoothness of the logit map on bounded sets. Since $\mathcal{Q}(\mathcal{K}\times\mathcal{P}) \subseteq \mathcal{P}$ and $\mathcal{P}$ is finite dimensional and compact, $\mathcal{A}(\mathcal{K}\times\mathcal{P})$ is relatively compact in $E \times \mathbb{R}^L$.

D.2 Proof of Proposition 1 (uniqueness)

We prove uniqueness in three steps, starting with two preliminary lemmas. Throughout, the notation is analogous to Appendix D.1.

Lemma D.1.
Take $U \in \mathbb{R}^L$. Let

$$P_\ell(U) = \frac{\exp\{\vartheta U_\ell\}}{\sum_{l=1}^{L} \exp\{\vartheta U_l\}}.$$

Then, for any $U, U' \in \mathbb{R}^L$, $|P_\ell(U) - P_\ell(U')| \leq \frac{\vartheta}{2}\|U - U'\|_\infty$.

Proof. By the mean-value theorem,

$$P_\ell(U) - P_\ell(U') = \int_0^1 \nabla P_\ell\left(U' + t(U - U')\right) \cdot (U - U')dt.$$

Taking absolute values and using the triangular inequality,

$$|P_\ell(U) - P_\ell(U')| \leq \int_0^1 \left|\nabla P_\ell\left(U' + t(U - U')\right) \cdot (U - U')\right| dt \leq \int_0^1 \left\|\nabla P_\ell\left(U' + t(U - U')\right)\right\|_1 \|U - U'\|_\infty dt,$$

where the second inequality follows from Hölder's. We have

$$\left|\frac{\partial P_\ell(U)}{\partial U_\ell}\right| = \vartheta P_\ell(U)(1 - P_\ell(U)) \quad ; \quad \left|\frac{\partial P_\ell(U)}{\partial U_l}\right| \vartheta P_\ell(U) P_l(U), \quad \forall l \neq \ell.$$

Therefore, $\|\nabla P_\ell(U)\| = 2\vartheta P_\ell(U)(1 - P_\ell(U)) \leq \frac{\vartheta}{2}$. Since this bound holds for all utility vectors, it holds $t \to U' + t(U - U')$. The lemma follows. □

Lemma D.2.
Fix $p \in \mathcal{P}$. If $\vartheta\beta$ is small enough, there exists a unique allocation of young workers.

Proof. We prove uniqueness by showing that $\mathcal{M}$ is a contraction on $\mathcal{K}$ when $\vartheta\beta$ is small enough. Endow $E = C(\mathcal{S}^y, \mathbb{R}^L)$ with the sup norm. We show that there exists a constant $\kappa < 1$ such that, for all $\eta, \eta' \in \mathcal{K}$ and all $p \in \mathcal{P}$, $\|\mathcal{M}[\eta, p] - \mathcal{M}[\eta', p]\|_\infty \leq \kappa\|\eta - \eta'\|_\infty$. For the rest of the proof, I drop the p index since they are constant.

Fix $\eta, \eta' \in \mathcal{K}$. For any ℓ and any $s \in \mathcal{S}^y$, let $P_\ell(U(s;\eta))$ denote the logit probabilities of young workers, where $U_\ell(s;\eta) = sT_\ell - \alpha p_\ell + B_\ell + \beta O(s, \eta_\ell)$ and O is defined in (38). Therefore,

$$|\mathcal{M}_\ell[\eta](s) - \mathcal{M}_\ell[\eta'](s)| = n^y(s)|P_\ell(U(s;\eta)) - P_\ell(U(s;\eta'))| \leq \bar{n}^y |P_\ell(U(s;\eta)) - P_\ell(U(s;\eta'))|,$$

where $\bar{n}^y = \sup_{s \in \mathcal{S}^y} n^y(s)$. By Lemma D.1, $|P_\ell(U(s;\eta)) - P_\ell(U(s;\eta'))| \leq \frac{\vartheta}{2}\|U(s;\eta) - U(s;\eta')\|_\infty$. It remains to bound the difference in utility indices. Since the present component of utilities is independent of η (conditional on p): $|U_\ell(s;\eta) - U_\ell(s;\eta')| = \beta|O(s,\eta_\ell) - O(s,\eta'_\ell)|$. Let $\overline{\mathcal{H}} =$

$\sup_{(s,s_p)\in(\mathcal{S}^y)^2} |H(s,s_p)|$. Then

$$\begin{aligned}|O(s,\eta_\ell) - O(s,\eta'_\ell)| &= \left| \int_{\mathcal{S}^y} H(s,s_p) \left[\pi^{\eta}_\ell(s_p) - \pi^{\eta'}_\ell(s_p) \right] ds_p \right| \\ &\leq \overline{\mathcal{H}} \int_{\mathcal{S}^y} |\pi^{\eta}_\ell(s_p) - \pi^{\eta'}_\ell(s_p)| ds_p \\ &\leq \frac{2\overline{\mathcal{H}}}{\underline{\eta}} \|\eta_\ell - \eta'_\ell\|_\infty,\end{aligned}$$

where the last inequality follows from $\underline{\eta} \leq \eta_\ell(s) \leq \bar{\eta}$ for all $s \in \mathcal{S}^y$ since $\eta \in \mathcal{K}$. Since this inequality for any $s \in \mathcal{S}^y$ and ℓ, we get

$$\|U(s;\eta) - U(s;\eta')\|_\infty \leq \beta \frac{2\overline{\mathcal{H}}}{\underline{\eta}} \|\eta - \eta'\|_\infty.$$

Hence,

$$\|\mathcal{M}[\eta] - \mathcal{M}[\eta']\|_\infty \leq \frac{\bar{n}^y}{\underline{\eta}} \overline{\mathcal{H}} \vartheta \beta \|\eta - \eta'\|_\infty.$$

Lastly, since H is continuous on compact $(\mathcal{S}^y)^2$, $\overline{\mathcal{H}} < \infty$. It follows that $\mathcal{M}$ is indeed a contraction when $\vartheta\beta$ is small. Banach's fixed point theorem implies that $\mathcal{M}$ has at most one fixed point in $\mathcal{K}$ — which exists by Proposition 1. □

Equipped with Lemma D.2, I prove prove uniqueness in two steps. First, I show that, given housing prices p, the allocation of young workers is unique (Lemma D.2). Second, after substituting this unique allocation into the housing market clearing condition, I show the resulting operator is a contraction.

Given Lemma D.2, it remains to prove the second step. For every $p \in \mathcal{P}$, define the reduced housing price operator by

$$\widehat{\mathcal{Q}}_\ell(p) = \mathcal{Q}[\eta(p), p] = \frac{1}{\phi} \left(\frac{\alpha \widehat{N}_\ell(p)}{\bar{H}} \right)^{(1-\phi)/\phi},$$

where $\widehat{N}_\ell(p) \equiv N_\ell(\eta(p), p)$. A unique fixed point exists if, for any $p, p' \in \mathcal{P}$, there exists $\chi < 1$ such that

$$\left\| \widehat{\mathcal{Q}}(p) - \widehat{\mathcal{Q}}(p') \right\|_\infty \leq \chi \|p - p'\|_\infty,$$

where $\chi < 1$.

Define $q(x) = \frac{1}{\phi} \left(\frac{\alpha x}{\bar{H}} \right)^{(1-\phi)/\phi}$. This function is continuously differentiable and increasing on $[\underline{\eta}|\mathcal{S}^y|, 2]$. Let $C_Q = \sup_{x \in [\underline{\eta}|\mathcal{S}^y|, 2]} q'(x)$. The mean-value theorem implies

$$\|\widehat{\mathcal{Q}}(p) - \widehat{\mathcal{Q}}(p')\|_\infty \leq C_Q \|\widehat{N}(p) - \widehat{N}(p')\|_\infty. \tag{40}$$

By the triangular inequality:

$$\|N(\eta(p),p) - N(\eta(p'),p')\| \leq \|N(\eta(p),p) - N(\eta(p'),p)\| + \|N(\eta(p'),p) - N(\eta(p'),p')\|. \tag{41}$$

I first bound the second term, before returning to the first.

Fix $\eta \in \mathcal{K}$ and take two price vectors $(p, p') \in \mathcal{P}$. Since η is fixed, the change in population

depends on the old population. Define the probability that a young worker with skill s_y locates in ℓ in their old age when they meet someone s_p in their youth:

$$R_{\ell,p}(s_y, s_p) = \int P^o_{\ell,p}\left(e\gamma(s_y, s_p)\right) dF(e). \tag{42}$$

The old population in city ℓ can then be written as

$$N^o_\ell(\eta, p) = \sum_l^L \int_{\mathcal{S}^y}\int_{\mathcal{S}^y} \eta_l(s_y)\pi^\eta_l(s_p)R_{\ell,p}(s_y, s_p)ds_p ds_y.$$

The utility of old workers solely depend on housing prices. Hence, Lemma D.1 implies

$$\left|P^o_{\ell,p}(s_o) - P^o_{\ell,p'}(s_o)\right| \le \frac{\vartheta\alpha}{2}\|p - p'\|_\infty.$$

It follows that

$$\left|R_{\ell,p}(s_y, s_p) - R_{\ell,p'}(s_y, s_p)\right| \le \frac{\vartheta\alpha}{2}\|p - p'\|_\infty,$$

for every (s_y, s_p). Since $\sum_l^L \int_{\mathcal{S}^y}\int_{\mathcal{S}^y} \eta_l(s_y)\pi^\eta_l(s_p)ds_p ds_y = 1$, I obtain

$$|N_\ell(\eta, p) - N_\ell(\eta, p')| = |N^o_\ell(\eta, p) - N^o_\ell(\eta, p')| \le \frac{\vartheta\alpha}{2}\|p - p'\|_\infty. \tag{43}$$

I now bound the first term in (41). First, for any $(\eta, \eta') \in \mathcal{K}$, (42) implies that the number of old workers in ℓ satisfy

$$|N^o_\ell(\eta, p) - N^o_\ell(\eta', p)| \le L|\mathcal{S}^y|\|\eta - \eta'\|_\infty + \frac{2}{\underline{\eta}}\|\eta - \eta'\|_\infty.$$

Meanwhile, the young population obeys

$$|N^y_\ell(\eta) - N^y_\ell(\eta')| = \left|\int_{\mathcal{S}^y}\left[\eta_\ell(s_y) - \eta'_\ell(s_y)\right] ds_y\right| \le |\mathcal{S}^y|\|\eta - \eta'\|_\infty.$$

Consequently,

$$|N_\ell(\eta, p) - N_\ell(\eta', p)| \le \left((L+1)|\mathcal{S}^y| + \frac{2}{\underline{\eta}}\right)\|\eta - \eta'\|_\infty. \tag{44}$$

I now relate the change in local density η to housing prices. Since old workers' utilities only depend on housing prices, we have $\left|\mathcal{V}^o_p(s_o) - \mathcal{V}^o_{p'}(s_o)\right| \le \alpha\|p - p'\|_\infty$. It follows from the definition of H_p and Assumptions 1 and 2 that $\left|H_p(s_y, s_p) - H_{p'}(s_y, s_p)\right| \le \alpha\|p - p'\|_\infty$ for every $(s_y, s_p) \in (\mathcal{S}^y)^2$. Since π^η_ℓ integrates to one, we further have $\left|O_p(s_y, \eta_\ell) - O_{p'}(s_y, \eta_\ell)\right| \le \alpha\|p - p'\|_\infty$. Holding η fixed, the preceding bound gives $\|U(s_y; \eta, p) - U(s_y; \eta, p')\|_\infty \le \alpha(1+\beta)\|p - p'\|_\infty$. Lemma D.1 implies $|P_\ell(U) - P_\ell(U')| \le \frac{\vartheta}{2}\|U - U'\|_\infty$. Therefore,

$$\|\mathcal{M}[\eta, p] - \mathcal{M}[\eta, p']\|_\infty \le \frac{\bar{n}^y\vartheta\alpha(1+\beta)}{2}\|p - p'\|_\infty.$$

Using the equations for $\eta(p)$ and $\eta(p')$, we obtain

$$\begin{aligned}\|\eta(p)-\eta(p')\|_\infty &\leq \|\mathcal{M}[\eta(p),p]-\mathcal{M}[\eta(p'),p]\|_\infty + \|\mathcal{M}[\eta(p'),p]-\mathcal{M}[\eta(p'),p']\|_\infty \\ &\leq \kappa\|\eta(p)-\eta(p')\|_\infty + \frac{\bar{n}^y\vartheta\alpha(1+\beta)}{2}\|p-p'\|_\infty,\end{aligned}$$

where κ is the contraction constant in Lemma D.2. Re-arranging:

$$\|\eta(p)-\eta(p')\|_\infty \leq \frac{\bar{n}^y\vartheta\alpha(1+\beta)}{2(1-\kappa)}\|p-p'\|_\infty. \tag{45}$$

Plugging (45) into (44), and combining it with (43), I get

$$\|\widehat{N}(p)-\widehat{N}(p')\|_\infty \leq \left[\left((L+1)|\mathcal{S}^y|+\frac{2}{\underline{\eta}}\right)\frac{\bar{n}^y\vartheta\alpha(1+\beta)}{2(1-\kappa)}+\frac{\vartheta\alpha}{2}\right]\|p-p'\|_\infty. \tag{46}$$

Using it into (40):

$$\|\widehat{\mathcal{Q}}(p)-\widehat{\mathcal{Q}}(p')\|_\infty \leq C_Q\left[\left((L+1)|\mathcal{S}^y|+\frac{2}{\underline{\eta}}\right)\frac{\bar{n}^y\vartheta\alpha(1+\beta)}{2(1-\kappa)}+\frac{\vartheta\alpha}{2}\right]\|p-p'\|_\infty.$$

We know from Lemma D.2 that $\kappa = C_M\vartheta\beta$ for some constant C_M independent of ϑ. Hence, as $\vartheta\to 0$, the term in bracket tends toward zero. Therefore, for ϑ small enough, $\widehat{\mathcal{Q}}$ is a contraction on $\mathcal{P}$. Since $\widehat{\mathcal{Q}}(\mathcal{P})\subseteq\mathcal{P}$, Banach's implies that there exists a unique price vector $p^\star$ satisfying $p^\star = \widehat{\mathcal{Q}}(p^\star)$. Given $p^\star$, Lemma D.2 implies that the young allocation is uniquely given by $\eta(p^\star)$.

D.3 First Order Approximation

I study a first order approximation around $\log T_\ell \approx \log \bar{T}$. Throughout, I use $\partial_l \cdot_\ell = \partial \cdot_\ell / \partial \log T_l$ to denote the partial derivative of $\cdot_\ell$ w.r.t. $\log T_l$, evaluated at $\boldsymbol{T} = \bar{T}$.

At the point of approximation, the equilibrium is symmetric.

Proposition D.1.
Suppose $T_\ell = \bar{T}$ for all ℓ. Then, a symmetric equilibrium exists, and is unique if $\vartheta\beta$ is small enough.

Proof. Existence follows by guess and verify using $\pi^y_\ell = n^y$. Proposition 1 proves uniqueness. □

Let e_ℓ denote the employment share of location ℓ. The employment shares are common across young and old workers since learning opportunities are uniform across space at the symmetric equilibrium. The employment shares solve the fixed point:

$$e_\ell = \frac{\exp\{\vartheta(B_\ell - \alpha\bar{p}_\ell)\}}{\sum_{\ell'}\exp\{\vartheta(B_{\ell'}-\alpha\bar{p}_{\ell'})\}},$$

where symmetric housing prices are $\bar{p}_\ell = \frac{1}{\phi}\left(\frac{2\alpha e_\ell}{H}\right)^{(1-\phi)/\phi}$. Given these, the symmetric spatial allocation is described by $n^a_\ell(s) = e_\ell n^a(s) \equiv \bar{n}^a_\ell(s)$, $\pi^a_\ell(s) = n^a(s)$, and $N_\ell = 2e_\ell$. Define relative TFP as $\mathcal{T}_\ell \equiv \log T_\ell - \sum_{\ell'} e_{\ell'}\log T_{\ell'}$.

I start with the spatial allocation of old workers. Differentiating (5) w.r.t. $\log T_l$ and evaluating the derivative at $\boldsymbol{T} = \bar{T}$ yields

$$\partial_l n^o_\ell(s) = \theta\bar{n}^o_\ell(s)\left[sI_{\ell l} - P_{\ell l}\right], \tag{47}$$

where $I_{\ell l} = 1\{\ell = l\} - e_l$ and $P_{\ell l} \equiv \frac{\alpha}{\bar{T}} \left(\partial_l p_\ell - \sum_{\ell'} e_{\ell'} \partial_l p_{\ell'}\right)$. In the above, I have already guessed that there is no first-order effect of local TFPs on the aggregate skill distribution, $\partial_l n^o(s) = 0$. I verify this guess below. Hence, to first order,

$$n^o_\ell(s) \approx \bar{n}^o_\ell(s) \left(1 + \theta \left[s\, \mathcal{T}_\ell - \mathcal{P}_\ell\right]\right) \quad \text{where} \quad \mathcal{P}_\ell \equiv \sum_{\ell'} P_{\ell\ell'} \log T_{\ell'}.$$

Moving onto young workers, (6) implies

$$\partial_l n^y_\ell(s) = \theta \bar{n}^y_\ell(s) \left[s I_{\ell l} - P_{\ell l} + \frac{\beta}{\bar{T}} \left(\partial_l O_\ell(s) - \sum_{\ell'} e_{\ell'} \partial_l O_{\ell'}(s)\right)\right]. \tag{48}$$

It remains to compute the learning option value. Under symmetric local TFPs, $\mathcal{V}^o(s) = \bar{T}s + c$ for some constant c. Moreover, $\partial_l \mathcal{V}^o(s) = e_l \bar{T} s - \alpha \sum_{\ell'} e_{\ell'} \partial_l p_{\ell'}$. Differentiating O_ℓ, using $\int \partial_l \pi^y_\ell(s_p) ds_p = 0$, and taking the relative value of O_ℓ returns

$$\partial_l O_\ell(s) - \sum_{\ell'} e_{\ell'} \partial_l O_{\ell'}(s) = \bar{T} \int \gamma(s, s_p) \partial_l \pi^y_\ell(s_p) ds_p. \tag{49}$$

Meanwhile, the local skill densities move according to

$$\partial_l \pi^y_\ell(s_p) = \frac{1}{e_\ell} \left[\partial_l n^y_\ell(s_p) - n^y(s_p) \int \partial_l n^y_\ell(t) dt\right].$$

Combining this expression with (49), then plugging it into (48)

$$\partial_l n^y_\ell(s) = \theta \bar{n}^y_\ell(s) \left[s I_{\ell l} - P_{\ell l}\right] + \theta \beta n^y(s) \int \partial_l n^y_\ell(s_p) K(s, s_p) ds_p, \tag{50}$$

where the learning kernel is defined by $K(s, s_p) \equiv \gamma(s, s_p) - \mathbb{E}[\gamma(s, S^y)]$. In particular, the kernel satisfies $\int K(s, s_p) dN^y(s_p) = 0$.

Equation (50) defines an infinitely-dimensional linear fixed point in $\partial_l n^y_\ell(s)$. We guess and verify the form of its solution. Guess that

$$\partial_l n^y_\ell(s) = \theta \bar{n}^y_\ell(s) \left[I_{\ell l} \eta(s) - P_{\ell l}\right], \tag{51}$$

for some function η independent of ℓ and l. Plugging the guess into the right-hand side of (50),

$$= \theta \bar{n}^y_\ell(s) \left[s I_{\ell l} - P_{\ell l} + \theta \beta I_{\ell l} \int K(s, s_p) \eta(s_p) dN^y(s_p)\right].$$

This verifies the the guess with

$$\eta(s) = s + \theta \beta \Theta(s), \quad \text{where} \quad \Theta(s) = \int K(s, s_p) \eta(s_p) dN^y(s_p). \tag{52}$$

Given this solution, the local skill densities are governed by

$$\partial_l \pi^y_\ell(s) = \theta n^y(s) \left(\eta(s) - \mathbb{E}[\eta(S^y)]\right) I_{\ell l}. \tag{53}$$

Furthermore, to first order, the spatial allocation of young workers is

$$n_\ell^y(s) \approx \bar{n}_\ell^y(s)\left(1+\theta\left[\eta(s)\mathcal{T}_\ell - \mathcal{P}_\ell\right]\right).$$

It remains to solve for housing prices. Integrating (47) and (51) over skills gives

$$\partial_l N_\ell^o = \theta e_\ell\left[\mathbb{E}[S^o]I_{\ell l} - P_{\ell l}\right] \quad \text{and} \quad \partial_l N_\ell^y = \theta e_\ell\left[\mathbb{E}[\eta(S^y)]I_{\ell l} - P_{\ell l}\right].$$

Hence $\partial_l N_\ell = 2\theta e_\ell\left[\mu I_{\ell l} - P_{\ell l}\right]$, where $\mu \equiv \frac{1}{2}\left(\mathbb{E}[S^o] + \mathbb{E}[\eta(S^y)]\right)$. Differentiating (8) and evaluating it at $\boldsymbol{T} = \bar{T}$:

$$\partial_l p_\ell = \theta\left(\frac{1-\phi}{\phi}\right)\bar{p}_\ell\left[\mu I_{\ell l} - P_{\ell l}\right]. \tag{54}$$

Since $P_{\ell l}$ depends on the price derivatives $\partial_l p_\ell$, (54) defines a fixed point. Define

$$\lambda_\ell \equiv \frac{\vartheta\alpha\left(\frac{1-\phi}{\phi}\right)\bar{p}_\ell}{1+\vartheta\alpha\left(\frac{1-\phi}{\phi}\right)\bar{p}_\ell} \in (0,1).$$

Plugging the expression for $P_{\ell l}$ into (54), summing (54) across locations and re-arranging, we obtain

$$P_{\ell l} = \mu\left[\lambda_\ell I_{\ell l} + (1-\lambda_\ell)\sum_{\ell'}\omega_{\ell'} I_{\ell' l}\right],$$

where the weights are $\omega_\ell \equiv \frac{e_\ell(1-\lambda_\ell)}{\sum_{\ell'} e_{\ell'}(1-\lambda_{\ell'})}$. Hence, to first order, relative housing prices are given by

$$\mathcal{P}_\ell = \mu\left[\lambda_\ell \mathcal{T}_\ell + (1-\lambda_\ell)\sum_l \omega_l \mathcal{T}_l\right]. \tag{55}$$

Under homogeneous amenities, $\bar{p}_\ell \propto 1$, $e_\ell \propto 1$, and (55) simplifies to $\mathcal{P}_\ell = \mu\lambda\mathcal{T}_\ell$, thereby leading to (13) and (14) in the main text after the normalization $\mu = 1$.

I conclude the derivation by verifying that there is no first-order effect on the skill distribution of old workers. Differentiating the skill density:

$$\partial_l n^o(s_o) = \sum_\ell \int\int \frac{1}{\gamma(s_y,s_p)} f\left(\frac{s_o}{\gamma(s_y,s_p)}\right)\left[\partial_l n_\ell^y(s_y) n^y(s_p) + e_\ell n^y(s_y)\partial_l \pi_\ell^y(s_p)\right] ds_p ds_y.$$

By feasibility, $\sum_\ell \partial_l n_\ell^y(s) = \partial_l n^y(s) = 0$. Furthermore, $\partial_l \pi_\ell^y(s) = \frac{1}{e_\ell}\left[\partial_l n_\ell^y(s) - n^y(s)\partial_l N_\ell^y\right]$, so that feasibility also implies $\sum_\ell e_\ell \partial_l \pi_\ell^y(s) = 0$. Combined, $\partial_l n^o(s) = 0$, and there is no first-order effect.

D.4 Existence and Uniqueness of the Solution to the Learning Integral Equation

The first order approximation is well-defined so long as the infinitely-dimensional fixed point (52) has a solution. Lemma D.3 proves that this integral equation has a unique solution provided either $\vartheta\beta$ or the learning spillovers are not too large. Lemma D.4 further bound the solution. For the two

lemmas to follow, define the operator $T : L^2(dN^y) \to L^2(dN^y)$ by

$$Th(s) = \int K(s,t)h(t)dN^y(t).$$

Lemma D.3.
Equation (52) *has a unique solution if*

$$\theta\beta \left(\int \int \left| \gamma(s, s_p) - \mathbb{E}[\gamma(s, S^y)] \right|^2 dN^y(s_p) dN^y(s) \right)^{1/2} < 1. \tag{56}$$

This solution is continuous and bounded on $[\underline{s}, \bar{s}]$.

Proof. Throughout the proof, we use $L^2(dN^y)$ norms. Omit the y superscript. Consider the Fredholm equations of the second kind $\eta^f = f + \theta\beta T\eta^f$. This fixed point has a unique solution in $L^2(dN)$ if T is a bounded operator and $\theta\beta\|T\| < 1$. Since γ is continuous on compact support $\mathcal{S}^y$, $\|K\| < \infty$ and $\|S^y\| < \infty$. Cauchy–Schwarz then implies $\|Th\| \leq \|K\|\|h\|$ for all $h \in L^2(dN)$. Hence T is a bounded operator and $\theta\beta\|T\| \leq \theta\beta\|K\|$. Condition (56) is sufficient for $\theta\beta\|T\| < 1$ to hold. Taking $f(s) = s$, (52) has a unique solution. It remains to show continuity and boundedness. Since K is bounded and continuous in its first argument, and since $\eta \in L^2(dN)$, $s \mapsto T\eta(s) = \int K(s,t)\eta(t)dN(t)$ is bounded and continuous. Hence so is η. □

Lemma D.4 (Bound)**.**
The solution η *to* (52) *satisfies*

$$\|\eta - \mathbb{E}[\eta(S^y)]\| \leq \frac{\mathbb{S}d[S^y]}{1 - \theta\beta\|K\|}.$$

The inequality holds along the sequence $\eta_0(s) = s$ *and* $\eta_{m+1}(s) = s + \theta\beta T\eta_m(s)$ *for* $m \geq 0$.

Proof. Omit the y superscript. Consider the solution η. Subtracting $\mathbb{E}[S^y]$ from (52): $\eta - \mathbb{E}[S^y] = s - \mathbb{E}[S^y] + \theta\beta T\eta$. Since T is centered in its second argument, we get $\eta - \mathbb{E}[S^y] = s - \mathbb{E}[S^y] + \theta\beta T(\eta - \mathbb{E}[S^y])$. Taking norms and using $\|Th\| \leq \|K\|\|h\|$ yields $\|\eta - \mathbb{E}[S^y]\| \leq \mathbb{S}\mathrm{d}[S^y] + \theta\beta\|K\|\|\eta - \mathbb{E}[S^y]\|$, or $\|\eta - \mathbb{E}[S^y]\| \leq \frac{\mathbb{S}\mathrm{d}[S^y]}{1-\theta\beta\|K\|}$ afer re-arranging. Since $\mathbb{E}[\eta(S^y)]$ minimizes the $L^2(dN)$ distance between η and a constant, $\|\eta - \mathbb{E}[\eta(S^y)]\| \leq \|\eta - \mathbb{E}[S^y]\|$. The argument for the sequences proceeds similarly. □

D.5 Second Order Approximation

We now consider a second order approximation around $\boldsymbol{T} = \bar{T}$ to characterize between-city wage variance. We use similar notations as for the first-order in Section D.3.

Let $\log W_\ell^y$ denote the log wage of a randomly drawn young worker in city ℓ. The expectation of this random variable is $\mathbb{E}[\log W_\ell^y] = \log T_\ell + \int \log s\, \pi_\ell^y(s)ds$. The between-city wage variance is

$$\mathbb{V}\mathrm{ar}[\mathbb{E}(\log W_\ell^y)] = \sum_\ell \left(\mathbb{E}[\log W_\ell^y] - \mathbb{E}[\log W^y] \right)^2 N_\ell^y,$$

where $\mathbb{E}[\log W^y] \equiv \sum_\ell N_\ell^y \mathbb{E}[\log W_\ell^y]$. At $\boldsymbol{T} = \bar{T}$, $\mathbb{E}[\log W_\ell^y] = \mathbb{E}[\log W^y]$ for every ℓ. Hence the first derivative is zero. Differentiating twice and evaluating at $\boldsymbol{T} = \bar{T}$ leaves only the product of the first

derivatives of city average wages:

$$\partial^2_{ll'}\mathbb{V}\text{ar}[\mathbb{E}(\log W^y_\ell)] = 2\sum_\ell e_\ell \left[\partial_{l'}\mathbb{E}[\log W^y_\ell] - \partial_{l'}\mathbb{E}[\log W^y]\right]\left[\partial_l\mathbb{E}[\log W^y_\ell] - \partial_l\mathbb{E}[\log W^y]\right]. \quad (57)$$

The derivative of the local wage is $\partial_l\mathbb{E}[\log W^y_\ell] = e_l + I_{\ell l}\left[1 + \theta\mathbb{C}\text{ov}[\log S^y, \eta(S^y)]\right]$. The derivative of the aggregate wage is

$$\partial_l\mathbb{E}[\log W^y] = \sum_\ell \partial_l N^y_\ell \mathbb{E}[\log W^y_\ell] + \sum_\ell e_\ell \partial_l\mathbb{E}[\log W^y_\ell] = e_l.$$

It follows that $\partial_l\mathbb{E}[\log W^y_\ell] - \partial_l\mathbb{E}[\log W^y] = I_{\ell l}\left[1 + \theta\mathbb{C}\text{ov}[\log S^y, \eta(S^y)]\right]$. Substituting into (57) gives

$$\partial^2_{ll'}\mathbb{V}\text{ar}[\mathbb{E}(\log W^y_\ell)] = 2e_l I_{ll'}\left[1 + \theta\mathbb{C}\text{ov}[\log S^y, \eta(S^y)]\right]^2.$$

To second order, the between-city wage inequality is therefore

$$\mathbb{V}\text{ar}[\mathbb{E}(\log W^y_\ell)] \approx \mathbb{V}\text{ar}_{e_\ell}[\log T_\ell]\left[1 + \theta\mathbb{C}\text{ov}[\log S^y, \eta(S^y)]\right]^2.$$

D.6 Planner's Problem

Let $\mathbb{E}_\ell[V^{y\star}_t(s,\boldsymbol{\varepsilon}) \mid s]$ denote the expected utility of young workers with skill s working in city ℓ at time t, where the expectation is taken across the unobserved idiosyncratic preferences $\boldsymbol{\varepsilon}$. Likewise, define $\mathbb{E}_\ell[V^{o\star}_t(s,\boldsymbol{\varepsilon}) \mid s]$ for old workers. Further, suppose that, within a cohort, the planner places equal weights across workers and cities. Across cohorts, the planner discounts the utilities of future generations at rate β_S. Lastly, the planner places weight ω^L_ℓ on the landowners in city ℓ. Then, a utilitarian planner has has social welfare function:

$$\int \sum_\ell \mathbb{E}_\ell[V^{o\star}_0(s,\boldsymbol{\varepsilon}) \mid s] n^{o\star}_{0\ell}(s)ds + \sum_{t\geq 0}\beta^t_S\left\{\int\sum_\ell \mathbb{E}_\ell[V^{y\star}_t(s,\boldsymbol{\varepsilon}) \mid s] n^{y\star}_{t\ell}(s)ds + \sum_\ell \omega^L_\ell c^{L\star}_{t\ell}\right\}.$$

This function generalizes (17) for non-unitary weights across cohorts and workers vs. land-owners.

The expected utility of workers must be consistent with their location choices. Housing consumption is a technological requirement rather a preference. Accordingly, housing do not enter workers' utility. By the property of extreme value type one distribution, the expected utility of old workers working in city ℓ is identical to their *ex-ante* expected utility:

$$\mathbb{E}_\ell[V^{o\star}_t(s,\boldsymbol{\varepsilon}) \mid s] = \mathbb{E}[V^{o\star}_t(s,\boldsymbol{\varepsilon}) \mid s] = \frac{1}{\vartheta}\log\left(\sum_\ell \mathrm{e}^{\vartheta\left(c^{o\star}_{t\ell}(s) + B^o_\ell\right)}\right) \equiv \mathcal{V}^{o\star}_t(s).$$

Similarly for the young,

$$\mathcal{V}^{y\star}_t(s) = \frac{1}{\vartheta}\log\left(\sum_\ell \mathrm{e}^{\vartheta\left(c^{y\star}_{t\ell}(s) + B^y_\ell + \beta O^\star_{t\ell}(s)\right)}\right),$$
$$O^\star_{t\ell}(s) = \int\int \mathcal{V}^{o\star}_{t+1}[e\gamma(s,s_p)]dF(e)\pi^{y\star}_{t\ell}(s_p)ds_p.$$

Hence, the social welfare function simplifies to

$$\int \mathcal{V}_0^{o\star}(s) \sum_\ell n_{0\ell}^{o\star}(s) ds + \sum_{t\geq 0} \beta_S^t \left\{ \int \mathcal{V}_t^{y\star}(s) \sum_\ell n_{t\ell}^{y\star}(s) ds + \sum_\ell \omega_\ell^L c_{t\ell}^{L\star} \right\}.$$

The problem of the planner is to maximize this objective function by choosing a sequence $\left\{ \boldsymbol{c}_t^{y\star}, \boldsymbol{c}_t^{o\star}, \boldsymbol{c}_t^{L\star}, \boldsymbol{n}_t^{y\star}, \boldsymbol{n}_t^{o\star}, n_{t+1}^{o\star}, \boldsymbol{K}_t^\star \right\}_{t=0}^\infty$ taking the initial old skill distribution n_0^o as given. When choosing this allocation, the planner faces the following constraints, where the variables in parentheses refer to the associated multipliers:[29]

1. Aggregate numéraire consumption feasibility (κ_t):

$$\int \sum_\ell \Big\{ \left(c_{t\ell}^{y\star}(s) - T_\ell s \right) n_{t\ell}^{y\star}(s) + \left(c_{t\ell}^{o\star}(s) - T_\ell s \right) n_{t\ell}^{o\star}(s) \Big\} ds + \sum_\ell \left(c_{t\ell}^{L\star} + K_{t\ell}^\star \right) \leq 0, \qquad \forall t.$$

2. Housing feasibility in city ℓ ($\rho_{t\ell}$):

$$\alpha \int \left(n_{t\ell}^{y\star}(s) + n_{t\ell}^{o\star}(s) \right) ds \leq \left(K_{t\ell}^\star \right)^\phi \bar{H}^{1-\phi}, \qquad \forall (t, \ell). \tag{58}$$

3. Young labor supply ($\lambda_{t\ell}^y(s)$):

$$n_{t\ell}^{y\star}(s) \leq n^y(s) \left(\frac{\mathrm{e}^{\vartheta \left(c_{t\ell}^{y\star}(s) + B_\ell^y + \beta O_{t\ell}^\star(s) \right)}}{\sum_{\ell'} \mathrm{e}^{\vartheta \left(c_{t\ell'}^{y\star}(s) + B_{\ell'}^y + \beta O_{t\ell'}^\star(s) \right)}} \right), \quad \forall (s, t, \ell). \tag{59}$$

4. Old labor supply ($\lambda_{t\ell}^o(s)$):

$$n_{t\ell}^{o\star}(s) \leq n_t^{o\star}(s) \left(\frac{\mathrm{e}^{\vartheta \left(c_{t\ell}^{o\star}(s) + B_\ell^o \right)}}{\sum_{\ell'} \mathrm{e}^{\vartheta \left(c_{t\ell'}^{o\star}(s) + B_{\ell'}^o \right)}} \right), \quad \forall (s, t, \ell). \tag{60}$$

5. The law of motion for the skill distribution of old workers ($\nu_t(s)$):

$$n_{t+1}^{o\star}(s_o) = \sum_\ell \int \int \frac{1}{\gamma(s_y, s_p)} f \left(\frac{s_o}{\gamma(s_y, s_p)} \right) n_{t\ell}^{y\star}(s_y) \pi_{t\ell}^{y\star}(s_p) ds_p ds_y, \quad \forall (s_o, t). \tag{61}$$

[29] Implicit to these constraints, $\pi_{t\ell}^{y\star}$ is a function of $n_{t\ell}^{y\star}$, and $O_{t\ell}^\star(s)$ is a function of $\pi_{t\ell}^{y\star}$.

The Lagrangian associated with the planner's problem is

$$\mathcal{L} = \frac{1}{\vartheta}\int \log\left(\sum_{\ell} \mathrm{e}^{\vartheta\left(c^{o\star}_{0\ell}(s)+B^{o}_{\ell}\right)}\right)\sum_{\ell} n^{o\star}_{0\ell}(s)ds + \frac{1}{\vartheta}\sum_{t\geq 0}\beta_S^t \int \log\left(\sum_{\ell} \mathrm{e}^{\vartheta\left(c^{y\star}_{t\ell}(s)+B^{y}_{\ell}+\beta O^{\star}_{t\ell}(s)\right)}\right)\sum_{\ell} n^{y\star}_{t\ell}(s)ds +$$
$$\sum_{t\geq 0}\beta_S^t \sum_{\ell}\omega^L_{\ell} c^{L\star}_{t\ell} -$$
$$\sum_{t}\sum_{\ell}\int \lambda^{y}_{t\ell}(s)\left\{n^{y\star}_{t\ell}(s) - n^{y}(s)\left(\frac{\mathrm{e}^{\vartheta\left(c^{y\star}_{t\ell}(s)+B^{y}_{\ell}+\beta O^{\star}_{t\ell}(s)\right)}}{\sum_{\ell'}\mathrm{e}^{\vartheta\left(c^{y\star}_{t\ell'}(s)+B^{y}_{\ell'}+\beta O^{\star}_{t\ell'}(s)\right)}}\right)\right\}ds -$$
$$\sum_{t}\sum_{\ell}\int \lambda^{o}_{t\ell}(s)\left\{n^{o\star}_{t\ell}(s) - n^{o\star}_{t}(s)\left(\frac{\mathrm{e}^{\vartheta\left(c^{o\star}_{t\ell}(s)+B^{o}_{\ell}\right)}}{\sum_{\ell'}\mathrm{e}^{\vartheta\left(c^{o\star}_{t\ell'}(s)+B^{o}_{\ell'}\right)}}\right)\right\}ds -$$
$$\sum_{t}\int \nu_t(s_o)\left\{n^{o\star}_{t+1}(s_o) - \sum_{\ell}\int\int \frac{1}{\gamma(s_y,s_p)} f\left(\frac{s_o}{\gamma(s_y,s_p)}\right) n^{y\star}_{t\ell}(s_y)\pi^{y\star}_{t\ell}(s_p)ds_p ds_y\right\}ds_o -$$
$$\sum_{t}\kappa_t\left[\int\sum_{\ell}\left\{\left(c^{y\star}_{t\ell}(s) - T_\ell s\right)n^{y\star}_{t\ell}(s) + \left(c^{o\star}_{t\ell}(s) - T_\ell s\right)n^{o\star}_{t\ell}(s)\right\}ds + \sum_{\ell}\left(c^{L\star}_{t\ell} + K^{\star}_{t\ell}\right)\right] -$$
$$\sum_{t}\sum_{\ell}\rho_{t\ell}\left[\alpha\int\left(n^{y\star}_{t\ell}(s) + n^{o\star}_{t\ell}(s)\right)ds - \left(K^{\star}_{t\ell}\right)^{\phi}\bar{H}^{1-\phi}\right].$$

The necessary optimality conditions for this maximization problem are derived with respect to the functions $s \to c^{y\star}_{t\ell}(s)$, $s \to c^{o\star}_{t\ell}(s)$, $s \to n^{y\star}_{t\ell}(s)$, $s \to n^{o\star}_{t\ell}(s)$, and $s \to n^{o\star}_{t+1}(s)$, and with respect to the vectors $c^{L\star}_{t\ell}$ and $K^{\star}_{t\ell}$. The worker side is an infinite-dimensional maximization problem, and I derive its necessary optimality conditions using functional derivatives. For the remainder of the derivation, I drop the star superscript for notational simplicity, and I anticipate that the labor supply constraints (59) and (60) and the housing feasibility constraints (58) bind at the optimum.

Landowners. The Lagrangian is linear in the consumption of the landowners. An interior solution requires $\kappa_t = \beta_S^t \omega^L_{\ell}$. This holds for every city iff $\omega^L_{\ell} = \omega^L$, and thus $\kappa_t = \beta_S^t \omega^L$. Furthermore, we can already anticipate that the worker's optimality condition will imply $\kappa_t = \beta_S^t$. Hence, an interior solution requires $\omega^L = 1$. I impose this restriction from now on.

The optimality condition w.r.t. $K_{t\ell}$ is $\kappa_t = \rho_{t\ell}\phi K^{\phi-1}_{t\ell}\bar{H}^{1-\phi}$. Meanwhile, (58) demands $\alpha N_{t\ell} = K^{\phi}_{t\ell}\bar{H}^{1-\phi}$, where $N_{t\ell} \equiv \int\left(n^{y}_{t\ell}(s) + n^{o}_{t\ell}(s)\right)ds > 0$ is total population. Combining the optimality and feasibility conditions:

$$p_{t\ell} \equiv \frac{\rho_{t\ell}}{\kappa_t} = \frac{1}{\phi}\left(\frac{\alpha N_{t\ell}}{\bar{H}}\right)^{(1-\phi)/\phi}. \tag{62}$$

Initial old workers. Suppose the planner contemplates an alternative consumption plan given by $\tilde{c}^{o}_{0l}(s) = c^{o}_{0l}(s) + \xi h(s)$ for some l and some arbitrary admissible function $h(s)$, while $\tilde{c}^{o}_{0\ell}(s) = c^{o}_{0\ell}(s)$ for $\ell \neq l$. Optimality requires that $d\mathcal{L}/d\xi = 0$ at $\xi = 0$, or

$$\int h(s)n^{o}_{0l}(s)ds + \vartheta\int h(s)n^{o}_{0l}(s)\left(\lambda^{o}_{0l}(s) - \sum_{\ell}\lambda^{o}_{0\ell}(s)\frac{n^{o}_{0\ell}(s)}{n^{o}_{0}(s)}\right)ds - \kappa_0\int h(s)n^{o}_{0l}(s)ds = 0.$$

Since this must hold for every admissible function h, its integrand must equal zero almost everywhere. In addition, an interior allocation is guaranteed by the idiosyncratic location preferences. Accordingly,

$$1+\vartheta\left(\lambda_{0l}^{o}(x)-\sum_{\ell}\lambda_{0\ell}^{o}(x)\frac{n_{0\ell}^{o}(x)}{n_{0}^{o}(x)}\right)=\kappa_{0}.$$

In the above, only $\lambda_{0l}^{o}(x)$ depends on l. Hence, $\lambda_{0l}^{o}(x)=\lambda_{0}^{o}(x)$, and we conclude that $\kappa_0=1$.

Proceeding in a similar fashion for the optimality condition regarding $n_{0l}^{o}(s)$:

$$c_{0l}^{o}(x)-(T_{l}x-\alpha p_{0l})=\mathcal{V}_{0}^{o}(x)-\lambda_{0}^{o}(x).$$

The right-hand side is independent of l. Therefore, conditional on the same housing prices, the optimal allocation of the initial old workers is the same as in the laissez-faire equilibrium.

Old workers. I move to the optimality conditions for $(\boldsymbol{n}_{t}^{o},n_{t\ell}^{o},c_{t\ell}^{o})$ for $t\geq 1$. The optimality condition w.r.t. c_{tl}^{o} is

$$\begin{aligned}\kappa_{t}\int n_{tl}^{o}(s)h(s)ds=\beta_{S}^{t-1}\beta\int\sum_{\ell}n_{t-1\ell}^{y}(s)D_{c_{tl}^{o}}O_{t-1\ell}[h](s)ds+\\ \vartheta\beta\sum_{\ell}\int\lambda_{t-1\ell}^{y}(s)n_{t-1\ell}^{y}(s)\left(D_{c_{tl}^{o}}O_{t-1\ell}[h](s)-\sum_{\ell'}\frac{n_{t-1\ell'}^{y}(s)}{n^{y}(s)}D_{c_{tl}^{o}}O_{t-1\ell'}[h](s)\right)ds+\\ \vartheta\int n_{tl}^{o}(s)h(s)\left(\lambda_{tl}^{o}(s)-\sum_{\ell}\lambda_{t\ell}^{o}(s)\frac{n_{t\ell}^{o}(s)}{n_{t}^{o}(s)}\right)ds,\end{aligned}$$

where $D_{c_{tl}^{o}}O_{t-1\ell}[h](s)$ denotes the functional derivative of $O_{t-1\ell}(s)$ with respect to c_{tl}^{o} in the direction h. Anticipating $\lambda_{t-1\ell}^{y}(s)=\lambda_{t-1}^{y}(s)$, the second line vanishes. From the expression for O, we have

$$D_{c_{tl}^{o}}O_{t-1\ell}[h](s)=\int\int h[e\gamma(s,s_{p})]\frac{n_{tl}^{o}[e\gamma(s,s_{p})]}{n_{t}^{o}[e\gamma(s,s_{p})]}dF(e)\pi_{t-1\ell}^{y}(s_{p})ds_{p}.$$

Combined, the optimality condition becomes

$$\begin{aligned}\kappa_{t}\int n_{tl}^{o}(s)h(s)ds=\beta_{S}^{t-1}\beta\int h(q)\frac{n_{tl}^{o}(q)}{n_{t}^{o}(q)}\sum_{\ell}\int\int\frac{1}{\gamma(s,s_{p})}f\left(\frac{q}{\gamma(s,s_{p})}\right)n_{t-1\ell}^{y}(s)\pi_{t-1\ell}^{y}(s_{p})ds_{p}dsdq+\\ \vartheta\int n_{tl}^{o}(s)h(s)\left(\lambda_{tl}^{o}(s)-\sum_{\ell}\lambda_{t\ell}^{o}(s)\frac{n_{t\ell}^{o}(s)}{n_{t}^{o}(s)}\right)ds,\end{aligned}$$

where I have used the change of variable $e\gamma(s,s_p)\to q$ in the first line. By (61),

$$\sum_{\ell}\int\int\frac{1}{\gamma(s,s_{p})}f\left(\frac{q}{\gamma(s,s_{p})}\right)n_{t-1\ell}^{y}(s)\pi_{t-1\ell}^{y}(s_{p})ds_{p}ds=n_{t}^{o}(q).$$

Plugging this into the optimality condition, focusing on the interior allocation guaranteed by the idiosyncratic location preferences, and using that this condition must hold for every admissible

function h, we obtain

$$\kappa_t = \beta_S^{t-1}\beta + \vartheta\left(\lambda_{tl}^o(x) - \sum_\ell \lambda_{t\ell}^o(x)\frac{n_{t\ell}^o(x)}{n_t^o(x)}\right).$$

Only $\lambda_{tl}^o(x)$ depends on l. Hence, $\lambda_{tl}^o(x) = \lambda_t^o(x)$ for all l, and $\kappa_t = \beta_S^{t-1}\beta$.

Turning to the optimality condition with respect to n_{tl}^o, note that $O_{t-1\ell}(s)$ is independent of n_{tl}^o. Proceeding as above and using $\lambda_{tl}^o(x) = \lambda_t^o(x)$:

$$c_{tl}^o(x) - (T_l x - \alpha p_{tl}) = -\frac{\lambda_t^o(x)}{\kappa_t}. \tag{63}$$

There as well, the last term is independent of l, and conditional on skills and housing prices, the optimal allocation of old workers is the same as in the laissez-faire equilibrium.

Finally, consider the optimality condition with respect to the aggregate old skill density n_{t+1}^o. Because it is a density, any admissible perturbation must satisfy $\int h(s)ds = 0$. Optimality demands:

$$0 = \sum_\ell \int \lambda_{t+1\ell}^o(s)h(s)\frac{n_{t+1\ell}^o(s)}{n_{t+1}^o(s)}ds - \int \nu_t(s)h(s)ds = \int \left[\lambda_{t+1}^o(s) - \nu_t(s)\right]h(s)ds,$$

where the second equality uses $\lambda_{t+1\ell}^o(s) = \lambda_{t+1}^o(s)$ for all ℓ. Therefore, there exists a scalar a_t such that $\lambda_{t+1}^o(s) - \nu_t(s) = a_t$ for almost every s.

Young workers. I conclude with the optimality conditions for $(n_{t\ell}^y, c_{t\ell}^y)$ for $t \geq 0$. For consumption, note that $O_{tl}(s)$ is independent of c_{tl}^y. Hence, the optimality condition w.r.t. c_{tl}^y is

$$\beta_S^t + \vartheta\left(\lambda_{tl}^y(x) - \sum_\ell \lambda_{t\ell}^y(x)\frac{n_{t\ell}^y(x)}{n^y(x)}\right) = \kappa_t.$$

Only $\lambda_{tl}^y(x)$ depends on l, and we conclude that $\lambda_{tl}^y(x) = \lambda_t^y(x)$ for all l, and $\kappa_t = \beta_S^t$. The optimality condition for old workers instead gives $\kappa_t = \beta_S^{t-1}\beta$ for $t \geq 1$. Consequently, an interior solution requires $\beta_S = \beta$. I impose this condition from now on, and $\kappa_t = \beta^t$.

Moving onto n_{tl}^y. Take $\tilde{n}_{tl}^y(s) = n_{tl}^y(s) + \xi h(s)$, the optimality condition is

$$0 = \beta^{t+1}\sum_\ell \int n_{t\ell}^y(s)D_{n_{tl}^y}O_{t\ell}[h](s)ds + \beta^t \int \left(\mathcal{V}_t^y(s) + sT_l - \alpha p_{tl} - c_{tl}^y(s)\right)h(s)ds - \int \lambda_{tl}^y(s)h(s)ds +$$
$$\int \nu_t(s_o)\int\int \frac{1}{\gamma(s_y,s_p)}f\left(\frac{s_o}{\gamma(s_y,s_p)}\right)\left\{h(s_y)\pi_{tl}^y(s_p) + \pi_{tl}^y(s_y)\left[h(s_p) - \pi_{tl}^y(s_p)\int h(\sigma)d\sigma\right]\right\}ds_p ds_y ds_o.$$

where I have already used $\lambda_{t\ell}^y(s) = \lambda_t^y(s)$ to simplify the expression. From the expression for $O_{t\ell}(s)$,

$$D_{n_{tl}^y}O_{t\ell}[h](s) = \frac{1\{\ell = l\}}{N_{tl}^y}\left(\int\int \mathcal{V}_{t+1}^o[e\gamma(s,s_p)]dF(e)h(s_p)ds_p - O_{tl}(s)\int h(\sigma)d\sigma\right),$$

where $N_{t\ell}^y$ is defined analogously to N_ℓ^y. Substituting this derivative into the optimality condition, rearranging the integrals, and using that the resulting condition must hold for every admissible

perturbation h, we obtain

$$\begin{aligned} c^y_{tl}(x) = xT_l - \alpha p_{tl} + \mathcal{V}^y_t(x) - \frac{\lambda^y_t(x)}{\beta^t} + \\ \beta \int \left(\int \mathcal{V}^o_{t+1}[e\gamma(s,x)]dF(e) - O_{tl}(s) \right) \pi^y_{tl}(s)ds + \\ \frac{1}{\beta^t} \int \int \nu_t[e\gamma(x,s_p)]dF(e)\pi^y_{tl}(s_p)ds_p + \\ \frac{1}{\beta^t} \int \left\{ \int \nu_t[e\gamma(s,x)]dF(e) - \int \int \nu_t[e\gamma(s,s_p)]dF(e)\pi^y_{tl}(s_p)ds_p \right\} \pi^y_{tl}(s)ds. \end{aligned} \tag{64}$$

From there, we proceed in two steps. We first show that the spatial allocation of young workers is independent of transfers within old workers and between young and old workers. We then obtain the final expression by simplifying (64) in the no-transfer case.

From the optimality condition for ν_t, we know that $\lambda^o_{t+1}(s) - \nu_t(s) = a_t$. Using this in (64), adding $\beta O_{tl}(x)$ to both sides, and rearranging the integrals yields

$$\begin{aligned} c^y_{tl}(x) + \beta O_{tl}(x) = C_t(x) + xT_l - \alpha p_{tl} + \\ \beta \int \int \left[\mathcal{V}^o_{t+1} + \frac{\lambda^o_{t+1}}{\beta^{t+1}} \right] [e\gamma(x,s_p)]dF(e)\pi^y_{tl}(s_p)ds_p + \\ \beta \int \left\{ \int \left[\mathcal{V}^o_{t+1} + \frac{\lambda^o_{t+1}}{\beta^{t+1}} \right] [e\gamma(s,x)]dF(e) - \right. \\ \left. \int \int \left[\mathcal{V}^o_{t+1} + \frac{\lambda^o_{t+1}}{\beta^{t+1}} \right] [e\gamma(s,s_p)]dF(e)\pi^y_{tl}(s_p)ds_p \right\} \pi^y_{tl}(s)ds, \end{aligned} \tag{65}$$

where $C_t(x) \equiv \mathcal{V}^y_t(x) - \frac{\lambda^y_t(x)}{\beta^t} - \frac{a_t}{\beta^t}$ is independent of the location.

From (63), $c^o_{t+1,l}(s) = T_l s - \alpha p_{t+1,l} - \frac{\lambda^o_{t+1}(s)}{\beta^{t+1}}$. Therefore, $\mathcal{V}^o_{t+1}(s) = -\frac{\lambda^o_{t+1}(s)}{\beta^{t+1}} + \frac{1}{\vartheta} \log \left(\sum_{\ell'} \mathrm{e}^{\vartheta \left(T_{\ell'} s - \alpha p_{t+1,\ell'} + B^o_{\ell'} \right)} \right)$. Hence, $\mathcal{V}^o_{t+1}(q) + \lambda^o_{t+1}(q)/\beta^{t+1}$ is independent of (λ^o_t, ν_t). Moreover, (λ^o_t, ν_t) do not affect the spatial allocation of old workers. Since $p_{t\ell}$ only depends on the local population, we also have $p_{t\ell}$ are independent of (λ^o_t, ν_t). Accordingly, (65) is independent of (λ^o_t, ν_t). Said differently, the optimal allocation of young workers is independent of transfers amongst old workers and across generations.

We can therefore focus without loss on the allocation without transfers across generations and within generations across skills. Suppose $c^o_{t+1,l}(s) = sT_l - \alpha p_{t+1,l}$. The optimality condition with respect to $n^o_{t+1,l}(s)$ implies $\lambda^o_{t+1}(s) = 0$. Plugged into the optimality condition for $\nu_t(s)$, this yields $\nu_t(s) = -a_t$. Since n^o_{t+1} is a density, its multiplier is defined up to an additive constant, and we can normalize $a_t = 0 = \nu_t(s)$. Plugging this into (64) returns (18) and (19) after imposing steady-state.

Equations (18), (19), (60), and (62) determine the optimal allocation as a fixed point in the local labor supplies and the shadow housing prices. Existence and uniqueness can be established as in the laissez-faire.

D.7 First Order Approximation to the Optimal Allocation

Consider the steady-state optimal allocation. I derive here its first-order approximation. Several proofs and derivations are similar to the laissez-faire and thus presented more succinctly here. I use similar notations as in the laissez-faire, and drop the star superscript for notational simplicity

unless when I am comparing the optimal allocation to the laissez-faire.

As in the laissez-faire, it is trivial to guess and verify that the optimal allocation is symmetric when $\boldsymbol{T} = \bar{T}$. Furthermore, at the symmetric point, the optimal allocation coincides with the laissez-faire.

Moving onto the approximation, the optimal old consumption satisfies $c_\ell^o(s) = T_\ell s - \alpha p_\ell$. This is identical to the laissez-faire expression conditional on $\boldsymbol{p}$. Hence, the optimal old allocation obeys

$$\partial_l n_\ell^o(s) = \theta \bar{n}_\ell^o(s) \left[s I_{\ell l} - P_{\ell l} \right]. \tag{66}$$

For young workers, (59) implies

$$\partial_l n_\ell^y(s) = \vartheta \bar{n}_\ell^y(s) \left\{ \partial_l c_\ell^y(s) + \beta \partial_l O_\ell(s) - \sum_{\ell'} e_{\ell'} \left(\partial_l c_{\ell'}^y(s) + \beta \partial_l O_{\ell'}(s) \right) \right\}. \tag{67}$$

At $\boldsymbol{T} = \bar{T}$, $\tau_\ell(x) = \tau(x)$ for every ℓ. Differentiating (18) and taking its difference with the national average:

$$\partial_l c_\ell^y(x) - \sum_{\ell'} e_{\ell'} \partial_l c_{\ell'}^y(x) = \bar{T} \left[x I_{\ell l} - P_{\ell l} \right] + \beta \bar{T} \int \partial_l \pi_\ell^y(s) \left(\gamma(s,x) - \mathbb{E}[\gamma(s, S^y)] - \mathbb{E}[\gamma(S^y, s)] \right) ds.$$

The net change in the learning option value is derived as in the laissez-faire. Combining these two expressions in (67):

$$\begin{aligned} \partial_l n_\ell^y(x) = {} & \theta \bar{n}_\ell^y(x) \left[x I_{\ell l} - P_{\ell l} \right] + \\ & \theta \beta n^y(x) \int \left(\partial_l n_\ell^y(s) - n^y(s) \int \partial_l n_\ell^y(q) dq \right) \left(\gamma(s,x) - \mathbb{E}[\gamma(s, S^y)] \right) ds - \\ & \theta \beta n^y(x) \int \left(\partial_l n_\ell^y(s) - n^y(s) \int \partial_l n_\ell^y(q) dq \right) \mathbb{E}[\gamma(S^y, s)] ds + \\ & \theta \beta n^y(x) \int \partial_l n_\ell^y(s) \left(\gamma(x,s) - \mathbb{E}[\gamma(x, S^y)] \right) ds. \end{aligned}$$

Guessing and verifying the functional form

$$\partial_l n_\ell^y(s) = \theta \bar{n}_\ell^y(s) \left[I_{\ell l} \eta^\star(s) - P_{\ell l} \right], \tag{68}$$

we obtain

$$\begin{aligned} \eta^\star(x) = x + {} & \theta \beta \int \left(\{ \gamma(x,s) - \mathbb{E}[\gamma(x, S^y)] \} - \left\{ \mathbb{E}[\gamma(S^y, s)] - \mathbb{E}[\gamma(S^y, \widetilde{S}^y)] \right\} \right) \eta^\star(s) dN^y(s) + \\ & \theta \beta \int \left(\{ \gamma(s,x) - \mathbb{E}[\gamma(S^y, x)] \} - \left\{ \mathbb{E}[\gamma(s, S^y)] - \mathbb{E}[\gamma(S^y, \widetilde{S}^y)] \right\} \right) \eta^\star(s) dN^y(s). \end{aligned} \tag{69}$$

It remains to solve for the shadow housing prices. Integrating (66) and (68) over skills and adding them gives $\partial_l N_\ell = 2\theta e_\ell \left[\mu^\star I_{\ell l} - P_{\ell l} \right]$, where $\mu^\star \equiv \frac{1}{2} \left(\mathbb{E}[S^o] + \mathbb{E}[\eta^\star(S^y)] \right)$. By the optimality condition for housing, the shadow price satisfies the same equation as the laissez-faire housing price.

Hence, using the same derivation as in the laissez-faire, we get

$$P_{\ell l} = \mu^\star \left[\lambda_\ell I_{\ell l} + (1-\lambda_\ell) \sum_{\ell'} \omega_{\ell'} I_{\ell' l} \right].$$

In addition to the allocation, it is also possible to approximate the transfers that decentralize the optimal allocation. Let $\tau(x) \equiv \sum_\ell e_\ell \tau_\ell(x)$ denote the employment-weighted average of the transfers, and $t_\ell^y(x) \equiv \tau_\ell(x) - \tau(x)$ the relative transfer to skill x in ℓ, At the symmetric benchmark, $\tau_\ell(x) = \tau(x)$ for every ℓ, so $t_\ell^y(x) = 0$. Differentiating (19):

$$dt_\ell^y(x) = \beta \bar{T} \int d\pi_\ell^y(s) \left(\gamma(s,x) - \mathbb{E}[\gamma(s,S^y)]\right) ds - \beta \bar{T} \int \mathbb{E}[\gamma(S^y,s)] d\pi_\ell^y(s) ds.$$

Meanwhile, $d\pi_\ell^y(s) = \theta n^y(s) \left(\eta^\star(s) - \mathbb{E}[\eta^\star(S^y)]\right) d\mathcal{T}_\ell$. Combined, we can thus decompose the transfers into a city tax, $\bar{\tau}_\ell$, and an additional skill-by-city subsidy, $d_\ell^y(x)$. Specifically,

$$t_\ell^y(x) \approx \bar{\tau}_\ell + d_\ell^y(x),$$

where

$$\bar{\tau}_\ell = -\bar{T}\theta\beta\mathcal{T}_\ell \int \mathbb{E}[\gamma(S^y,s)] \left(\eta^\star(s) - \mathbb{E}[\eta^\star(S^y)]\right) dN^y(s)$$

and

$$d_\ell^y(x) = \bar{T}\theta\beta\mathcal{T}_\ell \int \left(\gamma(s,x) - \mathbb{E}[\gamma(s,S^y)]\right) \left(\eta^\star(s) - \mathbb{E}[\eta^\star(S^y)]\right) dN^y(s).$$

As for the laissez-faire, the first-order approximation is well-defined if the integral equation (69) has a solution. We show this is the case provided $\theta\beta$ or learning complementarities are not too large. We also bound the solution. For these proofs, rewrite (69) as

$$\eta^\star(x) = x + \theta\beta T^\star \eta^\star(x), \tag{70}$$

where

$$T^\star f(x) = \int K^\star(x,s) f(s) dN(s)$$

and

$$\begin{aligned} K^\star(x,s) = \gamma(x,s) - \mathbb{E}[\gamma(x,S^y)] - \mathbb{E}[\gamma(S^y,s)] + \mathbb{E}[\gamma(S^y,\widetilde{S}^y)] + \\ \gamma(s,x) - \mathbb{E}[\gamma(S^y,x)] - \mathbb{E}[\gamma(s,S^y)] + \mathbb{E}[\gamma(S^y,\widetilde{S}^y)]. \end{aligned}$$

The kernel is symmetric and centered in both arguments:

$$K^\star(x,s) = K^\star(s,x), \qquad \int K^\star(x,s) dN(s) = 0, \qquad \int K^\star(x,s) dN(x) = 0. \tag{71}$$

Lemma D.5 (Existence and uniqueness of the optimal solution)**.**

The equation (70) *has a unique solution, bounded and continuous on* $[\underline{s}, \bar{s}]$*, if*

$$\theta\beta \left(\int \int |K^\star(x,s)|^2 \, dN(s) dN(x) \right)^{1/2} < 1.$$

Proof. Identical to Lemma D.3 after replacing K and T with $K^\star$ and $T^\star$. □

Lemma D.6 (A bound on the optimal solution)**.**
The solution to (70) *satisfies*

$$\mathbb{E}[\eta^\star(S^y)] = \mathbb{E}[S^y] \quad \textit{and} \quad \|\eta^\star - \mathbb{E}[\eta^\star(S^y)]\| \leq \frac{\mathbb{S}d[S^y]}{1 - \theta\beta\|K^\star\|}. \tag{72}$$

The same bound holds along the sequence $\eta_0^\star(s) = s$ *and* $\eta_{m+1}^\star(s) = s + \theta\beta T^\star \eta_m^\star(s)$ *for* $m \geq 0$.

Proof. Taking expectations in (70) and using (71) gives $\mathbb{E}[\eta^\star(S^y)] = \mathbb{E}[S^y]$. Moreover, because $T^\star 1 = 0$, $\eta^\star - \mathbb{E}[\eta^\star(S^y)] = \mathbb{E}[S^y]$. Taking $L^2(dN)$ norms and re-arranging yields the inequality in (72). The argument for the sequence proceeds similarly. □

Finally, I conclude by showing that the optimal allocation also satisfies SPAM provided between-skill learning complementarities are not too strong. As for Proposition 4, it is sufficient to show that $\eta^\star$ is weakly increasing.

Lemma D.7 (Optimal SPAM)**.**
Define

$$\mathcal{F}^\star = \left\{ f \in L^2(dN) : f \textit{ is weakly increasing and } \|f - \mathbb{E}[f(S^y)]\| \leq \frac{\mathbb{S}d[S^y]}{1 - \theta\beta\|K^\star\|} \right\}.$$

Suppose that, for every $f \in \mathcal{F}^\star$*,*

$$\theta\beta \left\{ \mathbb{C}ov[\gamma_1(x, S^y), f(S^y)] + \mathbb{C}ov[\gamma_2(S^y, x), f(S^y)] \right\} > -1 \qquad \textit{for every } x \in [\underline{s}, \bar{s}]. \tag{73}$$

Then $\eta^\star$ *is strictly increasing.*

Proof. The proof follows the same induction argument as Proposition 4. Consider the sequence $\eta_0^\star(s) = s$ and $\eta_{m+1}^\star(s) = s + \theta\beta T^\star \eta_m^\star(s)$. By Lemma D.6, every iterate satisfies $\|\eta_m^\star - \mathbb{E}[\eta_m^\star(S^y)]\| \leq \frac{\mathbb{S}d[S^y]}{1-\theta\beta\|K^\star\|}$. We show by induction that every $\eta_m^\star$ is strictly increasing. The initial iterate $\eta_0^\star(s) = s$ is strictly increasing and belongs to $\mathcal{F}^\star$. Suppose that $\eta_m^\star$ is weakly increasing. It then belongs to $\mathcal{F}^\star$. Differentiating $T^\star \eta_m^\star$ under the integral sign gives

$$\frac{d}{dx} T^\star \eta_m^\star(x) = \mathbb{C}\mathrm{ov}[\gamma_1(x, S^y), \eta_m^\star(S^y)] + \mathbb{C}\mathrm{ov}[\gamma_2(S^y, x), \eta_m^\star(S^y)].$$

Consequently, $\eta_{m+1}^{\star\prime}(x) = 1 + \theta\beta \left\{ \mathbb{C}\mathrm{ov}[\gamma_1(x, S^y), \eta_m^\star(S^y)] + \mathbb{C}\mathrm{ov}[\gamma_2(S^y, x), \eta_m^\star(S^y)] \right\}$. Condition (73) implies that $\eta_{m+1}^{\star\prime}(x) > 0$ for every x. Hence every iterate is strictly increasing. The conclusion that $\eta^\star$ is strictly increasing follows as in Lemma 4. □

The condition on γ required by Lemma D.7 differ from that of Proposition 4. However, they share similarities: if γ is supermodular, Lemma D.7 necessarily holds, while if γ is submodular, (73) imposes an implicit bound on how submodular the function can be. In particular, there exists a set of positive measure such that Proposition 4 and Lemma D.7 hold jointly.

D.8 Proof of Proposition 7

Since we are focusing on the oldest workers, I drop the age subscript. I also normalize $\bar{T} = 1$.

Amenities, migration costs, and taste shock dispersion. We first prove global identification of $\{\vartheta, \delta, \boldsymbol{B}, \boldsymbol{\tau}\}$ when $\boldsymbol{T}$ are known. We normalize $\frac{1}{L}\sum_\ell B_\ell = 0$ and $\kappa_{\ell\ell} = 0$. We also impose the functional form

$$\kappa_{l\ell} = 1\{l \neq \ell\}\left(\tau_l + \delta \log \text{dist}_{l\ell}\right), \tag{74}$$

where $\text{dist}_{l\ell}$ is known and symmetric. Old workers destination probabilities are

$$p_{l\ell}(s) = \frac{\exp\left\{\vartheta\left(sT_\ell + B_\ell - \alpha p_\ell - \kappa_{l\ell}\right)\right\}}{\sum_{\ell'}\exp\left\{\vartheta\left(sT_{\ell'} + B_{\ell'} - \alpha p_{\ell'} - \kappa_{l\ell'}\right)\right\}},$$

which do not depend on x since the oldest workers do not learn. Furthermore, since we know $\boldsymbol{T}$, skills are identified from wages: $s = w/T_\ell$. Comparing the likelihood to move to $\ell \neq l$ with that of staying in l:

$$\log\frac{p_{l\ell}(s)}{p_{ll}(s)} = \vartheta\left(sT_\ell - sT_l - \alpha[p_\ell - p_l]\right) + \text{FE}_{l\ell}, \tag{75}$$

where $\text{FE}_{l\ell} = \vartheta B_\ell - \vartheta B_l - \kappa_{l\ell}$.[30] The left-hand side is measured from migration flows across the (observed) skill distribution. On the right-hand side, we also know wages and housing prices. (75) thus identifies ϑ and $\text{FE}_{l\ell}$. Using the functional form assumption for the migration costs:

$$\frac{\text{FE}_{l\ell}}{\vartheta} = \text{FE}^o_l + \text{FE}^d_\ell - \delta \log \text{dist}_{l\ell},$$

where $\text{FE}^o_\ell = -(B_\ell + \tau_\ell)$ and $\text{FE}^d_\ell = B_\ell$. Altogether, this identifies $\{\delta, \boldsymbol{B}, \boldsymbol{\tau}\}$.

TFP. Local TFPs are identified from average wages net of selection. Average log wages in destination ℓ satisfy

$$\mathbb{E}_\ell[\log w] = \log T_\ell + \mathbb{E}_\ell[\log s]. \tag{76}$$

To first order, the average skill in ℓ is

$$\mathbb{E}_\ell[\log s] = \frac{\sum_l \Xi_l p^0_{l\ell}\mathbb{E}[\log w \mid l]}{\Xi_l \Phi_l p^0_{l\ell}} + \mathbb{C}\text{ov}^0_\ell\left(\log w,\, w\left[\vartheta \log T_\ell - \sum_{\ell'} p^0_{l\ell'}\vartheta \log T_{\ell'}\right]\right) + o(\|\log \boldsymbol{T}\|), \tag{77}$$

where $\Xi_l = \int \chi_l(s)\, ds$ is the mass of workers from origin l and I have used $s = w$ at $\boldsymbol{T} = 1$. The first term is known from migration flows, and the distribution of people and wages across origins. The covariance, which controls the amount of sorting in location ℓ, depends on $\vartheta \log T_\ell$. To first order, the log likelihood that someone with skill s moves from l to ℓ is

$$\log p_{l\ell}(w) = FE_{l\ell} + FE_l(w) + \vartheta w \log T_\ell + o(\|\log \boldsymbol{T}\|),$$

[30] The parameter α is identified from aggregate housing expenditure share (Appendix F.1).

where

$$FE_{l\ell} = \vartheta\left(B_\ell - \alpha p_\ell - \kappa_{l\ell}\right) - \log\left[\sum_{\ell'} \exp\left\{\vartheta\left(B_{\ell'} - \alpha p_{\ell'} - \kappa_{l\ell'}\right)\right\}\right],$$
$$FE_l(w) = -\vartheta w \sum_{\ell'} p^0_{l\ell'} \log T_{\ell'},$$

and I have again used $w = s$. With saturated origin-destination fixed effects and origin-by-wage controls, this identifies $\{\vartheta \log T_\ell\}_\ell$. Then, (76) and (77) identify $\boldsymbol{T}$.

E Quantitative Model

This section fully lays down the quantitative model.

E.1 Setup

Lifecycle. Individuals work for A periods. They give birth to one descendant, whose arrival is stochastic: the likelihood that an adult gives birth to a child when they are a years old is b_a. Newborns take the location of their parents at the same time of birth as their birthplace, denoted ℓ_b. Workers exit the labor force after A periods, and are not altruistic.

Learning. In addition to their birthplace, workers are endowed with two additional characteristics when they enter the labor force: their initial skills, s, and their learning abilities, x. The joint distribution of (s, x) is χ_b. Skills and learning abilities differ in two dimensions. First, skills enter production and affect the marginal products of labor. Second, skills evolve over time while learning abilities are constant.

Human capital accumulation is shaped by workers' own characteristics, as well as who is around them and where they live. Take an individual with skill s and learning abilities x who works in location ℓ. The skill of this worker in the next period is

$$\log s' = \log \gamma(s, s_p) + \log x + \log \mu_\ell + \log e, \tag{78}$$

where e is a learning shock drawn from the distribution F. s_p refers to the skill of the individual with whom this worker interacts. The function γ summarizes how a worker's own skill and that of their partner shape their learning. In addition, the location of a worker shapes human capital accumulation through the place effect μ_ℓ. Place effects are exogenous.

The way workers interact with each other is governed by three assumptions. First, interactions are "face-to-face", and thus take place within a location. Second, interactions are random. Third and last, the matching function has constant returns to scale. Combined, the probability to interact with a skill s_p in location ℓ is given the local skill density, $\pi_\ell(s_p)$.

Preferences. Workers are risk neutral and discount the future at rate β. Within a period, they consume two goods: a freely traded good, c, taken as the numeraire, and local housing, h. Housing is a necessity: each worker consumes α units of local housing. Conditional on residing in location ℓ, flow utility is

$$u^a_\ell(c, h, \boldsymbol{\varepsilon}) = c + h + B_{a\ell} + \varepsilon_\ell,$$

subject to $h = \alpha$. $B_{a\ell}$ denotes the amenities offered by location ℓ to workers with age a, and ε are idiosyncratic location preferences. These preferences are i.i.d. over time and workers.

Workers face the static budget constraint $c + p_\ell h \leq I$, where p_ℓ is the unit price of housing and I income. Workers do not have access to a savings device, which is wlog since they are risk neutral. They supply inelastically one unit of labor every period, and their income equals their wage, denoted w.

Workers choose where to live every period. They incur migration costs $\kappa_{l\ell}$ to move from l to ℓ. The timing of their decision is as follows:

1. Knowing their skills, learning abilities, and preferences, workers choose where to work.
2. They consume the numeraire good and α units of local housing, and enjoy the local amenities.
3. They accumulate human capital, which they can only use in the next period.

Importantly, workers do not know with whom they will interact, or their idiosyncratic learning shock ε, when choosing where to work.

Production. Each city is endowed with homogeneous competitive firms with TFP T_ℓ. The firm uses labor as sole input to produce the numeraire. It uses the constant returns to scale technology

$$Y_\ell(\boldsymbol{n}) = \int T_\ell s n_\ell(s) ds.$$

Free entry in each location holds. Labor markets are segmented by skills and location. Free entry implies $w_\ell(s) = T_\ell s$.

Housing. In every location, landowners supply the housing stock. They produce local housing using the production function $H_\ell = K^\phi \bar{H}^{1-\phi}$, where K is a numeraire input and $\bar{H}$ is a fixed factor. The ownership of the fixed factor leads to profits, which are used to consume the numeraire.

Functional forms. I impose the following functional forms. First, the learning function is

$$\log \gamma(s, s_p) = \xi(s) + \varphi(s) s_p.$$

Given this functional form, the function $\varphi(s)$ captures how the returns to interactions vary across skills. If $\varphi(s) > 0$, interacting with relatively more skilled workers spur further growth. If $\varphi'(s) > 0$, these gains are greater for high-skill themselves.

Second, the initial distribution over skills and learning abilities is log normal. The average skill of new entrants is μ_s, and I normalize the log leaning ability to zero. The variance-covariance matrix is denoted Σ, with variances σ_s^2 and σ_x^2 and correlation ρ_{sx}.

Migration costs follow gravity:

$$\kappa_{al\ell} = 1\{\ell \neq l\} \zeta_a \left(\tau_l + \delta \log \text{dist}_{l\ell}\right).$$

The parameter τ_l is an origin-specific fixed cost, δ is the distance semi elasticity, and ζ_a lets migration cost scale up and down with ages. I normalize $\zeta_A = 1$. The variable $\text{dist}_{l\ell}$ is the (symmetric) physical distance in km between (l, ℓ).

Lastly, the taste shocks are Gumbel distributed with inverse dispersion ϑ.

E.2 Steady State

Let $\chi_{al}(s, x)$ denote the measure of (s, x) who start the period in location l, conditional on age a. Let $\chi_a(s, x) = \sum_l \chi_{al}(s, x)$ be the aggregate distribution over (s, x). Lastly, let $n_{al\ell}(s, x)$

denote the measure of workers with characteristics (s,x) who move from l to ℓ at age a. There as well, by construction, $\chi_{al}(s,x) = \sum_\ell n_{al\ell}(s,x)$. The total measure of (s,x) workers in ℓ is $n_{a\ell}(s,x) = \sum_l n^a_{l\ell}(s,x)$. That of workers with skill s is $n_{a\ell}(s) = \int n_{a\ell}(s,x)dx$. The skill distribution in ℓ is therefore

$$\pi_\ell(s) = \frac{\sum_a n_{a\ell}(s)}{N_\ell}, \tag{79}$$

where $N_\ell = \sum_\ell N_{a\ell} = \sum_\ell \int n_{a\ell}(s)ds$ is the city's population.

Housing. We start with the housing block. Land-owners solve $\Pi_\ell = \max_K p_\ell K^\phi \bar{H}^{1-\phi} - K$. Optimality demands

$$H_\ell = \bar{H}\left(p_\ell\phi\right)^{\frac{\phi}{1-\phi}}, \quad \text{and} \quad \Pi_\ell = \bar{H}p_\ell^{\frac{1}{1-\phi}}\left(\phi^{\frac{\phi}{1-\phi}} - \phi^{\frac{1}{1-\phi}}\right).$$

On the workers side, each resident demands α units of housing. Aggregate housing demand in location ℓ is therefore αN_ℓ. The housing market clearing pins down housing prices:

$$p_\ell = \phi^{-1}\left(\frac{\alpha N_\ell}{\bar{H}}\right)^{\frac{1-\phi}{\phi}}. \tag{80}$$

Spatial allocation. Let the option value of learning in ℓ be denoted by

$$O_{a\ell}(s,x) = \int\int \mathcal{V}_{a+1\ell}[\varepsilon x \mu_\ell \gamma(s,s_p),x]dF(\varepsilon)\pi_\ell(s_p)ds_p, \tag{81}$$

where $\mathcal{V}_{a\ell}(s,x) = \mathbb{E}[V_{a\ell}(s,x,\boldsymbol{\varepsilon}) \mid s,x,\ell,a]$ is expected future utility prior to the realization of the taste shocks. Households solve

$$V_{al}(s,x,\boldsymbol{\varepsilon}) = \max_\ell \; w_\ell(s) - \alpha p_\ell + B_{a\ell} - \kappa_{al\ell} + \varepsilon_\ell + \beta O_{a\ell}(s,x).$$

Accordingly, the measure of workers in ℓ is given by

$$\frac{n_{al\ell}(s,x)}{\chi_{al}(s,x)} = \frac{\mathrm{e}^{\vartheta(w_\ell(s) - \alpha p_\ell + B_{a\ell} - \kappa_{al\ell} + \beta O_{a\ell}(s,x))}}{\sum_{\ell'}\mathrm{e}^{\vartheta\left(w_{\ell'}(s) - \alpha p_{\ell'} + B_{a\ell'} - \kappa_{al\ell'} + \beta O^a_{a\ell'}(s,x)\right)}}. \tag{82}$$

The oldest workers exist the labor in the next period and thus discard the learning option value. Their allocation is

$$\frac{n_{Al\ell}(s,x)}{\chi_{Al}(s,x)} = \frac{\mathrm{e}^{\vartheta(w_\ell(s) - \alpha p_\ell + B_{A\ell} - \kappa_{Al\ell})}}{\sum_{\ell'}\mathrm{e}^{\vartheta(w_{\ell'}(s) - \alpha p_{\ell'} + B_{A\ell'} - \kappa_{Al\ell'})}}. \tag{83}$$

Skill distributions. Given the learning process, the distributions $\{\chi_{al}(s,x)\}_a$ satisfy the law of motion

$$\chi_{a\ell}(s,x) = \int\int \frac{1}{\mu_\ell x \gamma(s_i,s_p)} f\left(\frac{s}{\mu_\ell x \gamma(s_i,s_p)}\right) n_{a-1}\ell(s_i,x)\pi_\ell(s_p)ds_p ds_i. \tag{84}$$

Furthermore, given the birth process, the distribution of new entrants is

$$\chi_{1\ell}(s,x) = \chi_b(s,x) \sum_a N_{a\ell} b_a. \tag{85}$$

The equilibrium of the model is jointly described by equations (79) to (85).

Definition E.1 (Steady state)**.**
A steady state equilibrium is a spatial allocation, $\{n_{al\ell}\}_{a,l,\ell}$, local skill densities, $\{\pi_\ell\}_\ell$, distribution of characteristics, $\{\chi_{al}\}_{al}$, and housing prices $\{p_\ell\}_\ell$, such that:

1. *Given the local skill densities, the spatial allocations satisfy* (82) *and* (83)*;*
2. *The local skill densities are consistent with the spatial allocation* (79)*;*
3. *The distribution of characteristics obey the law of motion* (84) *subject to the initial condition* (85) *and the spatial allocation of workers;*
4. *Housing prices clear the housing market* (80)*.*

E.3 Numerical Algorithm

I solve for the steady state numerically. The algorithm contains three loops.

In the most inner loop, I iterate backward once on the lifecycle given local skill densities, housing prices, and distribution over characteristics. Starting from A, compute (83), and obtain the expected utility over idiosyncratic preferences from $\mathcal{V}_{A\ell}(s,x) = \vartheta^{-1} \log\left(\sum_{\ell'} \mathrm{e}^{\vartheta[w_{\ell'}(s) - \alpha p_{\ell'} + B_{A\ell'} - \kappa_{Al\ell'}]}\right)$. Given $\mathcal{V}_{a+1\ell}(s,x)$ for all (s,x,ℓ), solve for the learning option value from (81). Given $O_{a\ell}(s,x)$, compute the spatial allocation for age a from (82). Compute expected utility from $\mathcal{V}_{a\ell}(s,x) = \vartheta^{-1} \log\left(\sum_{\ell'} \mathrm{e}^{\vartheta[w_{\ell'}(s) - \alpha p_{\ell'} + B_{a\ell'} - \kappa_{Al\ell'} + \beta O_{a\ell'}(s,x)]}\right)$. Stop when $a = 1$.

In the intermediate loop, given the distribution over characteristics, I solve for the local skill densities and housing prices by iterating jointly over (79) and (80).

Lastly, in the outer loop, I solve for the distribution over characteristics by iterating over (84) and (85).

F Estimation

F.1 Identification

This section discusses identification of the parameters not discussed in the main text.

Housing. Housing expenditure per individual is αp_ℓ. Total housing expenditure as a fraction of total expenditure is thus $\frac{\sum_\ell \alpha N_\ell p_\ell}{\sum_\ell w_\ell N_\ell}$. $\alpha = \widehat{\text{housing share}} \times \frac{\sum_\ell w_\ell N_\ell}{\sum_\ell N_\ell p_\ell}$. Regarding the housing supply, the housing market clearing condition (8) gives

$$\log p_\ell = -\log \phi + \frac{1-\phi}{\phi} \log \frac{\alpha}{\bar{H}} + \frac{1-\phi}{\phi} \log N_\ell, \tag{86}$$

which identifies $(\phi, \bar{H})$ given α and data on housing prices and employment share.

Learning technology. Suppose the learning technology obeys $\gamma(s, s_p) = \xi(s) + \varphi(s) s_p$. Section 4.2 proves identification of $\varphi(s)$. I now show identification of ξ. The skill control function F in (27) do not separately identify ξ from average learning abilities across skills, $\mathbb{E}[\tilde{x} \mid s]$. Rather, ξ is identified

similar to the identification of the persistence parameter in an autoregressive model with worker fixed effect. Let $r_{it+1} \equiv \tilde{s}_{it+1} - \tilde{\varphi}(s_{it})\bar{s}_{\ell_{it}} - \mu_{\ell_{it}}$, which we know. The law of motion implies

$$r_{it+1} = \xi(s_{it}) + \tilde{x}_i + \tilde{e}_{it+1} + \nu_{it+1}.$$

In first difference:

$$\Delta_t r_{it+1} = \Delta_t \xi(s_{it}) + \Delta_t \tilde{e}_{it+1} + \Delta_t \nu_{it+1}$$

As in standard AR1 model, s_{it} is correlated with (e_{it}, ν_{it}), and therefore OLS does not identify the above expression. However, the mean conditional on s_{it-1} is

$$\mathbb{E}[\Delta_t r_{it+1} \mid s_{it-1} = s] = \mathbb{E}[\xi(s_{it}) \mid s_{it-1} = s] - \xi(s_{it-1}), \tag{87}$$

since $\mathbb{E}[\varepsilon_{ik} \mid s_{it-1} = s] = \mathbb{E}[\nu_{ik} \mid s_{it-1} = s] = 0$ for all $k \geq 0$. In matrix format: $\boldsymbol{r} = (\boldsymbol{P} - \boldsymbol{I})\boldsymbol{\xi}$, where $\boldsymbol{P}$ is the transition operator defined by $(\boldsymbol{P}F)(s) = \int F(s') \Pr[s_{it} = s' | s_{it-1} = s] ds'$. Since $\boldsymbol{P}$ is a transition matrix, its rows sum to one, and the above system identifies $\boldsymbol{\xi}$ up to a constant. The level of $\boldsymbol{\xi}$ is identified from $\mathbb{E}[r_{it+1}] = \mathbb{E}[\tilde{\xi}_{it}]$.

Skills and abilities. The last parameters estimated are those governing the skill and learning abilities distribution, together with that behind learning shocks. I now prove their identification. Once skilled are observed, μ_s and σ_s are identified by the mean and standard deviation of 25 year olds. The parameters shaping learning abilities are obtained from individual-specific long-term skill growth. Specifically, taking the i-specific mean of (25) yields

$$\mathbb{E}_i[\tilde{s}_{it+1}] = \mathbb{E}_i[\xi(s_{it})] + \mathbb{E}_i[\varphi(s_{it})\bar{s}_{\ell_{it}}] + \tilde{x}_i + \mathbb{E}_i[\tilde{\mu}_{\ell_{it}}]. \tag{88}$$

Inverting this expression yields x_i, and thus $\sigma_x = \mathbb{S}\mathrm{d}[x_i]$. For the covariance between skills and learning abilities, (25) implies

$$\mathbb{C}\mathrm{ov}[\tilde{x}_i, \tilde{s}_{i0}] = \mathbb{E}\left(\mathbb{C}\mathrm{ov}[\tilde{s}_{it+1} - \xi(s_{it}) - \varphi(s_{it})\bar{s}_{\ell_{it}} - \tilde{\mu}_{\ell_{it}}, \tilde{s}_{i0} \mid \ell]\right), \tag{89}$$

which identifies ρ given (σ_s, σ_x). Finally, the dispersion in learning shocks is recovered from the dispersion in residual growth taking into account unobserved learning interactions. Let $\rho_{it+1} \equiv \tilde{s}_{it+1} - \xi(s_{it}) - \varphi(s_{it})\bar{s}_{\ell_{it}} - \tilde{\mu}_{\ell_{it}} - \tilde{x}_i = \varphi(s_{it})\left(s_{p_{it}} - \bar{s}_{\ell_{it}}\right) + \tilde{e}_{it+1}$. We have

$$\mathbb{V}\mathrm{ar}[\rho_{it+1} \mid \ell] = \mathbb{V}\mathrm{ar}[\varphi(s_{it})\left(s_{p_{it}} - \bar{s}_{\ell_{it}}\right) \mid \ell] + \sigma_\varepsilon^2 = \mathbb{E}[\varphi(s_{it})^2 \mid \ell]\mathbb{V}\mathrm{ar}[s_{it} \mid \ell] + \sigma_\varepsilon^2$$

where the second equality follows from the law of total variance. Accordingly:

$$\sigma_e^2 = \mathbb{E}(\mathbb{V}\mathrm{ar}[\rho_{it+1} \mid \ell]) - \mathbb{E}(\mathbb{E}[\varphi(s_{it})^2 \mid \ell]\mathbb{V}\mathrm{ar}[s_{it} \mid \ell]).$$

F.2 Implementation

This section discusses the implementation of the estimation.

Learning technology: noise migration. As explained in the main text, I operationalize (26) by discretizing the skill distribution into quintiles. Accordingly, I compute the migration matrix $\{\Pr[\ell_{it} = \ell' \mid \ell_i^b = l, s_{it} \in S_g]\}_{l,\ell}$ for each skill quintile S_g. Inverse migration matrices are filled with noise in finite sample (Dingel and Tintelnot, 2026), let alone when they need to be estimated

across the skill distribution. Brute-force inversion would thus muddy the estimation of the spillover function φ in (27).[31]

Instead of directly inverting the migration matrices, I use a Tikhonov regularization to reduce the measurement error. Since (27) is estimated separately for each skill quintile, I present the regularization dropping the skill index and setting $\xi = 0$. Let $y_l = \mathbb{E}[\tilde{s}_{it+1} \mid \ell_i^b = l]$ and $\boldsymbol{y} = \{y_l\}$. Likewise, collect the migration matrix into $\boldsymbol{P}$. Our object of interest is $\bar{\boldsymbol{y}} = \boldsymbol{P}^{-1}\boldsymbol{y}$. I obtain $\bar{\boldsymbol{y}}$ via the minimization

$$\bar{\boldsymbol{y}} = \arg\min_{\boldsymbol{o}} (\boldsymbol{y} - \hat{\boldsymbol{P}}\boldsymbol{o})'(\boldsymbol{y} - \hat{\boldsymbol{P}}\boldsymbol{o}) + \lambda(\boldsymbol{M}\boldsymbol{o})'\boldsymbol{W}(\boldsymbol{M}\boldsymbol{o}),$$

where $\boldsymbol{M} = \boldsymbol{I} - \boldsymbol{1}\boldsymbol{N}'$, $\boldsymbol{W} = \text{diag}(\boldsymbol{N})$, and $\boldsymbol{N}$ the workplace employment share. The parameter λ is thus a penalty on the variance of $\bar{\boldsymbol{y}}$. $\lambda = 0$ implies $\bar{\boldsymbol{y}} = \boldsymbol{P}^{-1}\boldsymbol{y}$, while $\lambda \to \infty$ implies $\bar{\boldsymbol{y}} = \boldsymbol{0}$.

While the estimates $\bar{\boldsymbol{y}}$ are thus dependent on λ, they can be bounded. An upper bound for the variance of $\bar{\boldsymbol{y}}$ is the between-city variance in future skills. Given the law of motion for s_{it+1}, the workplace conditional average is

$$\mathbb{E}[\tilde{s}_{it+1} \mid \ell] = \mathbb{E}[x_i \mid \ell] + \bar{y}_\ell.$$

Sorting imposes $\mathbb{C}\text{ov}[\mathbb{E}[x_i \mid \ell], \bar{y}_\ell] \geq 0$. Hence:

$$\mathbb{V}\text{ar}[\mathbb{E}[\tilde{s}_{it+1} \mid \ell]] \geq \mathbb{V}\text{ar}[\bar{y}_\ell].$$

A lower bound is obtained from the within-worker dispersion in skill growth. The (i, ℓ) conditional average is

$$\mathbb{E}[\tilde{s}_{it+1} \mid \ell, i] = x_i + \bar{y}_\ell.$$

Hence:

$$\mathbb{V}\text{ar}[\mathbb{E}(\tilde{s}_{it+1} \mid \ell, i) \mid i] = \mathbb{V}\text{ar}[\bar{y}_\ell \mid i],$$

and the law of total variance implies

$$\mathbb{V}\text{ar}[\bar{y}_\ell] \geq \mathbb{E}[\mathbb{V}\text{ar}\left[\mathbb{E}\left(\tilde{s}_{it+1} \mid i, \ell\right) \mid i\right]].$$

This provides a lower bound.

I thus set λ so that the variance of $\bar{y}_\ell$ lies within the bounds. Within these bounds, the average learning spillovers is always positive and statistically different from zero at the 5% (Figure J.5). Furthermore, the estimated φ is moderately increasing in the dispersion of $\bar{y}_\ell$ i.e., decreasing in λ. I pick the most conservative estimate, and set λ so that the variance of $\bar{y}_\ell$ is equal to its lower bound. With heterogeneous spillovers, I compute the bounds and determine the optimal λ separately for each skill quintile.

Learning technology: controls. Column 4 in Table 1 controls for school quality by birthplace, and column 5 adds industry fixed effects. I detail how both columns are implemented, starting with school quality. Here as well, I focus on the simpler case of $\xi = 0$ to simplify the presentation.

[31] I also operationzlie (87) by discretizing the skill distribution into quintiles. However, the transition matrix needed for the estimation of ξ is less noisy than that for φ because it is not a spatial transition matrix. In such a case, the linear system can be directly inverted.

Suppose the distribution of abilities vary by birthplaces according to:

$$(s_{i0}, x_i) \mid \ell_i^b = l \sim \mathcal{N}\left(\begin{bmatrix} \tau_\ell^s - \bar{\tau}^s \\ \boldsymbol{\gamma}'\boldsymbol{X}_\ell + \tau_\ell^x \end{bmatrix}, \boldsymbol{\Sigma}\right).$$

$\boldsymbol{\gamma}'\boldsymbol{X}_\ell$ denote observable local learning abilities, and τ_ℓ^x unobserved local learning abilities. Let $\tilde{\tau}_\ell^x \equiv \boldsymbol{\gamma}'\boldsymbol{X}_\ell + \tau_\ell^x$. Then, the birthplace-conditional mean is

$$\mathbb{E}[y_{it+1} \mid \ell_i^b = l] = \sum_\ell (\theta_\ell + \tilde{\tau}_l^x) \Pr[\ell_{it} = \ell \mid \ell_i^b = l],$$

where $\theta_\ell = \mathbb{E}_\ell[\varphi(s_{it})]\bar{s}_\ell + \tilde{\mu}_\ell$ is our object of interest. Accordingly, we identify $\hat{\boldsymbol{\theta}} = \boldsymbol{\theta} + \boldsymbol{P}^{-1}(\boldsymbol{\gamma}'\boldsymbol{X} + \boldsymbol{\tau}^x)$. This expression tells us how to control for observed differences in learning abilities across birthplacs. Given a vector of controls $\boldsymbol{X}$ by birthplace, the workplace-level controls used in (27) are $\boldsymbol{P}^{-1}\boldsymbol{X}$. The assumption is then that our shift-share instrument is orthogonal to $\boldsymbol{P}^{-1}\boldsymbol{\tau}^x$.

Moving on to industries, suppose the law of motion (25) features industry effects ($\boldsymbol{\xi}$):

$$\tilde{s}_{it+1} = \varphi\bar{s}_{\ell_{it}} + \tilde{\mu}_{\ell_{it}} + \tilde{\xi}_{\iota_{it}} + \tilde{x}_i + \tilde{e}_{it+1} + \nu_{it+1},$$

where ι_{it} is the industry in which i works at time t. We asume $\mathbb{E}[e_{it+1} \mid \iota] = \mathbb{E}[\nu_{it+1} \mid \iota] = 0$, but allow for correlation between industry and location choices, as well as between learning abilities and industry choices. Averaging across birthplaces and industries:

$$\mathbb{E}[\tilde{s}_{it+1} \mid \ell_i^b = l, \iota] = \sum_\ell \Pr[\ell \mid \ell_i^b = l, \iota_{it} = \iota]\theta_\ell + \mathbb{E}[\tilde{x}_i \mid \iota] + \tilde{\xi}_\iota.$$

Meanwhile, the industry average is

$$\mathbb{E}[\tilde{s}_{it+1} \mid \iota] = \sum_\ell \Pr[\ell \mid \iota_{it} = \iota]\theta_\ell + \mathbb{E}[\tilde{x}_i \mid \iota] + \tilde{\xi}_\iota.$$

Differencing the two:

$$\mathbb{E}[\tilde{s}_{it+1} \mid \ell_i^b = l, \iota] - \mathbb{E}[\tilde{s}_{it+1} \mid \iota] = \sum_\ell \Big(\Pr[\ell \mid \ell_i^b = l, \iota_{it} = \iota] - \Pr[\ell \mid \iota_{it} = \iota]\Big)\theta_\ell$$

Updating the migration matrix to $\Pr[\ell \mid \ell_i^b = l, \iota_{it} = \iota] - \Pr[\ell \mid \iota_{it} = \iota]$, we can then proceed as before.

Abilities and learning shocks. The identification of σ_x and σ_e rely on $\mathbb{E}_i[\tilde{e}_{it+1}] = 0$. This holds for a lifecyle with $A \to \infty$. In our case, $A = 6$, and the asymptotic may not kick in. Separately estimating (σ_e, σ_x) when T is small is trickier. In fact, for $T = 2$, the two components are not separately identified. However, this discussion also makes clear that separately estimating the two is not of central importance to our counterfactuals.[32] In practice, I estimate σ_x from the data by restricting my attention to workers with $A \geq 3$, and I calibrate σ_e via indirect inference to match in the model the empirical aggregate skill variance (see also Appendix F.3).

[32] In particular, the learning technology is independent of the precision of the estimates of $(\sigma_\varepsilon, \sigma_x)$.

F.3 Indirect inference

As explained in Section 4.3, the remaining parameters — $\{B_{a\ell}\}_{\ell,a<A}$, $\{\zeta_a\}_{a<A}$, $\{\mu_\ell\}_\ell$, — are calibrated using indirect inference. In addition, we also calibrate the variance of learning shocks by indirect inference to deal with measurement errors in panels with small T. Lastly, since the model is over-identified, we also calibrate the level of the function ξ. I calibrate these parameters jointly by targeting the following moments:

1. Age-specific employment share of each workplace, $\{N_{a\ell}\}_{\ell,a<A}$;
2. Age-specific migration rates, $\{\sum_{l,\ell\neq l} N_{al\ell}\}_{a<A}$;
3. Average skill growth by workplace, $\{\mathbb{E}_\ell[\log s'/s]\}_\ell$;
4. Average skill, $\mathbb{E}[s]$;
5. Log skill variance, $\mathbb{V}\text{ar}[\log s]$.

I compute these moments within the model, and minimize the distance between the vector of empirical moments and that of simulated data. The distance is measured in either percent or log points depending on the moments being targeted. I compute the equilibrium for each vector of guess parameters, and iterate over the parameters. I stop when the absolute difference is less than 5×10^{-3}.

G Welfare

Definition. I measure welfare in terms of lifetime income equivalence. That is, by how much would we have to increase a worker's income in every remaining period to keep them indifferent between two equilibria? Specifically, let $\bar{\cdot}$ denote variable $\cdot$ in the second equilibrium, and define

$$V_l^a(s,x,\boldsymbol{\varepsilon};t) = \max_\ell \left\{u_{l\ell}^a(s,x) + \varepsilon_\ell + t + \beta O_\ell^a(s,x;t)\right\}, \tag{90}$$

$$O_\ell^a(s,x;t) = \int\int \mathbb{E}_{\boldsymbol{\varepsilon}'}\left[V_\ell^{a+1}\left(ex\mu_\ell\gamma(s,s_p),x,\boldsymbol{\varepsilon}';t\right)\right] dF(e)\pi_\ell(s_p)ds_p, \tag{91}$$

subject to the terminal condition $V_l^A(s,x,\boldsymbol{\varepsilon};t) = \max_\ell\left\{u_{l\ell}^A(s,x) + \varepsilon_\ell + t\right\}$. For a worker with characteristics (l,a,s,x), define $t_l^a(s,x)$ by

$$\mathbb{E}_{\boldsymbol{\varepsilon}}\left[V_l^a\left(s,x,\boldsymbol{\varepsilon};t_l^a(s,x)\right)\right] = \mathbb{E}_{\boldsymbol{\varepsilon}}\left[\bar{V}_l^a(s,x,\boldsymbol{\varepsilon};0)\right]. \tag{92}$$

$t_l^a(s,x)$ is the extra income that must be added in every remaining period in the first equilibrium to make the worker indifferent between the two regimes. Importantly, since t is independent of the destination location and enters additively in (90), it not does not affect the equilibrium.

Properties. This metric has several useful properties. First, it enters lifetime utility linearly. Define $\chi_a \equiv \sum_{h=0}^{A-a}\beta^h$. At the terminal age, $V_l^A(s,x,\boldsymbol{\varepsilon};t) = V_l^A(s,x,\boldsymbol{\varepsilon};0) + t$. Suppose then that $V_\ell^{a+1}(s',x',\boldsymbol{\varepsilon}';t) = V_\ell^{a+1}(s',x',\boldsymbol{\varepsilon}';0) + \chi_{a+1}t$. It then follows from (91) that $O_\ell^a(s,x;t) = O_\ell^a(s,x;0) + \chi_{a+1}t$. Substituting this expression into (90) yields $V_l^a(s,x,\boldsymbol{\varepsilon};t) = V_l^a(s,x,\boldsymbol{\varepsilon};0) + \chi_a t$, where $\chi_a = 1 + \beta\chi_{a+1}$. Hence, $\mathbb{E}_{\boldsymbol{\varepsilon}}\left[V_l^a(s,x,\boldsymbol{\varepsilon};t)\right] = \mathbb{E}_{\boldsymbol{\varepsilon}}\left[V_l^a(s,x,\boldsymbol{\varepsilon};0)\right] + \chi_a t$.

Second, given linearity, the metric equal to the scaled changes in expected lifetime utility:

$$t_l^a(s,x) = \frac{\mathbb{E}_{\boldsymbol{\varepsilon}}\left[\bar{V}_l^a(s,x,\boldsymbol{\varepsilon};0) - V_l^a(s,x,\boldsymbol{\varepsilon};0)\right]}{\chi_a}. \tag{93}$$

Third, linearity also implies that the welfare metric aggregates. For any Borel set $\mathcal{A}$ of characteristics, define $t(\mathcal{A})$ by $\mathbb{E}\left[V_l^a(s,x,\boldsymbol{\varepsilon};t(\mathcal{A}))\mid\mathcal{A}\right]=\mathbb{E}\left[\bar{V}_l^a(s,x,\boldsymbol{\varepsilon};0)\mid\mathcal{A}\right]$. Then,

$$t(\mathcal{A})=\frac{\mathbb{E}\left[\chi_a t_l^a(s,x)\mid\mathcal{A}\right]}{\mathbb{E}\left[\chi_a\mid\mathcal{A}\right]}. \tag{94}$$

Benchmarking. I express lifetime income equivalent welfare changes in terms of aggregate lifetime income. Specifically, define the net present value of aggregate lifetime income as $I\equiv\mathbb{E}\left[\sum_{a=1}^{A}\beta^a\mathcal{W}_a\right]$. I report in the main text $t(\mathcal{A})/I$ for various $\mathcal{A}$ that only contains new entrants.

H Data Sources and Sample Construction

H.1 Déclaration Annuelle de Données Sociales

The primary data source is the Déclaration Annuelle de Données Sociales (DADS). The DADS is a declaration procedure which must be completed by all businesses that employ staff. This document serves both fiscal and social administrative purposes, and contains information pertaining to their establishments and employees, annually and for every establishment. All employers and their employees are covered by the DADS declaration.

The French statistical agency (*INSEE*) transforms these documents into datasets. The most granular of these (DADS Postes) is at the establishment × worker × year level, and contains information on both the establishments and the worker. The DADS Postes is a short panel: for each year t and each worker i, the dataset report the information for (i,t) and $(i,t-1)$. Worker IDs are reshuffled every year.

Babet et al. (2025) develop an algorithm that turns the short panel into a long panel by matching workers on observable characteristics over time. Each year, workers are matched on sex, firm and establishment identifier, hours worked, start and end date, job duration, and municipality of residence. Among observations with identical characteristics, the end match is generated from the smallest absolute difference in gross wages. Non-unique matches are discarded. Multiple matches are seldom (0.4% of observations) and the algorithm manages to match more than 98% of observations.

I use the algorithm they provide. I define hourly wages as gross earnings over hours worked, and express all nominal variables in 2018 euros. I apply following sample criterion:

1. Workers employed and living in metropolitan France;
2. Age between 25 and 55;
3. Private sector;
4. Not employed in agriculture, education, and health;
5. Full-time contract with less than 12 hours per day worked;
6. Hourly gross wage higher than 80% of the minimum wage;
7. Hourly gross wage less than top 0.1% each year;
8. Never unemployed for more than 3 consecutive years.

I then aggregate the dataset by age groups of 5 years. Within each age group, I average continuous variables, and I aggregate categorical variable by their modal value. For categorical variables with ties, I take the observations associated with the highest hourly wage. I compute observation weights so that the age distribution is uniform, as in the model. Finally, I normalize nominal variables by the average wage.

All labor market statistics are computed using the final dataset.

H.2 Other Data Sources

Housing prices. I measure residential housing prices via rents. I obtain data on rents from the "Carte des Loyers" dataset (Rental Map). The dataset covers the 2011 — 2019 period. It provides the average rent per square meter at the municipal level after controlling for housing characteristics. I compute municipal prices as the average between apartment and home rental, weighted by each unit market share. I compute commuting zone residential housing prices as the population-weighted average of the municipal prices.

Distance. Distance is measured as the driving distance in kilometers between the most populous municipality within each commuting zone. Driving distances are obtained from Open Street Map.

School quality. The school quality data are obtained from the French Ministry of Education. For elementary school, the data reports the percentage of students reaching a satisfactory mastery level in French and mathematics as tested at the beginning of the school year. The data covers all grades (1st to 5th grade). For middle school and high school, the data reports the mean score in French and mathematics, available for the 6th, 8th, and 10th grades. Both datasets are reported at the département level. I map département into commuting zone according to the département population share in each commuting zone. I complement these data with success rate at exams at the end middle school (Diplome National du Brevet) and high school (Baccalauréat). These statistics are at the school level, which I aggregate at the commuting zones as enrollment-weighted mean. All test scores are standardized.

I Additional Tables

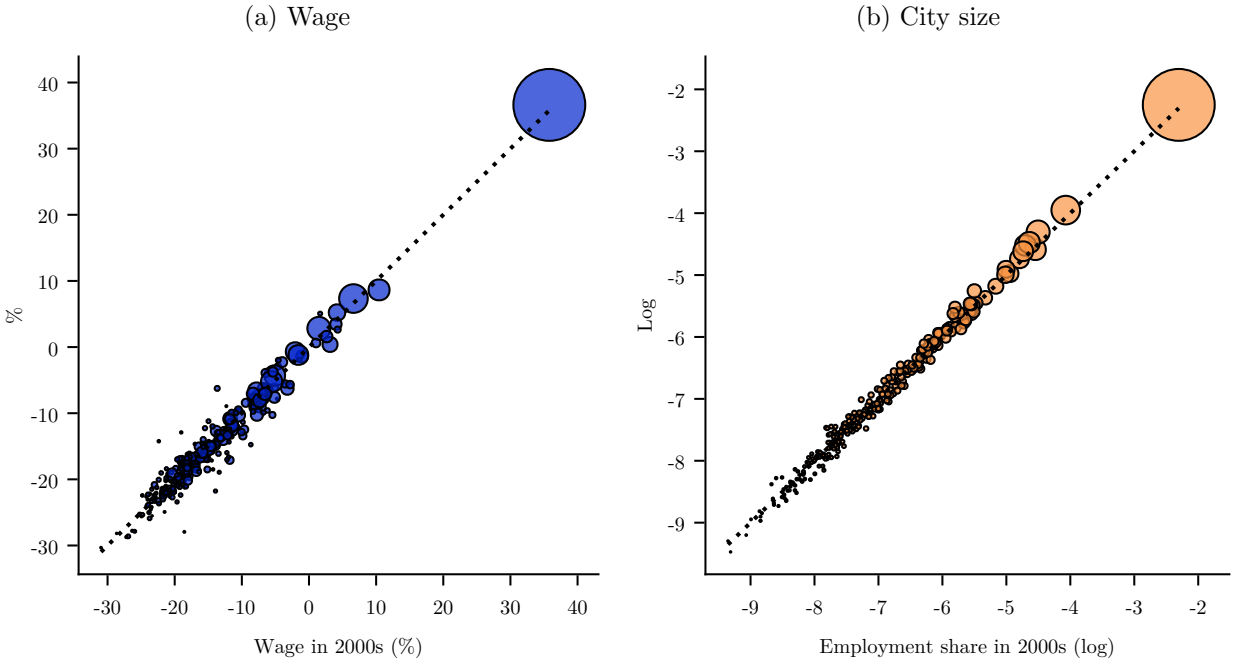
Table I.1: Summary statistics

	Weighted	Unweighted
Number of workers	12,792,847	12,792,847
Number of observations	2.75	2.77
Average age	38.9	38.5
Fraction of women (%)	35.2	34.2
Average wage (2018 euros)	20.3	20.4
Log wage standard deviation	0.43	0.43
Migration rate (%)	12.2	12.6

Data source: DADS Postes, cleaned and aggregated as detailed in Section 3.1. Weights constructed to yield uniform age distribution and synthetic balanced panel.

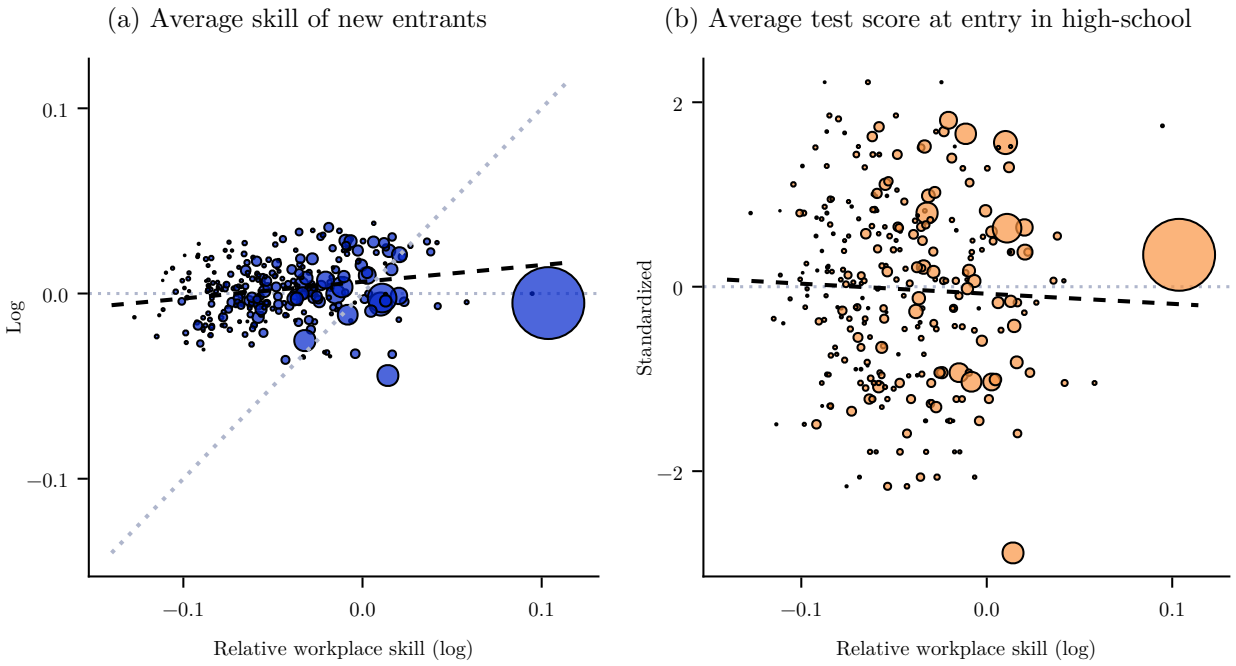
J Additional Figures

Figure J.1: Economic activity over space and time



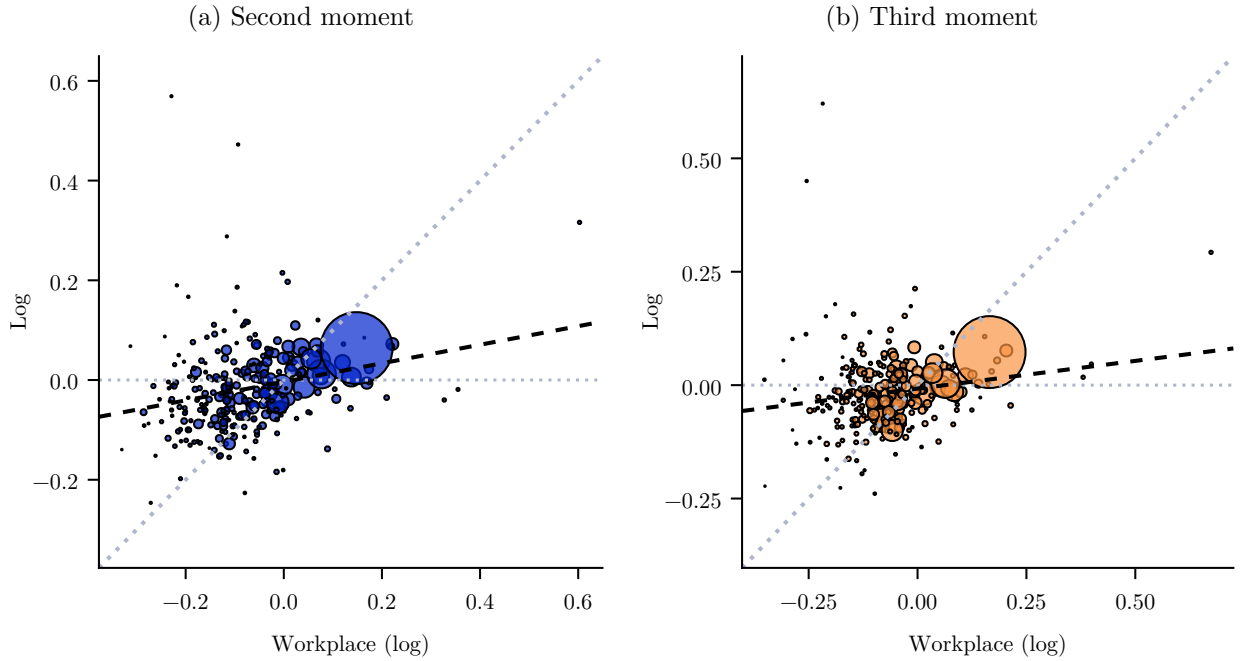
Both panels display statistics computed in the 2000s on the x-axis versus the same metric in the 2010s on the y-axis. Dotted black line is the 45 degree line. Left panel computes local relative average wages. Right panel computes local employment share. The size of the markers is proportional to the city size.

Figure J.2: Birthplace characteristics



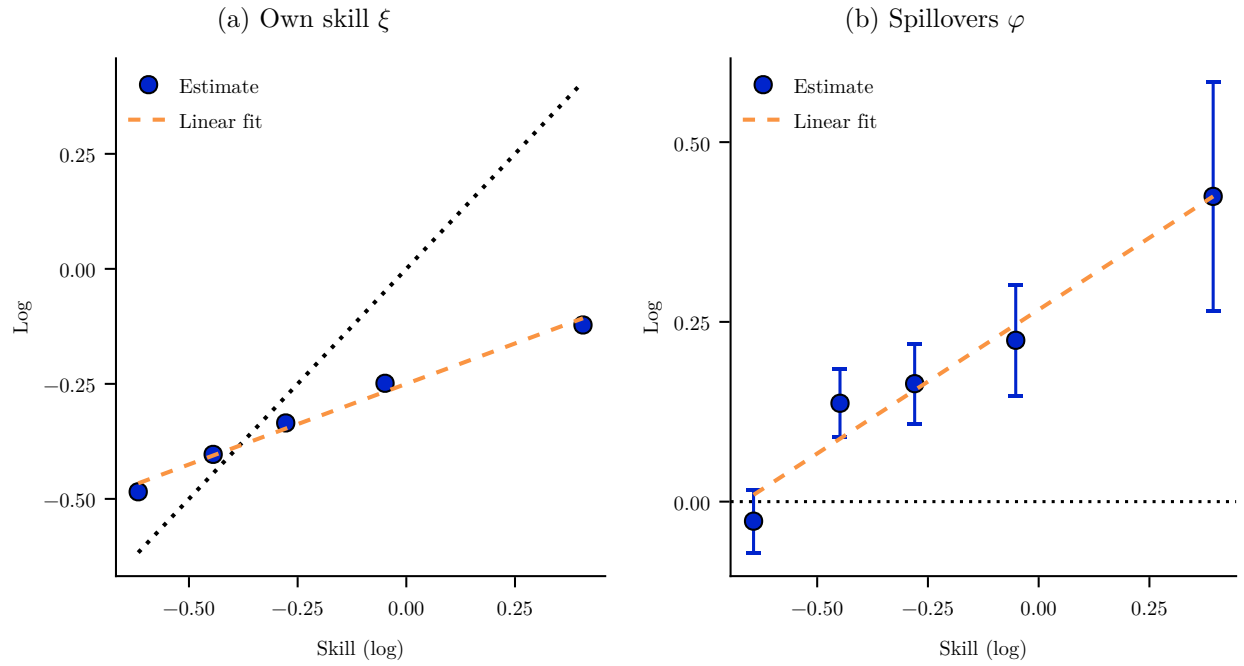
Left: average skill of workers under 30 by birthplace CZ. Right: average test score at the beginning of high-school (seconde grade) by birthplace CZ. Both statistics plotted against the relative average skill of workers employed in the CZ. Skills computed as wages deflated by local TFP. TFP estimated as detailed in Section 4.1. Test scores computed as explained in Section H. The size of the markers is proportional to the city size.

Figure J.3: Learning abilities across birthplaces



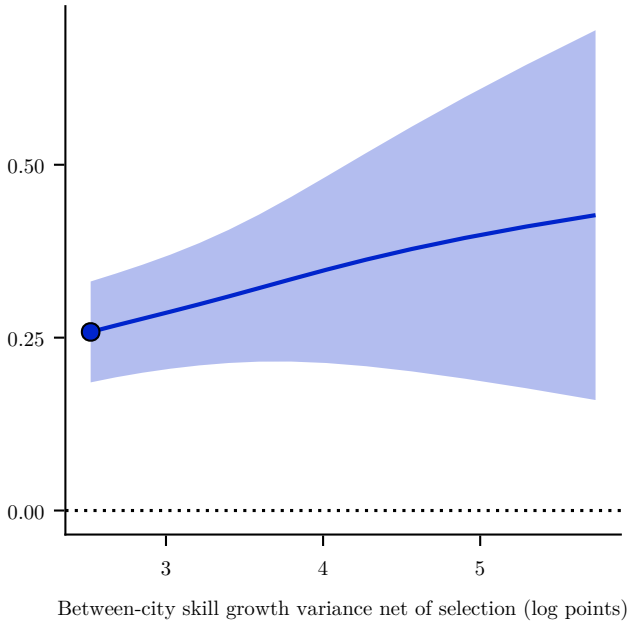
Left: second moment of the log learning abilities by birthplace (y-axis) and workplace (x-axis). Right: third moment of the log learning abilities by birthplace (y-axis) and workplace (x-axis). In both panels, learning abilities inferred from the model as detailed in Appendix F. Moments normalized and standardized by the aggregate second and third moments respectively. Grey dotted lines show 45% degree line. Dashed black line display linear fit. The size of the markers is proportional to the size size of the workplace.

Figure J.4: Learning technology



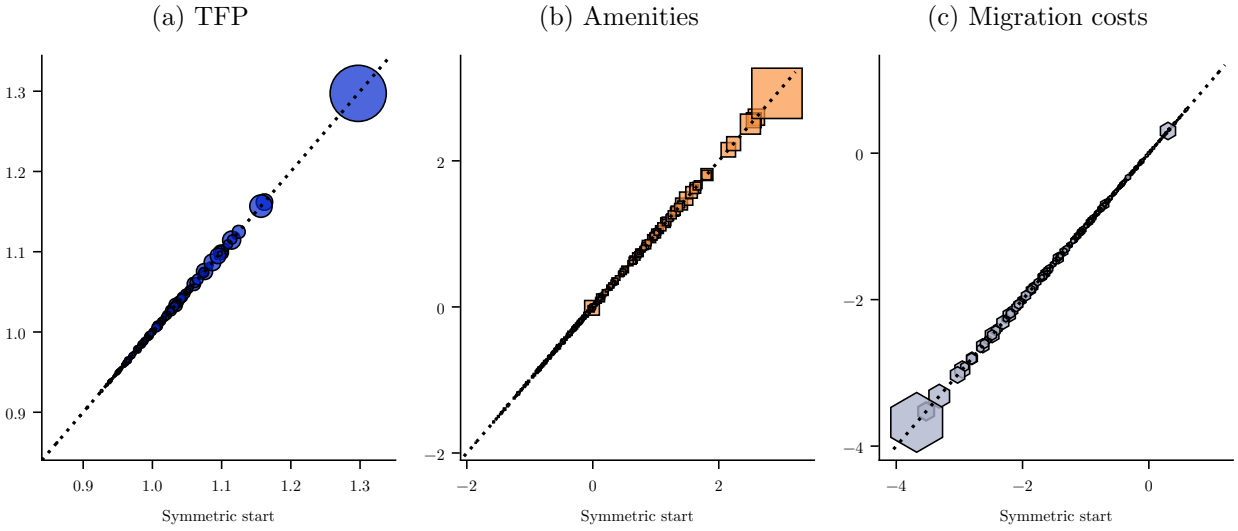
Left: non-parametric estimates of the own-skill function ξ in the learning technology γ. Estimation detailed in Appendix F. Dashed orange line shows the linear approximation $\xi(s) = \xi_0 + \xi_1 \log s$ used in the model. Dotted black line is 45 degree line. Right: non-parametric estimates of the spillover function ξ in the learning technology γ. Estimation detailed in Section 4.2. Dashed orange line shows linear approximation $\varphi(s) = \varphi_0 + \varphi_1 \log s$ used in the model. Both sets of estimates estimated by skill quintile, and projected against the average log skill of each quintile.

Figure J.5: Average learning spillovers



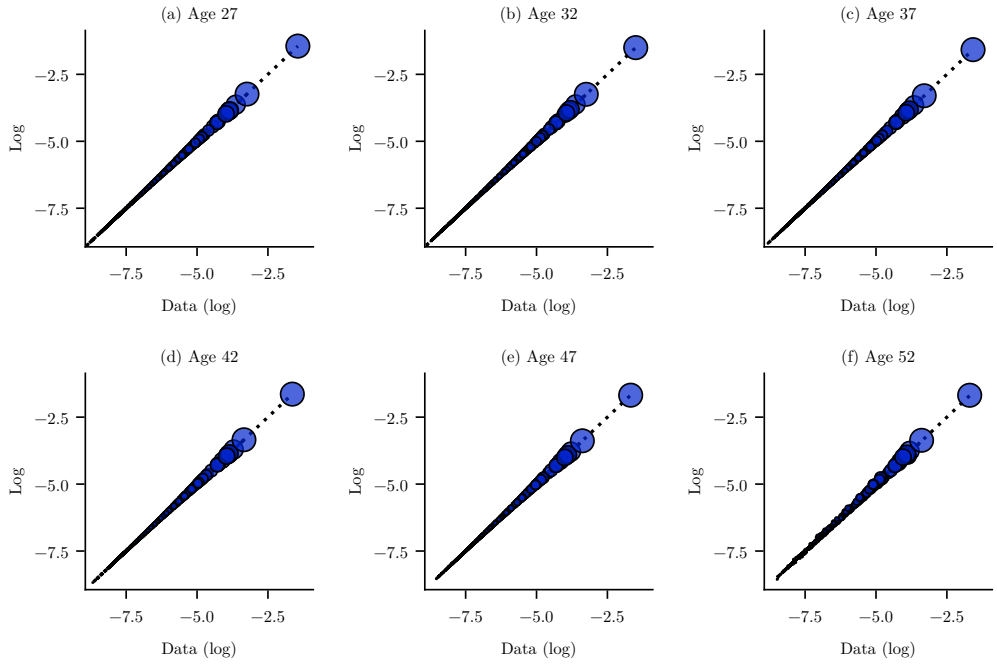
Average learning spillover $\mathbb{E}[\varphi(s_{it})]$ as a function of the underlying between-city variance in skill growth net of selection. Skill growth net of selection inferred from the birthplace design (Section 4.2). Inverion performed by Tikhonov regularization as explained in Appendix F.2. Different penalty parameter λ generate different between-city variance. Between-city variance bounded as explained in Appendix F.2. Blue circle refers to the baseline point estimate. Shaded blue region shows 95% confidence interval.

Figure J.6: Local fundamentals



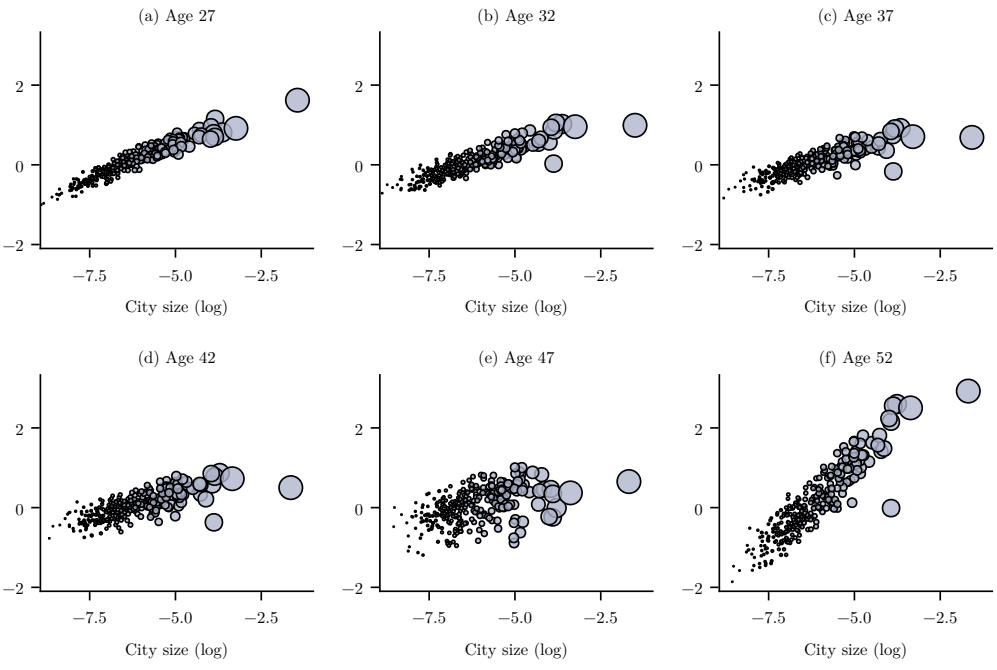
All panels display the local fundamentals estimated using the fixed point algorithm of Section 4.1. x- and y-axis present the fundamentals with the homogeneous and asymmetric TFP starts, respectively. (a): local TFP, mean normalized to unity. (b): amenities for oldest workers, mean normalized to zero. (c): origin-specific migration costs. The size of the markers is proportional to the city size.

Figure J.7: Model fit: city size by age



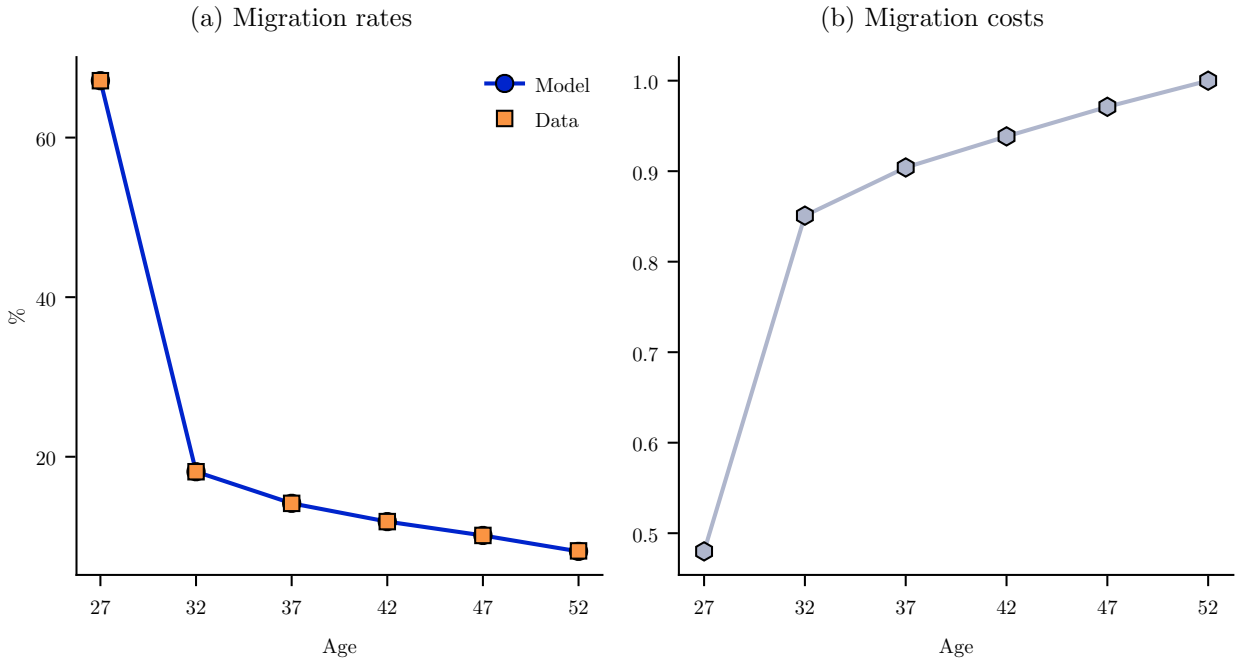
All panels display the employment share of each city by age group. x-axis and y-axis are the employment share in the data and model, respectively. The size of the markers is proportional to the city size.

Figure J.8: Local amenities by age



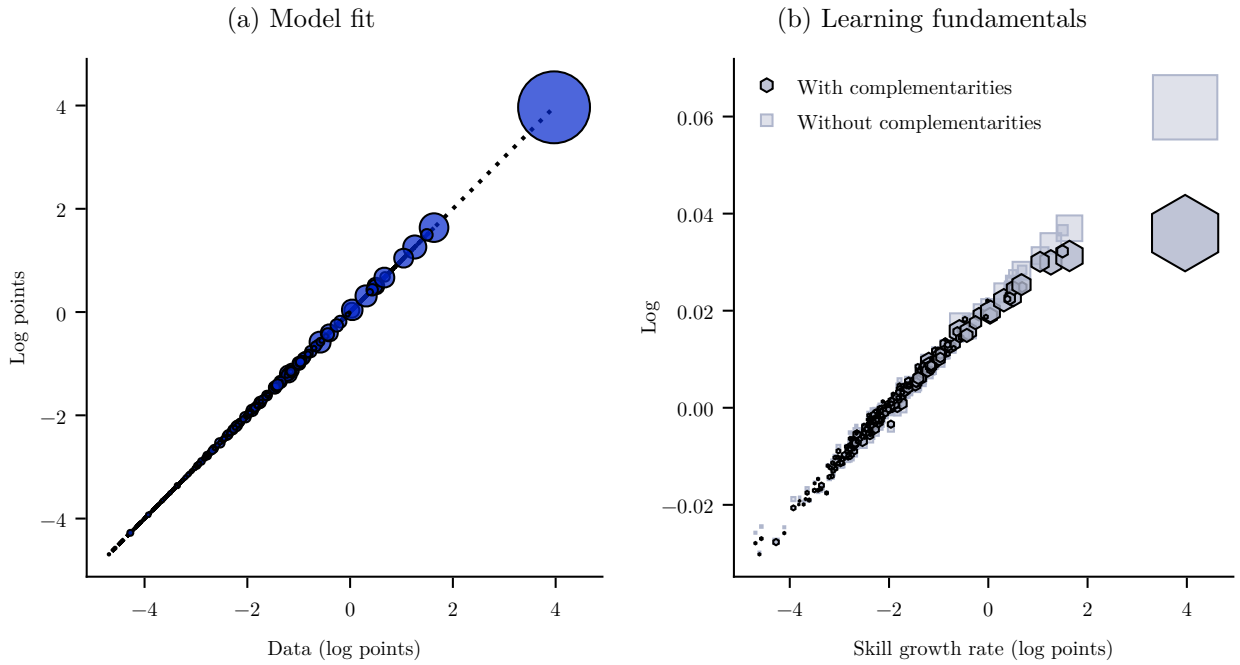
All panels display the local amenities against cities' employment share, by age group. Amenities are estimated as detailed in Section 4.1 for the oldest workers, and following Section F for the other age groups. The size of the markers is proportional to the city size.

Figure J.9: Migrations by age



Left: migration rates by age group, in the data (orange square) and in the model (blue circles). Right: relative migration costs ($\boldsymbol{\zeta}$). Migration costs normalized to unity for the oldest workers, and calibrated following Section F for the other age groups.

Figure J.10: Local learning fundamentals



Left: skill growth rate by location, in the data on the x-axis and in the model on the y-axis. Right: local learning fundamentals against the average skill growth. Learning fundamentals calibrated following Section F with mean normalized to unity. (log) mean normalized to zero. The size of the markers is proportional to the city size.